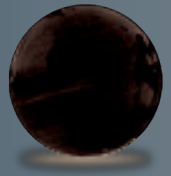


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Econometric Analysis

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William H. Greene



EIGHTH EDITION
GLOBAL EDITION
ECONOMETRIC ANALYSIS



William H. Greene

The Stern School of Business

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Authorized adaptation from the United States edition, entitled *Econometric Analysis*, 8th Edition, ISBN 978-0-13-446136-6 by William H. Greene, published by Pearson Education © 2018.

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British Library Cataloguing-in-Publication Data

A catalogue record for this book is available from the British Library

ISBN 10: 1-292-23113-0

ISBN 13: 978-1-292-23113-6

eBook ISBN 13: 978-1-292-23115-0

Typeset in Times Ten LT Std by SPi Global

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PREFACE



ECONOMETRIC ANALYSIS

Econometric Analysis is a broad introduction to the field of econometrics. This field grows continually. A (not complete) list of journals devoted at least in part to econometrics now includes: *Econometric Reviews*; *Econometric Theory*; *Econometrica*; *Econometrics*; *Econometrics and Statistics*; *The Econometrics Journal*; *Empirical Economics*; *Foundations and Trends in Econometrics*; *The Journal of Applied Econometrics*; *The Journal of Business and Economic Statistics*; *The Journal of Choice Modelling*; *The Journal of Econometric Methods*; *The Journal of Econometrics*; *The Journal of Time Series Analysis*; *The Review of Economics and Statistics*. Constructing a textbook-style survey to introduce the topic at a graduate level has become increasingly ambitious. Nonetheless, that is what I seek to do here. This text attempts to present, at an entry graduate level, enough of the topics in econometrics that a student can comfortably move on from here to practice or to more advanced study. For example, the literature on “Treatment Effects” is already vast, rapidly growing, complex in the extreme, and occasionally even contradictory. But, there are a few bedrock principles presented in Chapter 8 that (I hope) can help the interested practitioner or student get started as they wade into this segment of the literature. The book is intended as a bridge between an introduction to econometrics and the professional literature.

The book has two objectives. The first is to introduce students to *applied econometrics*, including basic techniques in linear regression analysis and some of the rich variety of models that are used when the linear model proves inadequate or inappropriate. Modern software has made complicated modeling very easy to put into practice. The second objective is to present sufficient *theoretical background* so that the reader will (1) understand the advanced techniques that are made so simple in modern software and (2) recognize new variants of the models learned about here as merely natural extensions that fit within a common body of principles. This book contains a substantial amount of theoretical material, such as that on the GMM, maximum likelihood estimation, and asymptotic results for regression models.

One overriding purpose has motivated all eight editions of *Econometric Analysis*. The vast majority of readers of this book will be users, not developers, of econometrics. I believe that it is not sufficient to teach econometrics by reciting (and proving) the theories of estimation and inference. Although the often-subtle theory is extremely important, the application is equally crucial. To that end, I have provided hundreds of worked numerical examples and extracts from applications in the received empirical literature in many fields. My purpose in writing this work, and in my continuing efforts to update it, is to show readers how to *do* econometric analysis. But, I also believe that readers want (and need) to know what is going on behind the curtain when they use ever more sophisticated modern software for ever more complex econometric analyses.

I have taught econometrics at the level of *Econometric Analysis* at NYU for many years. I ask my students to learn how to use a (any) modern econometrics program as part of their study. I've lost track of the number of my students who recount to me their disappointment in a previous course in which they were taught how to use software, but not the theory and motivation of the techniques. In October, 2014, Google Scholar published its list of the 100 most cited works over all fields and all time. (www.nature.com/polopoly_fs/721245!/file/GoogleScholartop100.xlsx). *Econometric Analysis*, the only work in econometrics on the list, ranked number 34 with 48,100 citations. (As of this writing, November 2016, the number of citations to the first 7 editions in all languages approaches 60,000.) I take this extremely gratifying result as evidence that there are readers in many fields who agree that the practice of econometrics calls for an understanding of *why*, as well as *how* to use the tools in modern software. This book is for them.

THE EIGHTH EDITION OF *ECONOMETRIC ANALYSIS*

This text is intended for a one-year graduate course for social scientists. Prerequisites should include calculus, mathematical statistics, and an introduction to econometrics at the level of, say, Gujarati and Porter's (2011) *Basic Econometrics*, Stock and Watson's (2014) *Introduction to Econometrics*, Kennedy's (2008) *Guide to Econometrics*, or Wooldridge's (2015) *Introductory Econometrics: A Modern Approach*. I assume, for example, that the reader has already learned about the basics of econometric methodology including the fundamental role of economic and statistical assumptions; the distinctions between cross-section, time-series, and panel data sets; and the essential ingredients of estimation, inference, and prediction with the multiple linear regression model. Self-contained (for our purposes) summaries of the matrix algebra, mathematical statistics, and statistical theory used throughout the book are given in Appendices A through D. I rely heavily on matrix algebra throughout. This may be a bit daunting to some early on but matrix algebra is an indispensable tool and I hope the reader will come to agree that it is a means to an end, not an end in itself. With matrices, the unity of a variety of results will emerge without being obscured by a curtain of summation signs. Appendix E and Chapter 15 contain a description of numerical methods that will be useful to practicing econometricians (and to us in the later chapters of the book).

Estimation of advanced nonlinear models is now as routine as least squares. I have included five chapters on estimation methods used in current research and five chapters on applications in micro- and macroeconometrics. The nonlinear models used in these fields are now the staples of the applied econometrics literature. As a consequence, this book also contains a fair amount of material that will extend beyond many first courses in econometrics. Once again, I have included this in the hope of laying a foundation for study of the professional literature in these areas.

PLAN OF THE BOOK

The arrangement of the book is as follows:

Part I begins the formal development of econometrics with its fundamental pillar, the *linear multiple regression model*. Estimation and inference with the linear least squares estimator are analyzed in Chapters 2 through 6. The *nonlinear regression model* is introduced

in Chapter 7 along with quantile, semi- and nonparametric regression, all as extensions of the familiar linear model. *Instrumental variables estimation* is developed in Chapter 8.

Part II presents three major extensions of the regression model. Chapter 9 presents the consequences of relaxing one of the main assumptions of the linear model, homoscedastic nonautocorrelated disturbances, to introduce the *generalized regression model*. The focus here is on heteroscedasticity; autocorrelation is mentioned, but a detailed treatment is deferred to Chapter 20 in the context of time-series data. Chapter 10 introduces systems of regression equations, in principle, as the approach to modeling simultaneously a set of random variables and, in practical terms, as an extension of the generalized linear regression model. Finally, *panel data methods*, primarily fixed and random effects models of heterogeneity, are presented in Chapter 11.

The second half of the book is devoted to topics that extend the linear regression model in many directions. Beginning with Chapter 12, we proceed to the more involved methods of analysis that contemporary researchers use in analysis of “real-world” data. Chapters 12 to 16 in Part III present different estimation methodologies. Chapter 12 presents an overview by making the distinctions between *parametric*, *semiparametric* and *nonparametric methods*. The leading application of semiparametric estimation in the current literature is the *generalized method of moments (GMM) estimator* presented in Chapter 13. This technique provides the platform for much of modern econometrics. *Maximum likelihood estimation* is developed in Chapter 14. *Monte Carlo* and *simulation-based methods* such as *bootstrapping* that have become a major component of current research are developed in Chapter 15. Finally, *Bayesian methods* are introduced in Chapter 16.

Parts IV and V develop two major subfields of econometric methods, *microeconometrics*, which is typically based on cross-section and panel data, and *macroeconometrics*, which is usually associated with analysis of time-series data. In Part IV, Chapters 17 to 19 are concerned with models of *discrete choice*, *censoring*, *truncation*, *sample selection*, *duration* and the analysis of *counts of events*. In Part V, Chapters 20 and 21, we consider two topics in time-series analysis, models of *serial correlation* and regression models for *nonstationary data*—the usual substance of macroeconomic analysis.

REVISIONS

With only a couple exceptions noted below, I have retained the broad outline of the text. I have revised the presentation throughout the book (including this preface) to streamline the development of topics, in some cases (I hope), to improve the clarity of the derivations. Major revisions include:

- I have moved the material related to “causal inference” forward to the early chapters of the book – these topics are now taught earlier in the graduate sequence than heretofore and I’ve placed them in the context of the models and methods where they appear rather than as separate topics in the more advanced sections of the seventh edition. Difference in difference regression as a method, and regression discontinuity designs now appear in Chapter 6 with the discussion of functional forms and in the context of extensive applications extracted from the literature. The analysis of treatment effects has all been moved from Chapter 19 (on censoring and truncation) to Chapter 8 on endogeneity under the heading of “Endogenous

- Dummy Variables.” Chapter 8, as a whole, now includes a much more detailed treatment of instrumental variable methods.
- I have added many new examples, some as extracts from applications in the received literature, and others as worked numerical examples. I have drawn applications from many different fields including industrial organization, transportation, health economics, popular culture and sports, urban development and labor economics.
 - Chapter 10 on systems of equations has been shifted (yet further) from its early emphasis on formal simultaneous linear equations models to systems of regression equations and the leading application, the single endogenous variable in a two equation recursive model – this is the implicit form of the regression model that contains one “endogenous” variable.
 - The use of robust estimation and inference methods has been woven more extensively into the general methodology, in practice and throughout this text. The ideas of robust estimation and inference are introduced immediately with the linear regression model in Chapters 4 and 5, rather than as accommodations to nonspherical disturbances in Chapter 9. The role that a robust variance estimator will play in the Wald statistic is developed immediately when the result is first presented in Chapter 5.
 - Chapters 4 (Least Squares), 6 (Functional Forms), 8 (Endogeneity), 10 (Equation Systems) and 11 (Panel Data) have been heavily revised to emphasize both contemporary econometric methods and the applications.
 - I have moved Appendices A-F to the Companion Web site, at www.pearsonglobaleditions.com, that accompanies this text. Students can access them at no cost.

The first semester of study in a course based on *Econometric Analysis* would focus on Chapters 1-6 (the linear regression model), 8 (endogeneity and causal modeling), and possibly some of 11 (panel data). Most of the revisions in the eighth edition appear in these chapters.

SOFTWARE AND DATA

There are many computer programs that are widely used for the computations described in this book. All were written by econometricians or statisticians, and in general, all are regularly updated to incorporate new developments in applied econometrics. A sampling of the most widely used packages and Web sites where you can find information about them are

EViews	www.eviews.com	(QMS, Irvine, CA)
Gauss	www.aptech.com	(Aptech Systems, Kent, WA)
LIMDEP	www.limdep.com	(Econometric Software, Plainview, NY)
MATLAB	www.mathworks.com	(Mathworks, Natick, MA)
NLOGIT	www.nlogit.com	(Econometric Software, Plainview, NY)
R	www.r-project.org/	(The R Project for Statistical Computing)
RATS	www.estima.com	(Estima, Evanston, IL)
SAS	www.sas.com	(SAS, Cary, NC)
Shazam	econometrics.com	(Northwest Econometrics Ltd., Gibsons, Canada)
Stata	www.stata.com	(Stata, College Station, TX)

A more extensive list of computer software used for econometric analysis can be found at the resource Web site, <http://www.oswego.edu/~economic/econsoftware.htm>.

With only a few exceptions, the computations described in this book can be carried out with any of the packages listed. *NLOGIT* was used for the computations in most of the applications. This text contains no instruction on using any particular program or language. Many authors have produced *RATS*, *LIMDEP/NLOGIT*, *EViews*, *SAS*, or *Stata* code for some of the applications, including, in a few cases, in the documentation for their computer programs. There are also quite a few volumes now specifically devoted to econometrics associated with particular packages, such as Cameron and Trivedi's (2009) companion to their treatise on microeconometrics.

The data sets used in the examples are also available on the Web site for the text, <http://people.stern.nyu.edu/wgreene/Text/econometricanalysis.htm>. Throughout the text, these data sets are referred to "Table Fn.m," for example Table F4.1. The "F" refers to Appendix F available on the Companion web site which contains descriptions of the data sets. The actual data are posted in generic ASCII and portable formats on the Web site with the other supplementary materials for the text. There are now thousands of interesting Web sites containing software, data sets, papers, and commentary on econometrics. It would be hopeless to attempt any kind of a survey. One code/data site that is particularly agreeably structured and well targeted for readers of this book is the data archive for the *Journal of Applied Econometrics (JAE)*. They have archived all the nonconfidential data sets used in their publications since 1988 (with some gaps before 1995). This useful site can be found at <http://qed.econ.queensu.ca/jae/>. Several of the examples in the text use the *JAE* data sets. Where we have done so, we direct the reader to the *JAE*'s Web site, rather than our own, for replication. Other journals have begun to ask their authors to provide code and data to encourage replication. Another easy-to-navigate site for aggregate data on the U.S. economy is <https://datahub.io/dataset/economagic>.

ACKNOWLEDGMENTS

It is a pleasure to express my appreciation to those who have influenced this work. I remain grateful to Arthur Goldberger (dec.), Arnold Zellner (dec.), Dennis Aigner, Bill Becker, and Laurits Christensen for their encouragement and guidance. After eight editions of this book, the number of individuals who have significantly improved it through their comments, criticisms, and encouragement has become far too large for me to thank each of them individually. I am grateful for their help and I hope that all of them see their contribution to this edition. Any number of people have submitted tips about the text. You can find many of them listed in the errata pages on the text Web site, <http://people.stern.nyu.edu/wgreene/Text/econometricanalysis.htm>, in particular: David Hoaglin, University of Massachusetts; Randall Campbell, Mississippi State University; Carter Hill, Louisiana State University; and Tom Doan, Estima Corp. I would also like to thank two colleagues who have worked on translations of *Econometric Analysis*, Marina Turuntseva (the Russian edition) and Umit Senesen (the Turkish translation). I must also acknowledge the mail I've received from hundreds of readers and practitioners from the world over who have given me a view into topics and questions that practitioners are interested in, and have provided a vast trove of helpful material for my econometrics courses.

I also acknowledge the many reviewers of my work whose careful reading has vastly improved the book through this edition: Scott Atkinson, University of Georgia; Badi Baltagi, Syracuse University; Neal Beck, New York University; William E. Becker (Ret.), Indiana University; Eric J. Belasko, Texas Tech University; Anil Bera, University of Illinois; John Burkett, University of Rhode Island; Leonard Carlson, Emory University; Frank Chaloupka, University of Illinois at Chicago; Chris Cornwell, University of Georgia; Craig Depken II, University of Texas at Arlington; Frank Diebold, University of Pennsylvania; Edward Dwyer, Clemson University; Michael Ellis, Wesleyan University; Martin Evans, Georgetown University; Vahagn Galstyan, Trinity College Dublin; Paul Glewwe, University of Minnesota; Ed Greenberg, Washington University at St. Louis; Miguel Herce, University of North Carolina; Joseph Hilbe, Arizona State University; Dr. Uwe Jensen, Christian-Albrecht University; K. Rao Kadiyala, Purdue University; William Lott, University of Connecticut; Thomas L. Marsh, Washington State University; Edward Mathis, Villanova University; Mary McGarvey, University of Nebraska–Lincoln; Ed Melnick, New York University; Thad Mirer, State University of New York at Albany; Cyril Pasche, University of Geneva; Paul Ruud, University of California at Berkeley; Sherrie Rhine, Federal Deposit Insurance Corp.; Terry G. Seaks (Ret.), University of North Carolina at Greensboro; Donald Snyder, California State University at Los Angeles; Steven Stern, University of Virginia; Houston Stokes, University of Illinois at Chicago; Dmitrios Thomakos, Columbia University; Paul Wachtel, New York University; Mary Beth Walker, Georgia State University; Mark Watson, Harvard University; and Kenneth West, University of Wisconsin. My numerous discussions with Bruce McCullough of Drexel University have improved Appendix E and at the same time increased my appreciation for numerical analysis. I am especially grateful to Jan Kiviet of the University of Amsterdam, who subjected my third edition to a microscopic examination and provided literally scores of suggestions, virtually all of which appear herein. Professor Pedro Bacao, University of Coimbra, Portugal, and Mark Strahan of Sand Hill Econometrics and Umit Senesen of Istanbul Technical University did likewise with the sixth and seventh editions.

I would also like to thank the many people at Pearson Education who have put this book together with me: Adrienne D' Ambrosio, Neeraj Bhalla, Sugandh Juneja, and Nicole Suddeth and the composition team at SPi Global.

For over 25 years since the first edition, I've enjoyed the generous support and encouragement of many people, some close to me, especially my family, and many not so close. I'm especially grateful for the help, support and priceless encouragement of my wife, Sherrie Rhine, whose unending enthusiasm for this project has made it much less daunting, and much more fun.

William H. Greene
February 2017

ECONOMETRICS



1.1 INTRODUCTION

This book will present an introductory survey of econometrics. We will discuss the fundamental ideas that define the methodology and examine a large number of specific models, tools, and methods that econometricians use in analyzing data. This chapter will introduce the central ideas that are the paradigm of econometrics. Section 1.2 defines the field and notes the role that theory plays in motivating econometric practice. Sections 1.3 and 1.4 discuss the types of applications that are the focus of econometric analyses. The process of econometric modeling is presented in Section 1.5 with a classic application, Keynes's consumption function. A broad outline of the text is presented in Section 1.6. Section 1.7 notes some specific aspects of the presentation, including the use of numerical examples and the mathematical notation that will be used throughout the text.

1.2 THE PARADIGM OF ECONOMETRICS

In the first issue of *Econometrica*, the Econometric Society stated that its main object shall be to promote studies that aim at a unification of the theoretical-quantitative and the empirical-quantitative approach to economic problems and that are penetrated by constructive and rigorous thinking similar to that which has come to dominate the natural sciences. . . . But there are several aspects of the quantitative approach to economics, and no single one of these aspects taken by itself, should be confounded with econometrics. Thus, econometrics is by no means the same as economic statistics. Nor is it identical with what we call general economic theory, although a considerable portion of this theory has a definitely quantitative character. Nor should econometrics be taken as synonymous [sic] with the application of mathematics to economics. Experience has shown that each of these three viewpoints, that of statistics, economic theory, and mathematics, is a necessary, but not by itself a sufficient, condition for a real understanding of the quantitative relations in modern economic life. It is the *unification* of all three that is powerful. And it is this unification that constitutes econometrics.

The Society responded to an unprecedented accumulation of statistical information. It saw a need to establish a body of principles that could organize what would otherwise become a bewildering mass of data. Neither the pillars nor the objectives of econometrics have changed in the years since this editorial appeared. Econometrics concerns itself with the

application of mathematical statistics and the tools of statistical inference to the empirical measurement of relationships postulated by an underlying theory.

It is interesting to observe the response to a contemporary, likewise unprecedented accumulation of massive amounts of quantitative information in the form of “Big Data.” Consider the following assessment of what Kitchin (2014) sees as a paradigm shift in the analysis of data.

This article examines how the availability of Big Data, coupled with new data analytics, challenges established epistemologies across the sciences, social sciences and humanities, and assesses the extent to which they are engendering paradigm shifts across multiple disciplines. In particular, it critically explores new forms of empiricism that declare ‘the end of theory,’ the creation of data-driven rather than knowledge-driven science, and the development of digital humanities and computational social sciences that propose radically different ways to make sense of culture, history, economy and society. It is argued that: (1) Big Data and new data analytics are disruptive innovations which are reconfiguring in many instances how research is conducted; and (2) there is an urgent need for wider critical reflection within the academy on the epistemological implications of the unfolding data revolution, a task that has barely begun to be tackled despite the rapid changes in research practices presently taking place.

We note the suggestion that data-driven analytics are proposed to replace theory (and econometrics as envisioned by Frisch) for providing the organizing principles to guide empirical research. (We will examine an example in Chapter 18 where we consider analyzing survey data with ordered choice models. Also, see Varian (2014) for a more balanced view.) The focus is driven partly by the startling computational power that would have been unavailable to Frisch. It seems likely that the success of this new paradigm will turn at least partly on the questions pursued. Whether the interesting features of an underlying data-generating process can be revealed by appealing to the data themselves without a theoretical platform seems to be a prospect raised by the author. The article does focus on the role of an underlying theory in empirical research—this is a central pillar of econometric methodology. As of this writing, the success story of Big Data analysis is still being written.

The crucial role that econometrics plays in economics has grown over time. The Nobel Prize in Economics has recognized this contribution with numerous awards to econometricians, including the first which was given to (the same) Ragnar Frisch in 1969. Lawrence Klein in 1980, Trygve Haavelmo in 1989, James Heckman and Daniel McFadden in 2000, Robert Engle and Clive Granger in 2003. Christopher Sims in 2011 and Lars Hansen in 2013 were recognized for their empirical research. The 2000 prize was noteworthy in that it celebrated the work of two scientists whose research was devoted to the marriage of behavioral theory and econometric modeling.

Example 1.1 *Behavioral Models and the Nobel Laureates*

The pioneering work by both James Heckman and Dan McFadden rests firmly on a theoretical foundation of utility maximization.

For Heckman’s, we begin with the standard theory of household utility maximization over consumption and leisure. The textbook model of utility maximization produces a demand for leisure time that translates into a supply function of labor. When home production (i.e., work

in the home as opposed to the outside, formal labor market) is considered in the calculus, then desired *hours* of (formal) labor can be negative. An important conditioning variable is the *reservation wage*—the wage rate that will induce formal labor market participation. On the demand side of the labor market, we have firms that offer market wages that respond to such attributes as age, education, and experience. What can we learn about labor supply behavior based on observed market wages, these attributes, and observed hours in the formal market? Less than it might seem, intuitively because our observed data omit half the market—the data on formal labor market activity are not randomly drawn from the whole population.

Heckman's observations about this implicit truncation of the distribution of hours or wages revolutionized the analysis of labor markets. Parallel interpretations have since guided analyses in every area of the social sciences. The analysis of policy interventions such as education initiatives, job training and employment policies, health insurance programs, market creation, financial regulation, and a host of others is heavily influenced by Heckman's pioneering idea that when participation is part of the behavior being studied, the analyst must be cognizant of the impact of common influences in both the presence of the intervention and the outcome. We will visit the literature on sample selection and treatment/program evaluation in Chapters 5, 6, 8 and 19.

Textbook presentations of the theories of demand for goods that produce utility, because they deal in continuous variables, are conspicuously silent on the kinds of discrete choices that consumers make every day—what brand of product to choose, whether to buy a large commodity such as a car or a refrigerator, how to travel to work, whether to rent or buy a home, where to live, what candidate to vote for, and so on. Nonetheless, a model of *random utility* defined over the alternatives available to the consumer provides a theoretically sound platform for studying such choices. Important variables include, as always, income and relative prices. What can we learn about underlying preference structures from the discrete choices that consumers make? What must be assumed about these preferences to allow this kind of inference? What kinds of statistical models will allow us to draw inferences about preferences? McFadden's work on how commuters choose to travel to work, and on the underlying theory appropriate to this kind of modeling, has guided empirical research in discrete consumer choices for several decades. We will examine McFadden's models of discrete choice in Chapter 18.

1.3 THE PRACTICE OF ECONOMETRICS

We can make a useful distinction between *theoretical econometrics* and *applied econometrics*. Theorists develop new techniques for estimation and hypothesis testing and analyze the consequences of applying particular methods when the assumptions that justify those methods are not met. Applied econometricians are the users of these techniques and the analysts of data (real world and simulated). The distinction is far from sharp; practitioners routinely develop new analytical tools for the purposes of the study that they are involved in. This text contains a large amount of econometric theory, but it is directed toward applied econometrics. We have attempted to survey techniques, admittedly some quite elaborate and intricate, that have seen wide use in the field.

Applied econometric methods will be used for estimation of important quantities, analysis of economic outcomes such as policy changes, markets or individual behavior, testing theories, and for forecasting. The last of these is an art and science in itself that is the subject of a vast library of sources. Although we will briefly discuss some

aspects of forecasting, our interest in this text will be on estimation and analysis of models. The presentation, where there is a distinction to be made, will contain a blend of microeconomic and macroeconomic techniques and applications. It is also necessary to distinguish between *time-series analysis* (which is not our focus) and methods that primarily use *time-series data*. The former is, like forecasting, a growth industry served by its own literature in many fields. While we will employ some of the techniques of time-series analysis, we will spend relatively little time developing first principles.

1.4 MICROECONOMETRICS AND MACROECONOMETRICS

The connection between underlying behavioral models and the modern practice of econometrics is increasingly strong. Another distinction is made between *microeconomics* and *macroeconomics*. The former is characterized by its analysis of cross section and panel data and by its focus on individual consumers, firms, and micro-level decision makers. Practitioners rely heavily on the theoretical tools of microeconomics including utility maximization, profit maximization, and market equilibrium. The analyses are directed at subtle, difficult questions that often require intricate formulations. A few applications are as follows:

- What are the likely effects on labor supply behavior of proposed negative income taxes? [Ashenfelter and Heckman (1974)]
- Does attending an elite college bring an expected payoff in expected lifetime income sufficient to justify the higher tuition? [Kreuger and Dale (1999) and Kreuger (2000)]
- Does a voluntary training program produce tangible benefits? Can these benefits be accurately measured? [Angrist (2001)]
- Does an increase in the minimum wage lead to reduced employment? [Card and Krueger (1994)]
- Do smaller class sizes bring real benefits in student performance? [Hanushek (1999), Hoxby (2000), and Angrist and Lavy (1999)]
- Does the presence of health insurance induce individuals to make heavier use of the health care system—is moral hazard a measurable problem? [Riphahn et al. (2003)]
- Did the intervention addressing anticompetitive behavior of a group of 50 boarding schools by the UK Office of Fair Trading produce a measurable impact on fees charged? [Pesaresi, Flanagan, Scott, and Tragear (2015)]

Macroeconomics is involved in the analysis of time-series data, usually of broad aggregates such as price levels, the money supply, exchange rates, output, investment, economic growth, and so on. The boundaries are not sharp. For example, an application that we will examine in this text concerns spending patterns of municipalities, which rests somewhere between the two fields. The very large field of financial econometrics is concerned with long time-series data and occasionally vast panel data sets, but with a sharply focused orientation toward models of individual behavior. The analysis of market returns and exchange rate behavior is neither exclusively macro- nor microeconomic. [We will not be spending any time in this text on financial econometrics. For those with an interest in this field, we would recommend the celebrated work by Campbell, Lo, and Mackinlay (1997), or for a more time-series-oriented approach, Tsay (2005).]

Macroeconomic model builders rely on the interactions between economic agents and policy makers. For example:

- Does a monetary policy regime that is strongly oriented toward controlling inflation impose a real cost in terms of lost output on the U.S. economy? [Cecchetti and Rich (2001)]
- Did 2001's largest federal tax cut in U.S. history contribute to or dampen the concurrent recession? Or was it irrelevant?

Each of these analyses would depart from a formal model of the process underlying the observed data.

The techniques used in econometrics have been employed in a widening variety of fields, including political methodology, sociology,¹ health economics, medical research (e.g., how do we handle attrition from medical treatment studies?) environmental economics, economic geography, transportation engineering, and numerous others. Practitioners in these fields and many more are all heavy users of the techniques described in this text.

1.5 ECONOMETRIC MODELING

Econometric analysis usually begins with a statement of a theoretical proposition. Consider, for example, a classic application by one of Frisch's contemporaries:

Example 1.2 *Keynes's Consumption Function*

From Keynes's (1936) *General Theory of Employment, Interest and Money*:

We shall therefore define what we shall call the propensity to consume as the functional relationship f between X , a given level of income, and C , the expenditure on consumption out of the level of income, so that $C = f(X)$.

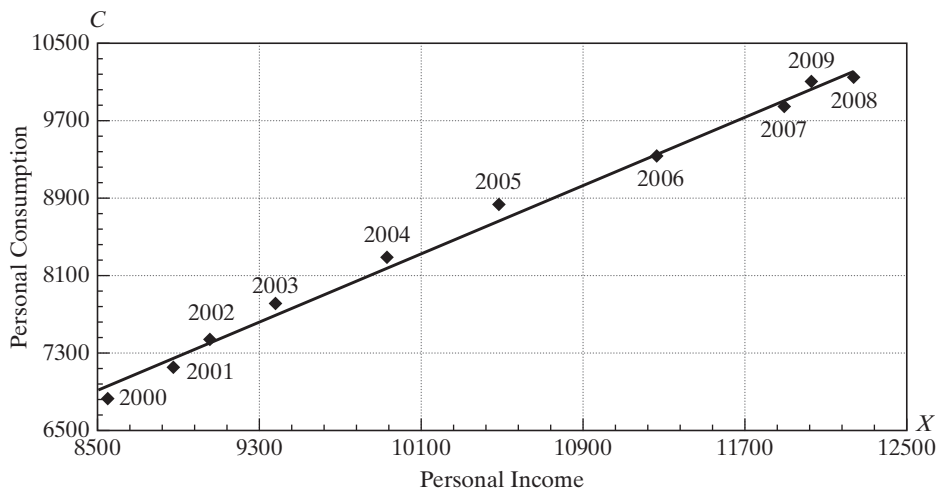
The amount that the community spends on consumption depends (i) partly on the amount of its income, (ii) partly on other objective attendant circumstances, and (iii) partly on the subjective needs and the psychological propensities and habits of the individuals composing it. The fundamental psychological law upon which we are entitled to depend with great confidence, both a priori from our knowledge of human nature and from the detailed facts of experience, is that men are disposed, as a rule and on the average, to increase their consumption as their income increases, but not by as much as the increase in their income. That is, . . . dC/dX is positive and less than unity.

But, apart from short period changes in the level of income, it is also obvious that a higher absolute level of income will tend as a rule to widen the gap between income and consumption. . . . These reasons will lead, as a rule, to a greater proportion of income being saved as real income increases.

The theory asserts a relationship between consumption and income, $C = f(X)$, and claims in the second paragraph that the marginal propensity to consume (MPC), dC/dX , is between zero and one.² The final paragraph asserts that the average propensity to consume (APC), C/X , falls as income rises, or $d(C/X)/dX = (MPC - APC)/X < 0$. It follows that $MPC < APC$. The

¹ See, for example, Long (1997) and DeMaris (2004).

² Modern economists are rarely this confident about their theories. More contemporary applications generally begin from first principles and behavioral axioms, rather than simple observation.

FIGURE 1.1 Aggregate U.S. Consumption and Income Data, 2000–2009.

most common formulation of the consumption function is a linear relationship, $C = \alpha + X\beta$, that satisfies Keynes's "laws" if β lies between zero and one and if α is greater than zero.

These theoretical propositions provide the basis for an econometric study. Given an appropriate data set, we could investigate whether the theory appears to be consistent with the observed "facts." For example, we could see whether the linear specification appears to be a satisfactory description of the relationship between consumption and income, and, if so, whether α is positive and β is between zero and one. Some issues that might be studied are (1) whether this relationship is stable through time or whether the parameters of the relationship change from one generation to the next (a change in the average propensity to save, $1 - APC$, might represent a fundamental change in the behavior of consumers in the economy); (2) whether there are systematic differences in the relationship across different countries, and, if so, what explains these differences; and (3) whether there are other factors that would improve the ability of the model to explain the relationship between consumption and income. For example, Figure 1.1 presents aggregate consumption and personal income in constant dollars for the United States for the 10 years of 2000–2009. (See Appendix Table F1.1.) Apparently, at least superficially, the data (the facts) are consistent with the theory. The relationship appears to be linear, albeit only approximately, the intercept of a line that lies close to most of the points is positive and the slope is less than one, although not by much. (However, if the line is fit by linear least squares regression, the intercept is negative, not positive.) Moreover, observers might disagree on what is meant by relationship in this description.

Economic theories such as Keynes's are typically sharp and unambiguous. Models of demand, production, labor supply, individual choice, educational attainment, income and wages, investment, market equilibrium, and aggregate consumption all specify precise, *deterministic relationships*. Dependent and independent variables are identified, a functional form is specified, and in most cases, at least a qualitative statement is made about the directions of effects that occur when independent variables in the model change. The model is only a simplification of reality. It will include the salient features of the relationship of interest but will leave unaccounted for influences that might well be present but are regarded as unimportant.

Correlations among economic variables are easily observable through descriptive statistics and techniques such as linear regression methods. The ultimate goal of the econometric model builder is often to uncover the deeper causal connections through elaborate structural, behavioral models. Note, for example, Keynes's use of the behavior of a *representative consumer* to motivate the behavior of macroeconomic variables, such as income and consumption. Heckman's model of labor supply noted in Example 1.1 is framed in a model of individual behavior. Berry, Levinsohn, and Pakes's (1995) detailed model of equilibrium pricing in the automobile market is another.

No model could hope to encompass the myriad essentially random aspects of economic life. It is thus also necessary to incorporate stochastic elements. As a consequence, observations on a variable will display variation attributable not only to differences in variables that are explicitly accounted for in the model, but also to the randomness of human behavior and the interaction of countless minor influences that are not. It is understood that the introduction of a random disturbance into a deterministic model is not intended merely to paper over its inadequacies. It is essential to examine the results of the study, in an *ex post* analysis, to ensure that the allegedly random, unexplained factor is truly unexplainable. If it is not, the model is, in fact, inadequate.³ The stochastic element endows the model with its statistical properties. Observations on the variable(s) under study are thus taken to be the outcomes of a random process. With a sufficiently detailed stochastic structure and adequate data, the analysis will become a matter of deducing the properties of a probability distribution. The tools and methods of mathematical statistics will provide the operating principles.

A model (or theory) can never truly be confirmed unless it is made so broad as to include every possibility. But it may be subjected to ever more rigorous scrutiny and, in the face of contradictory evidence, refuted. A deterministic theory will be invalidated by a single contradictory observation. The introduction of stochastic elements into the model changes it from an exact statement to a probabilistic description about expected outcomes and carries with it an important implication. Only a preponderance of contradictory evidence can convincingly invalidate the probabilistic model, and what constitutes a preponderance of evidence is a matter of interpretation. Thus, the probabilistic model is less precise but at the same time, more robust.⁴

The process of econometric analysis departs from the specification of a theoretical relationship. We initially proceed on the optimistic assumption that we can obtain precise measurements on all the variables in a correctly specified model. If the ideal conditions are met at every step, the subsequent analysis will be routine. Unfortunately, they rarely are. Some of the difficulties one can expect to encounter are the following:

- The data may be badly measured or may correspond only vaguely to the variables in the model. "The interest rate" is one example.

³ In the example given earlier, the estimated constant term in the linear least squares regression is negative. Is the theory wrong, or is the finding due to random fluctuation in the data? Another possibility is that the theory is broadly correct, but the world changed between 1936 when Keynes devised his theory and 2000–2009 when the data (outcomes) were generated. Or, perhaps linear least squares is not the appropriate technique to use for this model, and that is responsible for the inconvenient result (the negative intercept).

⁴ See Keuzenkamp and Magnus (1995) for a lengthy symposium on testing in econometrics.

- Some of the variables may be inherently unmeasurable. “Expectations” is a case in point.
- The theory may make only a rough guess as to the correct form of the model, if it makes any at all, and we may be forced to choose from an embarrassingly long menu of possibilities.
- The assumed stochastic properties of the random terms in the model may be demonstrably violated, which may call into question the methods of estimation and inference procedures we have used.
- Some relevant variables may be missing from the model.
- The conditions under which data are collected lead to a sample of observations that is systematically unrepresentative of the population we wish to study.

The ensuing steps of the analysis consist of coping with these problems and attempting to extract whatever information is likely to be present in such obviously imperfect data. The methodology is that of mathematical statistics and economic theory. The product is an econometric model.

1.6 PLAN OF THE BOOK

Our objective in this survey is to develop in detail a set of tools, then use those tools in applications. The following set of applications will include many that readers will use in practice. But it is not exhaustive. We will attempt to present our results in sufficient generality that the tools we develop here can be extended to other kinds of situations and applications not described here.

One possible approach is to organize (and orient) the areas of study by the type of data being analyzed—cross section, panel, discrete data, then time series being the obvious organization.

Alternatively, we could distinguish at the outset between micro- and macroeconometrics.⁵ Ultimately, all of these will require a common set of tools, including, for example, the multiple regression model, the use of moment conditions for estimation, instrumental variables (IV), and maximum likelihood estimation. With that in mind, the organization of this book is as follows: The first half of the text develops fundamental results that are common to all the applications. The concept of multiple regression and the linear regression model in particular constitutes the underlying platform of most modeling, even if the linear model itself is not ultimately used as the empirical specification. This part of the text concludes with developments of IV estimation and the general topic of panel data modeling. The latter pulls together many features of modern econometrics, such as, again, IV estimation, modeling heterogeneity, and a rich variety of extensions of the linear model. The second half of the text presents a variety

⁵ An excellent reference on the former that is at a more advanced level than this text is Cameron and Trivedi (2005). There does not appear to be available a counterpart, large-scale pedagogical survey of macroeconometrics that includes both econometric theory and applications. The numerous more focused studies include books such as Bardsen et al. (2005).

of topics. Part III is an overview of estimation methods. Finally, Parts IV and V present results from microeconometrics and macroeconometrics, respectively. The broad outline is as follows:

I. Regression Modeling

Chapters 2 through 6 present the multiple linear regression model. We will discuss specification, estimation, and statistical inference. This part develops the ideas of estimation, robust analysis, functional form, and principles of model specification.

II. Generalized Regression, Instrumental Variables, and Panel Data

Chapter 7 extends the regression model to nonlinear functional forms. The method of instrumental variables is presented in Chapter 8. Chapters 9 and 10 introduce the generalized regression model and systems of regression models. This section ends with Chapter 11 on panel data methods.

III. Estimation Methods

Chapters 12 through 16 present general results on different methods of estimation including GMM, maximum likelihood, and simulation-based methods. Various estimation frameworks, including non- and semiparametric and Bayesian estimation, are presented in Chapters 12 and 16.

IV. Microeconomic Methods

Chapters 17 through 19 are about microeconometrics, discrete choice modeling, limited dependent variables, and the analysis of data on events—how many occur in a given setting and when they occur. Chapters 17 through 19 are devoted to methods more suited to cross sections and panel data sets.

V. Macroeconomic Methods

Chapters 20 and 21 focus on time-series modeling and macroeconometrics.

VI. Background Materials

Appendices A through E present background material on tools used in econometrics including matrix algebra, probability and distribution theory, estimation, and asymptotic distribution theory. Appendix E presents results on computation. The data sets used in the numerical examples are described in Appendix F. The actual data sets and other supplementary materials can be downloaded from the author's Web site for the text: <http://people.stern.nyu.edu/wgreene/Text/>.

1.7 PRELIMINARIES

Before beginning, we note some specific aspects of the presentation in the text.

1.7.1 NUMERICAL EXAMPLES

There are many numerical examples given throughout the discussion. Most of these are either self-contained exercises or extracts from published studies. In general, their purpose is to provide a limited application to illustrate a method or model. The reader can replicate them with the data sets provided. This will generally not entail attempting to replicate the full published study. Rather, we use the data sets to provide applications that relate to the published study in a limited fashion that also focuses on a particular

technique, model, or tool. Thus, Riphahn, Wambach, and Million (2003) provide a very useful, manageable (though relatively large) laboratory data set that the reader can use to explore some issues in health econometrics. The exercises also suggest more extensive analyses, again in some cases based on published studies.

1.7.2 SOFTWARE AND REPLICATION

There are now many powerful computer programs that can be used for the computations described in this text. In most cases, the examples presented can be replicated with any modern package, whether the user is employing a high level integrated program such as *Stata*, *SAS*, or *NLOGIT*, or writing his own programs in languages such as *R*, *MATLAB*, or *Gauss*. The notable exception will be exercises based on simulation. Because, essentially, every package uses a different random number generator, it will generally not be possible to replicate exactly the examples in this text that use simulation (unless you are using the same computer program with the same settings that we are). Nonetheless, the differences that do emerge in such cases should be largely attributable to minor random variation. You will be able to replicate the essential results and overall features in these applications with any of the software mentioned. We will return to this general issue of replicability at a few points in the text, including in Section 15.2 where we discuss methods of generating random samples for simulation-based estimators.

1.7.3 NOTATIONAL CONVENTIONS

We will use vector and matrix notation and manipulations throughout the text. The following conventions will be used: A scalar variable will be denoted with an italic lowercase letter, such as y or x_{nK} . A column vector of scalar values will be denoted by a

boldface lowercase letter, such as $\boldsymbol{\beta} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_K \end{bmatrix}$ and, likewise, for $\mathbf{x}$ and $\mathbf{b}$. The dimensions

of a column vector are always denoted as those of a matrix with one column, such as $K \times 1$ or $n \times 1$ and so on. A matrix will always be denoted by a boldface uppercase

letter, such as the $n \times K$ matrix, $\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1K} \\ x_{21} & x_{22} & \cdots & x_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nK} \end{bmatrix}$. Specific elements in a

matrix are always subscripted so that the first subscript gives the row and the second gives the column. Transposition of a vector or a matrix is denoted with a prime. A row vector is obtained by transposing a column vector. Thus, $\boldsymbol{\beta}' = [\beta_1, \beta_2, \dots, \beta_K]$. The product of a row and a column vector will always be denoted in a form such as $\boldsymbol{\beta}'\mathbf{x} = \beta_1x_1 + \beta_2x_2 + \cdots + \beta_Kx_K$. The elements in a matrix, $\mathbf{X}$, form a set of vectors. In terms of its columns, $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_K]$ —each column is an $n \times 1$ vector. The one possible, unfortunately unavoidable source of ambiguity is the notation necessary to denote a row of a matrix such as $\mathbf{X}$. The elements of the i th row of $\mathbf{X}$ are the row vector, $\mathbf{x}_i' = [x_{i1}, x_{i2}, \dots, x_{iK}]$. When the matrix, such as $\mathbf{X}$, refers to a data matrix, we

will prefer to use the “ i ” subscript to denote observations, or the rows of the matrix and “ k ” to denote the variables, or columns. As we note unfortunately, this would seem to imply that $\mathbf{x}_i$, the transpose of $\mathbf{x}'_i$, would be the i th column of $\mathbf{X}$, which will conflict with our notation. However, with no simple alternative notation available, we will maintain this convention, with the understanding that $\mathbf{x}'_i$, *always* refers to the row vector that is the i th row of an $\mathbf{X}$ matrix. A discussion of the matrix algebra results used in the text is given in Appendix A. A particularly important set of arithmetic results about summation and the elements of the matrix product, $\mathbf{X}'\mathbf{X}$, appears in Section A.2.8.

THE LINEAR REGRESSION MODEL



2.1 INTRODUCTION

Econometrics is concerned with *model building*. An intriguing point to begin the inquiry is to consider the question, “What is the model?” The statement of a “model” typically begins with an observation or a proposition that movement of one variable “is caused by” movement of another, or “a variable varies with another,” or some qualitative statement about a relationship between a variable and one or more **covariates** that are expected to be related to the interesting variable in question. The model might make a broad statement about behavior, such as the suggestion that individuals’ usage of the health care system depends on, for example, perceived health status, demographics (e.g., income, age, and education), and the amount and type of insurance they have. It might come in the form of a verbal proposition, or even a picture (e.g., a flowchart or **path diagram** that suggests directions of influence). The econometric model rarely springs forth in full bloom as a set of equations. Rather, it begins with an *idea* of some kind of relationship. The natural next step for the econometrician is to translate that idea into a set of equations, with a notion that some feature of that set of equations will answer interesting questions about the variable of interest. To continue our example, a more definite statement of the relationship between insurance and health care demanded might be able to answer *how* does health care system utilization depend on insurance coverage? Specifically, is the relationship “positive”—all else equal, is an insured consumer more likely to demand more health care than an uninsured one—or is it “negative”? And, ultimately, one might be interested in a more precise statement, “How much more (or less)?” This and the next several chapters will build the framework that model builders use to pursue questions such as these using data and econometric methods.

From a purely statistical point of view, the researcher might have in mind a variable, y , broadly “demand for health care, H ,” and a vector of covariates, $\mathbf{x}$ (income, I , insurance, T), and a joint probability distribution of the three, $p(H, I, T)$. Stated in this form, the “relationship” is not posed in a particularly interesting fashion—what is the statistical process that produces health care demand, income, and insurance coverage? However, it is true that $p(H, I, T) = p(H|I, T)p(I, T)$, which decomposes the probability model for the joint process into two outcomes, the joint distribution of income and insurance coverage in the population, $p(I, T)$, and the distribution of “demand for health care” for a specific income and insurance coverage, $p(H|I, T)$. From this perspective, the conditional distribution, $p(H|I, T)$, holds some particular interest, while $p(I, T)$, the distribution of income and insurance coverage in the population, is perhaps of secondary, or no interest. (On the other hand, from the same perspective, the conditional “demand” for insurance coverage, given income, $p(T|I)$, might also be interesting.) Continuing this line of thinking,

the model builder is often interested not in joint variation of all the variables in the model, but in **conditional variation** of one of the variables related to the others.

The idea of the conditional distribution provides a useful starting point for thinking about a relationship between a variable of interest, a “y,” and a set of variables, “x,” that we think might bear some relationship to it. There is a question to be considered now that returns us to the issue of “What is the model?” What feature of the conditional distribution is of interest? The model builder, thinking in terms of features of the conditional distribution, often gravitates to the expected value, focusing attention on $E[y|\mathbf{x}]$, that is, the **regression function**, which brings us to the subject of this chapter. For the preceding example, this might be natural if y were “number of doctor visits” as in an application examined at several points in the chapters to follow. If we were studying incomes, I , however, which often have a highly skewed distribution, then the mean might not be particularly interesting. Rather, the **conditional median**, for given ages, $M[I|\mathbf{x}]$, might be a more interesting statistic. Still considering the distribution of incomes (and still conditioning on age), other quantiles, such as the 20th percentile, or a poverty line defined as, say, the 5th percentile, might be more interesting yet. Finally, consider a study in finance, in which the variable of interest is asset returns. In at least some contexts, means are not interesting at all—it is variances, and conditional variances in particular, that are most interesting.

The point is that we begin the discussion of the regression model with an understanding of what we mean by “the model.” For the present, we will focus on the conditional mean, which is usually the feature of interest. Once we establish how to analyze the regression function, we will use it as a useful departure point for studying other features, such as quantiles and variances. The **linear regression model** is the single most useful tool in the econometrician’s kit. Although to an increasing degree in contemporary research it is often only the starting point for the full investigation, it remains the device used to begin almost all empirical research. And it is the lens through which relationships among variables are usually viewed. This chapter will develop the linear regression model in detail. Here, we will detail the fundamental assumptions of the model. The next several chapters will discuss more elaborate specifications and complications that arise in the application of techniques that are based on the simple models presented here.

2.2 THE LINEAR REGRESSION MODEL

The **multiple linear regression model** is used to study the relationship between a **dependent variable** and one or more **independent variables**. The generic form of the linear regression model is

$$\begin{aligned} y &= f(x_1, x_2, \dots, x_K) + \varepsilon \\ &= x_1\beta_1 + x_2\beta_2 + \dots + x_K\beta_K + \varepsilon, \end{aligned} \tag{2-1}$$

where y is the dependent or **explained variable** and $x_1, \dots, x_K$ are the independent or **explanatory variables**. (We will return to the meaning of “independent” shortly.) One’s theory will specify $f(x_1, x_2, \dots, x_K)$. This function is commonly called the **population regression equation** of y on $x_1, \dots, x_K$. In this setting, y is the **regressand** and $x_k, k = 1, \dots, K$ are the **regressors** or covariates. The underlying theory will specify the dependent and independent variables in the model. It is not always obvious which is

appropriately defined as each of these—for example, a demand equation, $quantity = \beta_1 + price \times \beta_2 + income \times \beta_3 + \varepsilon$, and an inverse demand equation, $price = \gamma_1 + quantity \times \gamma_2 + income \times \gamma_3 + u$ are equally valid representations of a market. For modeling purposes, it will often prove useful to think in terms of “autonomous variation.” One can conceive of movement of the independent variables outside the relationships defined by the model while movement of the dependent variable is considered in response to some independent or exogenous stimulus.¹

The term ε is a random **disturbance**, so named because it “disturbs” an otherwise stable relationship. The disturbance arises for several reasons, primarily because we cannot hope to capture every influence on an economic variable in a model, no matter how elaborate. The net effect, which can be positive or negative, of these omitted factors is captured in the disturbance. There are many other contributors to the disturbance in an empirical model. Probably the most significant is errors of measurement. It is easy to theorize about the relationships among precisely defined variables; it is quite another matter to obtain accurate measures of these variables. For example, the difficulty of obtaining reasonable measures of profits, interest rates, capital stocks, or, worse yet, flows of services from capital stocks, is a recurrent theme in the empirical literature. At the extreme, there may be no observable counterpart to the theoretical variable. The literature on the permanent income model of consumption [e.g., Friedman (1957)] provides an interesting example.

We assume that each observation in a sample $(y_i, x_{i1}, x_{i2}, \dots, x_{iK}), i = 1, \dots, n$, is generated by an underlying process described by

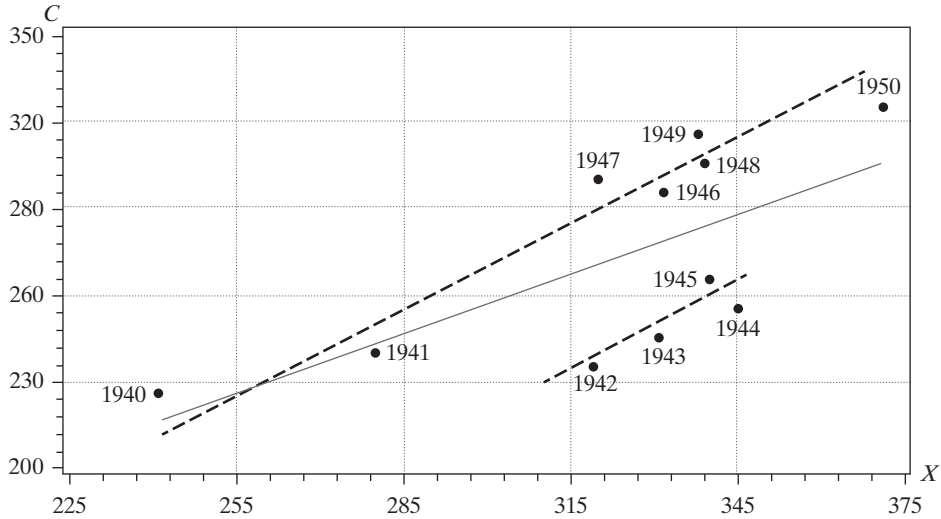
$$y_i = x_{i1}\beta_1 + x_{i2}\beta_2 + \dots + x_{iK}\beta_K + \varepsilon_i.$$

The observed value of y_i is the sum of two parts, the regression function and the disturbance, ε_i . Our objective is to estimate the unknown parameters of the model, use the data to study the validity of the theoretical propositions, and perhaps use the model to predict the variable y . How we proceed from here depends crucially on what we assume about the stochastic process that has led to our observations of the data in hand.

Example 2.1 Keynes’s Consumption Function

Example 1.2 discussed a model of consumption proposed by Keynes in his *General Theory* (1936). The theory that consumption, C , and income, X , are related certainly seems consistent with the observed “facts” in Figures 1.1 and 2.1. (These data are in Data Table F2.1.) Of course, the linear function is only approximate. Even ignoring the anomalous wartime years, consumption and income cannot be connected by any simple **deterministic relationship**. The linear part of the model, $C = \alpha + \beta X$, is intended only to represent the salient features of this part of the economy. It is hopeless to attempt to capture every influence in the relationship. The next step is to incorporate the inherent randomness in its real-world counterpart. Thus, we write $C = f(X, \varepsilon)$, where ε is a stochastic element. It is important not to view ε as a catchall for the inadequacies of the model. The model including ε appears adequate for the data not including the war years, but for 1942–1945, something systematic clearly seems to be missing. Consumption in these years could not rise to rates historically consistent with these levels of income because of wartime rationing. A model meant to describe consumption in this period would have to accommodate this influence.

¹ By this definition, it would seem that in our demand relationship, only income would be an independent variable while both price and quantity would be dependent. That makes sense—in a market, equilibrium price and quantity are determined at the same time, and do change only when something outside the market equilibrium changes.

FIGURE 2.1 Consumption Data, 1940–1950.

It remains to establish how the stochastic element will be incorporated in the equation. The most frequent approach is to assume that it is *additive*. Thus, we recast the equation in stochastic terms: $C = \alpha + \beta X + \varepsilon$. This equation is an empirical counterpart to Keynes's theoretical model. But, what of those anomalous years of rationing? If we were to ignore our intuition and attempt to fit a line to all these data—the next chapter will discuss at length how we should do that—we might arrive at the solid line in the figure as our best guess. This line, however, is obviously being distorted by the rationing. A more appropriate specification for these data that accommodates both the stochastic nature of the data and the special circumstances of the years 1942–1945 might be one that shifts straight down in the war years, $C = \alpha + \beta X + d_{\text{waryears}}\delta_w + \varepsilon$, where the new variable, d_{waryears} , equals one in 1942–1945 and zero in other years, and $\delta_w < 0$. This more detailed model is shown by the parallel dashed lines.

One of the most useful aspects of the multiple regression model is its ability to identify the separate effects of a set of variables on a dependent variable. Example 2.2 describes a common application.

Example 2.2 Earnings and Education

Many studies have analyzed the relationship between earnings and education. We would expect, on average, higher levels of education to be associated with higher incomes. The simple regression model

$$\text{earnings} = \beta_1 + \beta_2 \text{education} + \varepsilon,$$

however, neglects the fact that most people have higher incomes when they are older than when they are young, regardless of their education. Thus, β_2 will overstate the marginal impact of education. If age and education are positively correlated, then the regression model will associate all the observed increases in income with increases in education and none with, say, experience. A better specification would account for the effect of age, as in

$$\text{earnings} = \gamma_1 + \gamma_2 \text{education} + \gamma_3 \text{age} + \varepsilon.$$

It is often observed that income tends to rise less rapidly in the later earning years than in the early ones. To accommodate this possibility, we might further extend the model to

$$\text{earnings} = \delta_1 + \delta_2 \text{education} + \delta_3 \text{age} + \delta_4 \text{age}^2 + \varepsilon.$$

We would expect δ_3 to be positive and δ_4 to be negative.

The crucial feature of this model is that it allows us to carry out a conceptual experiment that might not be observed in the actual data. In the example, we might like to (and could) compare the earnings of two individuals of the same age with different amounts of education even if the data set does not actually contain two such individuals. How education should be measured in this setting is a difficult problem. The study of the earnings of twins by Ashenfelter and Krueger (1994), which uses precisely this specification of the earnings equation, presents an interesting approach. [Studies of twins and siblings have provided an interesting thread of research on the education and income relationship. Two other studies are Ashenfelter and Zimmerman (1997) and Bonjour, Cherkas, Haskel, Hawkes, and Spector (2003).] The experiment embodied in the earnings model thus far suggested is a comparison of two otherwise identical individuals who have different years of education. Under this interpretation, the impact of education would be $\partial E[\text{Earnings} | \text{Age}, \text{Education}] / \partial \text{Education} = \beta_2$. But, one might suggest that the experiment the analyst really has in mind is the truly unobservable impact of the additional year of education on a particular individual. To carry out the experiment, it would be necessary to observe the individual twice, once under circumstances that actually occur, Education_i , and a second time under the hypothetical (**counterfactual**) *circumstance*, $\text{Education}_i + 1$. It is convenient to frame this in a **potential outcomes model** [Rubin (1974)] for individual i :

$$\text{Potential Earning} = \begin{cases} y_{i0} & \text{if Education} = E_i, \\ y_{i1} & \text{if Education} = E_i + 1. \end{cases}$$

By this construction, all other effects would indeed be held constant, and $(y_{i1} - y_{i0})$ could reasonably be labeled the **causal effect** of the additional year of education. If we consider Education in this example as a **treatment**, then the real objective of the experiment is to measure the **effect of the treatment on the treated**. The ability to infer this result from nonexperimental data that essentially compares “otherwise similar individuals” will be examined in Chapters 8 and 19.

A large literature has been devoted to another intriguing question on this subject. Education is not truly independent in this setting. Highly motivated individuals will choose to pursue more education (e.g., by going to college or graduate school) than others. By the same token, highly motivated individuals may do things that, on average, lead them to have higher incomes. If so, does a positive β_2 that suggests an association between income and education really measure the causal effect of education on income, or does it reflect the result of some underlying effect on both variables that we have not included in the regression model? We will revisit the issue in Chapter 19.²

2.3 ASSUMPTIONS OF THE LINEAR REGRESSION MODEL

The linear regression model consists of a set of assumptions about how a data set will be produced by an underlying “data-generating process.” The theory will specify a relationship between a dependent variable and a set of independent variables. The

² This model lays yet another trap for the practitioner. In a cross section, the higher incomes of the older individuals in the sample might tell an entirely different, perhaps macroeconomic story (a cohort effect) from the lower incomes of younger individuals as time and their incomes evolve. It is not necessarily possible to deduce the characteristics of incomes of younger people in the sample *if they were older* by comparing the older individuals in the sample to the younger ones. A parallel problem arises in the analysis of treatment effects that we will examine in Chapter 8.

assumptions that describe the form of the model and relationships among its parts and imply appropriate estimation and inference procedures are listed in Table 2.1.

2.3.1 LINEARITY OF THE REGRESSION MODEL

Let the column vector $\mathbf{x}_k$ be the n observations on variable x_k , $k = 1, \dots, K$, in a random sample of n observations, and assemble these data in an $n \times K$ data matrix, $\mathbf{X}$. In most contexts, the first column of $\mathbf{X}$ is assumed to be a column of 1s so that β_1 is

TABLE 2.1 Assumptions of the Linear Regression Model

A1. Linearity: We list the assumptions as a description of the joint distribution of y and a set of independent variables, $(x_1, x_2, \dots, x_K) = \mathbf{x}$. The model specifies a linear relationship between y and $\mathbf{x}$; $y = x_1\beta_1 + x_2\beta_2 + \dots + x_K\beta_K + \varepsilon = \mathbf{x}'\boldsymbol{\beta} + \varepsilon$. We will be more specific and assume that this is the regression function, $E[y|x_1, x_2, \dots, x_K] = E[y|\mathbf{x}] = \mathbf{x}'\boldsymbol{\beta}$. The difference between y and $E[y|\mathbf{x}]$ is the disturbance, ε .

A2. Full rank: There is no exact *linear* relationship among any of the independent variables in the model. One way to formulate this is to assume that $E[\mathbf{xx}'] = \mathbf{Q}$, a $K \times K$ matrix that has full rank K . In practical terms, we wish to be sure that for a random sample of n observations drawn from this process, $(y_1, \mathbf{x}_1'), \dots, (y_i, \mathbf{x}_i'), \dots, (y_n, \mathbf{x}_n')$, that the $n \times K$ matrix $\mathbf{X}$ with n rows $\mathbf{x}_i'$ always has rank K if $n \geq K$. This assumption will be necessary for estimation of the parameters of the model.

A3. Exogeneity of the independent variables: $E[\varepsilon|x_1, x_2, \dots, x_K] = E[\varepsilon|\mathbf{x}] = 0$. This states that the expected value of the disturbance in the regression is not a function of the independent variables observed. This means that the independent variables will not carry useful information for prediction of ε . The assumption is labeled **mean independence**. By the Law of Iterated Expectations (Theorem B.1), it follows that $E[\varepsilon] = 0$. An implication of the exogeneity assumption is that $E[y|x_1, x_2, \dots, x_K] = \sum_{k=1}^K x_k\beta_k$. That is, the linear function in A1 is the **conditional mean function**, or **regression** of y on $x_1, \dots, x_K$. In the setting of a random sample, we will also begin from an assumption that observations on ε in the sample are uncorrelated with information in other observations—that is, $E[\varepsilon_i|\mathbf{x}_1, \dots, \mathbf{x}_n] = 0$. This is labeled **strict exogeneity**. An implication will be, for each observation in a sample of observations, $E[\varepsilon_i|\mathbf{X}] = 0$, and for the sample as a whole, $E[\boldsymbol{\varepsilon}|\mathbf{X}] = \mathbf{0}$.

A4. Homoscedasticity: The disturbance in the regression has **conditional variance**, $\text{Var}[\varepsilon|\mathbf{x}] = \text{Var}[\varepsilon] = \sigma^2$. (The second equality follows from Theorem B.4.) This assumption limits the generality of the model, and we will want to examine how to relax it in the chapters to follow. Once again, considering a random sample, we will assume that the observations ε_i and ε_j are uncorrelated for $i \neq j$. With reference to a times-series setting, this will be labeled **nonautocorrelation**. The implication will be $E[\varepsilon_i\varepsilon_j|\mathbf{x}_i, \mathbf{x}_j] = 0$. We will strengthen this to $E[\varepsilon_i\varepsilon_j|\mathbf{X}] = 0$ for $i \neq j$ and $E[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'|\mathbf{X}] = \sigma^2\mathbf{I}$.

A5. Data generation: The data in $(x_1, x_2, \dots, x_K)$ (that is, the process by which $\mathbf{x}$ is generated) may be any mixture of constants and random variables. The crucial elements for present purposes are the exogeneity assumption, A3, and the variance and covariance assumption, A4. Analysis can be done conditionally on the observed $\mathbf{X}$, so whether the elements in $\mathbf{X}$ are fixed constants or random draws from a stochastic process will not influence the results. In later, more advanced treatments, we will want to be more specific about the possible relationship between ε_i and $\mathbf{x}_j$. Nothing is lost by assuming that the n observations in hand are a **random sample** of independent, identically distributed draws from a joint distribution of $(y, \mathbf{x})$. In some treatments to follow, such as panel data, some observations will be correlated by construction. It will be necessary to revisit the assumptions at that point, and revise them as necessary.

A6. Normal distribution: The disturbances are normally distributed. This is a convenience that we will dispense with after some analysis of its implications. The normality assumption is useful for defining the computations behind statistical inference about the regression, such as confidence intervals and hypothesis tests. For practical purposes, it will be useful then to extend those results and in the process develop a more flexible approach that does not rely on this specific assumption.

the constant term in the model. Let $\mathbf{y}$ be the n observations, $y_1, \dots, y_n$, and let $\boldsymbol{\varepsilon}$ be the column vector containing the n disturbances. The model in (2-1) as it applies to each of and all n observations can now be written

$$\mathbf{y} = \mathbf{x}_1\beta_1 + \dots + \mathbf{x}_K\beta_K + \boldsymbol{\varepsilon}, \quad (2-2)$$

or in the form of Assumption A1,

$$\text{ASSUMPTION A1: } \mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}. \quad (2-3)$$

A NOTATIONAL CONVENTION

Henceforth, to avoid a possibly confusing and cumbersome notation, we will use a boldface $\mathbf{x}$ to denote a column or a row of $\mathbf{X}$. Which of these applies will be clear from the context. In (2-2), $\mathbf{x}_k$ is the k th column of $\mathbf{X}$. Subscript k will usually be used to denote columns (variables). It will often be convenient to refer to a single observation in (2-3), which we would write

$$y_i = \mathbf{x}_i'\boldsymbol{\beta} + \varepsilon_i. \quad (2-4)$$

Subscripts i, j , and t will generally be used to denote rows (observations) of $\mathbf{X}$. In (2-4), $\mathbf{x}_i'$ is a row vector that is the i th $1 \times K$ row of $\mathbf{X}$.

Our primary interest is in estimation and inference about the parameter vector $\boldsymbol{\beta}$. Note that the simple regression model in Example 2.1 is a special case in which $\mathbf{X}$ has only two columns, the first of which is a column of 1s. The assumption of linearity of the regression model includes the additive disturbance. For the regression to be linear in the sense described here, it must be of the form in (2-1) either in the original variables or after some suitable transformation. For example, the model

$$y = Ax^\beta e^\varepsilon$$

is linear (after taking logs on both sides of the equation), whereas

$$y = Ax^\beta + \varepsilon$$

is not. The observed dependent variable is thus the sum of two components, a deterministic element $\alpha + \beta x$ and a random variable ε . It is worth emphasizing that neither of the two parts is directly observed because α and β are unknown.

The linearity assumption is not so narrow as it might first appear. In the regression context, *linearity* refers to the manner in which the parameters and the disturbance enter the equation, not necessarily to the relationship among the variables. For example, the equations $y = \alpha + \beta x + \varepsilon$, $y = \alpha + \beta \cos(x) + \varepsilon$, $y = \alpha + \beta/x + \varepsilon$, and $y = \alpha + \beta \ln x + \varepsilon$ are all linear in some function of x by the definition we have used here. In the examples, only x has been transformed, but y could have been as well, as in $y = Ax^\beta e^\varepsilon$, which is a linear relationship in the logs of x and y ; $\ln y = \alpha + \beta \ln x + \varepsilon$. The variety of functions is unlimited. This aspect of the model is used in a number of commonly used functional forms. For example, the **loglinear model** is

$$\ln y = \beta_1 + \beta_2 \ln x_2 + \beta_3 \ln x_3 + \dots + \beta_K \ln x_K + \varepsilon.$$

This equation is also known as the **constant elasticity** form, as in this equation, the elasticity of y with respect to changes in x_k is $\partial \ln y / \partial \ln x_k = \beta_k$, which does not vary with x_k . The loglinear form is often used in models of demand and production. Different values of β_k produce widely varying functions.

Example 2.3 The U.S. Gasoline Market

Data on the U.S. gasoline market for the years 1953–2004 are given in Table F2.2 in Appendix F. We will use these data to obtain, among other things, estimates of the income, own price, and cross-price elasticities of demand in this market. These data also present an interesting question on the issue of holding “all other things constant,” that was suggested in Example 2.2. In particular, consider a somewhat abbreviated model of per capita gasoline consumption:

$$\ln(G/pop) = \beta_1 + \beta_2 \ln(Income/pop) + \beta_3 \ln price_G + \beta_4 \ln P_{newcars} + \beta_5 \ln P_{usedcars} + \varepsilon.$$

This model will provide estimates of the income and price elasticities of demand for gasoline and an estimate of the elasticity of demand with respect to the prices of new and used cars. What should we expect for the sign of β_4 ? Cars and gasoline are complementary goods, so if the prices of new cars rise, *ceteris paribus*, gasoline consumption should fall. Or should it? If the prices of new cars rise, then consumers will buy fewer of them; they will keep their used cars longer and buy fewer new cars. If older cars use more gasoline than newer ones, then the rise in the prices of new cars would lead to higher gasoline consumption than otherwise, not lower. We can use the multiple regression model and the gasoline data to attempt to answer the question.

A **semilog** model is often used to model growth rates:

$$\ln y_t = \mathbf{x}'_t \boldsymbol{\beta} + \delta t + \varepsilon_t.$$

In this model, the autonomous (at least not explained by the model itself) proportional, per period growth rate is $\partial \ln y / \partial t = \delta$. Other variations of the general form

$$f(y_t) = g(\mathbf{x}'_t \boldsymbol{\beta} + \varepsilon_t)$$

will allow a tremendous variety of functional forms, all of which fit into our definition of a linear model.

The linear regression model is sometimes interpreted as an approximation to some unknown, underlying function. (See Section A.8.1 for discussion.) By this interpretation, however, the linear model, even with quadratic terms, is fairly limited in that such an approximation is likely to be useful only over a small range of variation of the independent variables. The translog model discussed in Example 2.4, in contrast, has proven more effective as an approximating function.

Example 2.4 The Translog Model

Modern studies of demand and production are usually done with a **flexible functional form**. Flexible functional forms are used in econometrics because they allow analysts to model complex features of the production function, such as elasticities of substitution, which are functions of the second derivatives of production, cost, or utility functions. The linear model restricts these to equal zero, whereas the loglinear model (e.g., the Cobb–Douglas model) restricts the interesting elasticities to the uninteresting values of -1 or $+1$. The most popular flexible functional form is the **translog model**, which is often interpreted as a second-order approximation to an unknown functional form. [See Berndt and Christensen (1973).] One way to derive it is as follows. We first write $y = g(x_1, \dots, x_k)$. Then, $\ln y = \ln g(\dots) = f(\dots)$. Since by a trivial transformation $x_k = \exp(\ln x_k)$, we interpret the function as a function of the logarithms of the x 's. Thus, $\ln y = f(\ln x_1, \dots, \ln x_k)$.

Now, expand this function in a second-order Taylor series around the point $\mathbf{x} = [1, 1, \dots, 1]'$ so that at the expansion point, the log of each variable is a convenient zero. Then

$$\begin{aligned} \ln y &= f(\mathbf{0}) + \sum_{k=1}^K [\partial f(\cdot) / \partial \ln x_k]_{\ln \mathbf{x}=\mathbf{0}} \ln x_k \\ &\quad + \frac{1}{2} \sum_{k=1}^K \sum_{l=1}^K [\partial^2 f(\cdot) / \partial \ln x_k \partial \ln x_l]_{\ln \mathbf{x}=\mathbf{0}} \ln x_k \ln x_l + \varepsilon. \end{aligned}$$

The disturbance in this model is assumed to embody the familiar factors and the error of approximation to the unknown function. Because the function and its derivatives evaluated at the fixed value $\mathbf{0}$ are constants, we interpret them as the coefficients and write

$$\ln y = \beta_0 + \sum_{k=1}^K \beta_k \ln x_k + \frac{1}{2} \sum_{k=1}^K \sum_{l=1}^K \gamma_{kl} \ln x_k \ln x_l + \varepsilon.$$

This model is linear by our definition but can, in fact, mimic an impressive amount of curvature when it is used to approximate another function. An interesting feature of this formulation is that the loglinear model is a special case, when $\gamma_{kl} = 0$. Also, there is an interesting test of the underlying theory possible because if the underlying function were assumed to be continuous and twice continuously differentiable, then by Young's theorem it must be true that $\gamma_{kl} = \gamma_{lk}$. We will see in Chapter 10 how this feature is studied in practice.

Despite its great flexibility, the linear model will not accommodate all the situations we will encounter in practice. In Example 14.13 and Chapter 18, we will examine the regression model for doctor visits that was suggested in the introduction to this chapter. An appropriate model that describes the number of visits has conditional mean function $E[y|\mathbf{x}] = \exp(\mathbf{x}'\boldsymbol{\beta})$. It is tempting to linearize this directly by taking logs, because $\ln E[y|\mathbf{x}] = \mathbf{x}'\boldsymbol{\beta}$. But $\ln E[y|\mathbf{x}]$ is not equal to $E[\ln y|\mathbf{x}]$. In that setting, y can equal zero (and does for most of the sample), so $\mathbf{x}'\boldsymbol{\beta}$ (which can be negative) is not an appropriate model for $\ln y$ (which does not exist) or for y which cannot be negative. The methods we consider in this chapter are not appropriate for estimating the parameters of such a model. Relatively straightforward techniques have been developed for nonlinear models such as this, however. We shall treat them in detail in Chapter 7.

2.3.2 FULL RANK

Assumption A2 is that there are no exact *linear* relationships among the variables.

ASSUMPTION A2: $\mathbf{X}$ is an $n \times K$ matrix with rank K .

 (2-5)

Hence, $\mathbf{X}$ has full column rank; the columns of $\mathbf{X}$ are linearly independent and there are at least K observations. [See (A-42) and the surrounding text.] This assumption is known as an **identification condition**. To see the need for this assumption, consider an example.

Example 2.5 Short Rank

Suppose that a cross-section model specifies that consumption, C , relates to income as follows:

$$C = \beta_1 + \beta_2 \text{ nonlabor income} + \beta_3 \text{ salary} + \beta_4 \text{ total income} + \varepsilon,$$

where *total income* is exactly equal to *salary* plus *nonlabor income*. Clearly, there is an exact linear relationship among the variables in the model. Now, let

$$\beta'_2 = \beta_2 + a,$$

$$\beta'_3 = \beta_3 + a,$$

and

$$\beta'_4 = \beta_4 - a,$$

where a is any number. Then the exact same value appears on the right-hand side of C if we substitute β'_2 , β'_3 , and β'_4 for β_2 , β_3 , and β_4 . Obviously, there is no way to estimate the parameters of this model.

If there are fewer than K observations, then $\mathbf{X}$ cannot have **full rank**. Hence, we make the assumption that n is at least as large as K .

In the simple linear model with a constant term and a single x , the full rank assumption means that there must be variation in the regressor, x . If there is no variation in x , then all our observations will lie on a vertical line. This situation does not invalidate the other assumptions of the model; presumably, it is a flaw in the data set. The possibility that this suggests is that we *could* have drawn a sample in which there was variation in x , but in this instance, we did not. Thus, the model still applies, but we cannot learn about it from the data set in hand.

Example 2.6 An Inestimable Model

In Example 3.4, we will consider a model for the sale price of Monet paintings. Theorists and observers have different models for how prices of paintings at auction are determined. One (naïve) student of the subject suggests the model

$$\begin{aligned}\ln \text{Price} &= \beta_1 + \beta_2 \ln \text{Size} + \beta_3 \ln \text{Aspect Ratio} + \beta_4 \ln \text{Height} + \varepsilon \\ &= \beta_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \varepsilon,\end{aligned}$$

where $\text{Size} = \text{Width} \times \text{Height}$ and $\text{Aspect Ratio} = \text{Width}/\text{Height}$. By simple arithmetic, we can see that this model shares the problem found with the consumption model in Example 2.5—in this case, $x_2 - x_4 = x_3 + x_4$. So, this model is, like the previous one, not estimable—it is not identified. It is useful to think of the problem from a different perspective here (so to speak). In the linear model, it must be possible for the variables in the model to vary linearly independently. But, in this instance, while it is possible for any pair of the three covariates to vary independently, the three together cannot. The “model,” that is, the theory, is an entirely reasonable model as it stands. Art buyers might very well consider all three of these features in their valuation of a Monet painting. However, it is not possible to learn about that from the observed data, at least not with this linear regression model.

The full rank assumption is occasionally interpreted to mean that the variables in $\mathbf{X}$ must be able to vary independently from each other. This is clearly not the case in Example 2.6, which is a flawed model. But it is also not the case in the linear model

$$E[y|x,z] = \beta_1 + \beta_2 x + \beta_3 x^2 + \beta_4 z + \varepsilon.$$

There is nothing problematic with this model—nor with the model in Example 2.2 or the translog model in Example 2.4. Nonetheless, x and x^2 cannot vary independently. The resolution of this seeming contradiction is to sharpen what we mean by the variables in the model varying independently. First, it remains true that $\mathbf{X}$ must have full column rank to carry out the *linear* regression. But, independent variation of the variables in the model is a different concept. The columns of $\mathbf{X}$ are not necessarily the set of variables in the model. In the equation above, the “variables” are only x and z . The identification problem we consider here would state that it must be possible for z to vary independently

from x . If z is a deterministic function of x , then it is not possible to identify an effect in the model for variable z separately from that for x .

2.3.3 REGRESSION

The disturbance is assumed to have conditional expected value zero at every observation, which we write as

$$E[\varepsilon_i | \mathbf{X}] = 0. \quad (2-6)$$

For the full set of observations, we write Assumption A3 as

$$\text{ASSUMPTION A3: } E[\boldsymbol{\varepsilon} | \mathbf{X}] = \begin{bmatrix} E[\varepsilon_1 | \mathbf{X}] \\ E[\varepsilon_2 | \mathbf{X}] \\ \vdots \\ E[\varepsilon_n | \mathbf{X}] \end{bmatrix} = \mathbf{0}. \quad (2-7)$$

There is a subtle point in this discussion that the observant reader might have noted. In (2-7), the left-hand side states, in principle, that the mean of each ε_i *conditioned on all observations* $\mathbf{x}_j$ is zero. This strict exogeneity assumption states, in words, that no observations on $\mathbf{x}$ convey information about the expected value of the disturbance. It is conceivable—for example, in a time-series setting—that although $\mathbf{x}_i$ might provide no information about $E[\varepsilon_i | \cdot]$, $\mathbf{x}_j$ *at some other observation*, such as in the previous time period, might. Our assumption at this point is that there is no information about $E[\varepsilon_i | \cdot]$ contained in *any* observation $\mathbf{x}_j$. Later, when we extend the model, we will study the implications of dropping this assumption. [See Wooldridge (1995).] We will also assume that the disturbances convey no information about each other. That is, $E[\varepsilon_i | \varepsilon_1, \dots, \varepsilon_{i-1}, \varepsilon_{i+1}, \dots, \varepsilon_n] = 0$. In sum, at this point, we have assumed that the disturbances are purely random draws from some population.

The zero conditional mean implies that the unconditional mean is also zero, because by the **Law of Iterated Expectations** [Theorem B.1, (B-66)],

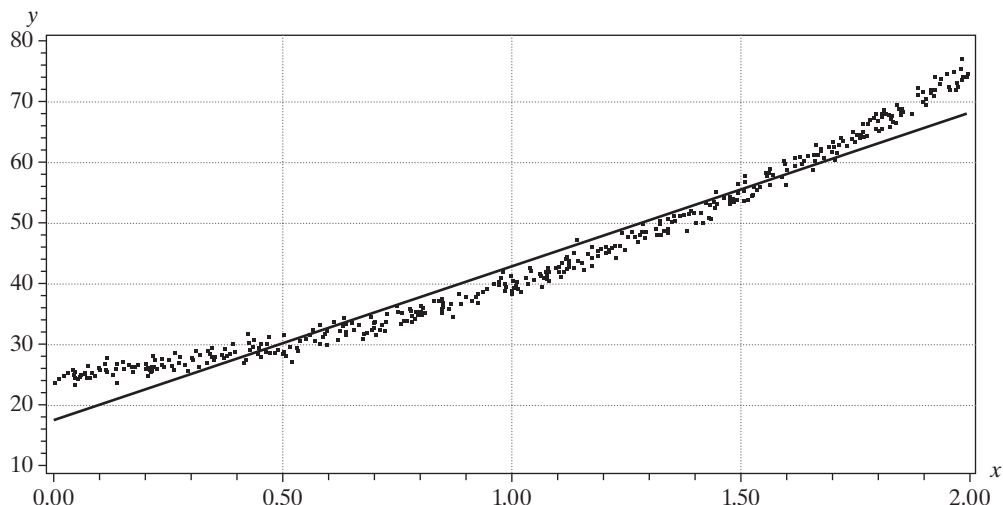
$$E[\varepsilon_i] = E_{\mathbf{x}}[E[\varepsilon_i | \mathbf{X}]] = E_{\mathbf{x}}[0] = 0.$$

For each ε_i , by Theorem B.2, $\text{Cov}[E[\varepsilon_i | \mathbf{X}], \mathbf{X}] = \text{Cov}[\varepsilon_i, \mathbf{X}]$, Assumption A3 implies that $\text{Cov}[\varepsilon_i, \mathbf{x}] = 0$ for all i . The converse is not true; $E[\varepsilon_i] = 0$ does not imply that $E[\varepsilon_i | \mathbf{x}_i] = 0$. Example 2.7 illustrates the difference.

Example 2.7 Nonzero Conditional Mean of the Disturbances

Figure 2.2 illustrates the important difference between $E[\varepsilon_i] = 0$ and $E[\varepsilon_i | x_i] = 0$. The overall mean of the disturbances in the sample is zero, but the mean for specific ranges of x is distinctly nonzero. A pattern such as this in observed data would serve as a useful indicator that the specification of the linear regression should be questioned. In this particular case, the true conditional mean function (which the researcher would not know in advance) is actually $E[y|x] = 25 + 5x(1 + 2x)$. The sample data are suggesting that a linear specification is not appropriate for these data. A quadratic specification would seem to be a good candidate. This modeling strategy is pursued in an application in Example 6.6.

In most cases, the zero overall mean assumption is not restrictive. Consider a two-variable model and suppose that the mean of ε is $\mu \neq 0$. Then $\alpha + \beta x + \varepsilon$ is the same

FIGURE 2.2 Disturbances with Nonzero Conditional Mean and Zero Unconditional Mean.

as $(\alpha + \mu) + \beta x + (\varepsilon - \mu)$. Letting $\alpha' = \alpha + \mu$ and $\varepsilon' = \varepsilon - \mu$ produces the original model. For an application, see the discussion of frontier production functions in Section 19.2.4. But if the original model does not contain a constant term, then assuming $E[\varepsilon_i] = 0$ could be substantive. This suggests that there is a potential problem in models without constant terms. As a general rule, regression models should not be specified without constant terms unless this is specifically dictated by the underlying theory.³ Arguably, if we have reason to specify that the mean of the disturbance is something other than zero, we should build it into the systematic part of the regression, leaving in the disturbance only the unknown part of ε . Assumption A3 also implies that

$$E[\mathbf{y}|\mathbf{X}] = \mathbf{X}\boldsymbol{\beta}. \quad (2-8)$$

Assumptions A1 and A3 comprise the *linear regression model*. The regression of $\mathbf{y}$ on $\mathbf{X}$ is the conditional mean, $E[\mathbf{y}|\mathbf{X}]$, so that without Assumption A3, $\mathbf{X}\boldsymbol{\beta}$ is *not* the conditional mean function.

The remaining assumptions will more completely specify the characteristics of the disturbances in the model and state the conditions under which the sample observations on $\mathbf{x}$ are obtained.

2.3.4 HOMOSCEDASTIC AND NONAUTOCORRELATED DISTURBANCES

The fourth assumption concerns the variances and covariances of the disturbances:

$$\text{Var}[\varepsilon_i|\mathbf{X}] = \sigma^2, \quad \text{for all } i = 1, \dots, n,$$

³ Models that describe first differences of variables might well be specified without constants. Consider $y_t - y_{t-1}$. If there is a constant term α on the right-hand side of the equation, then y_t is a function of αt , which is an explosive regressor. Models with linear time trends merit special treatment in the time-series literature. We will return to this issue in Chapter 21.

and

$$\text{Cov}[\varepsilon_i, \varepsilon_j | \mathbf{X}] = 0, \quad \text{for all } i \neq j.$$

Constant variance is labeled **homoscedasticity**. Consider a model that describes the profits of firms in an industry as a function of, say, size. Even accounting for size, measured in dollar terms, the profits of large firms will exhibit greater variation than those of smaller firms. The homoscedasticity assumption would be inappropriate here. Survey data on household expenditure patterns often display marked **heteroscedasticity**, even after accounting for income and household size.

Uncorrelatedness across observations is labeled generically nonautocorrelation. In Figure 2.1, there is some suggestion that the disturbances might not be truly independent across observations. Although the number of observations is small, it does appear that, on average, each disturbance tends to be followed by one with the same sign. This “inertia” is precisely what is meant by **autocorrelation**, and it is assumed away at this point. Methods of handling autocorrelation in economic data occupy a large proportion of the literature and will be treated at length in Chapter 20. Note that nonautocorrelation does not imply that observations y_i and y_j are uncorrelated. The assumption is that *deviations* of observations from their expected values are uncorrelated.

The two assumptions imply that

$$\begin{aligned} E[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' | \mathbf{X}] &= \begin{bmatrix} E[\varepsilon_1\varepsilon_1 | \mathbf{X}] & E[\varepsilon_1\varepsilon_2 | \mathbf{X}] & \cdots & E[\varepsilon_1\varepsilon_n | \mathbf{X}] \\ E[\varepsilon_2\varepsilon_1 | \mathbf{X}] & E[\varepsilon_2\varepsilon_2 | \mathbf{X}] & \cdots & E[\varepsilon_2\varepsilon_n | \mathbf{X}] \\ \vdots & \vdots & \ddots & \vdots \\ E[\varepsilon_n\varepsilon_1 | \mathbf{X}] & E[\varepsilon_n\varepsilon_2 | \mathbf{X}] & \cdots & E[\varepsilon_n\varepsilon_n | \mathbf{X}] \end{bmatrix} \\ &= \begin{bmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ & & \ddots & \\ 0 & 0 & \cdots & \sigma^2 \end{bmatrix}, \end{aligned}$$

which we summarize in Assumption A4:

$$\boxed{\text{ASSUMPTION A4: } E[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' | \mathbf{X}] = \sigma^2 \mathbf{I}.} \quad (2-9)$$

By using the variance decomposition formula in (B-69), we find

$$\text{Var}[\boldsymbol{\varepsilon}] = E[\text{Var}[\boldsymbol{\varepsilon} | \mathbf{X}]] + \text{Var}[E[\boldsymbol{\varepsilon} | \mathbf{X}]] = \sigma^2 \mathbf{I}.$$

Once again, we should emphasize that this assumption describes the information about the variances and covariances among the disturbances that is provided by the independent variables. For the present, we assume that there is none. We will also drop this assumption later when we enrich the regression model. We are also assuming that the disturbances themselves provide no information about the variances and covariances. Although a minor issue at this point, it will become crucial in our treatment of time-series applications. Models such as $\text{Var}[\varepsilon_t | \varepsilon_{t-1}] = \sigma^2 + \alpha \varepsilon_{t-1}^2$, a “GARCH” model (see Chapter 20), do not violate our conditional variance assumption, but do assume that $\text{Var}[\varepsilon_t | \varepsilon_{t-1}] \neq \text{Var}[\varepsilon_t]$.

2.3.5 DATA GENERATING PROCESS FOR THE REGRESSORS

It is common to assume that $\mathbf{x}_i$ is nonstochastic, as it would be in an experimental situation. Here the analyst chooses the values of the regressors and then observes y_i . This process might apply, for example, in an agricultural experiment in which y_i is yield and $\mathbf{x}_i$ is fertilizer concentration and water applied. The assumption of **nonstochastic regressors** at this point would be a mathematical convenience. With it, we could use the results of elementary statistics to obtain our results by treating the vector $\mathbf{x}_i$ simply as a known constant in the probability distribution of y_i . With this simplification, Assumptions A3 and A4 would be made unconditional and the counterparts would now simply state that the probability distribution of ε_i involves none of the constants in $\mathbf{X}$.

Social scientists are almost never able to analyze experimental data, and relatively few of their models are built around nonrandom regressors. Clearly, for example, in any model of the macroeconomy, it would be difficult to defend such an asymmetric treatment of aggregate data. Realistically, we have to allow the data on $\mathbf{x}_i$ to be random the same as y_i . So an alternative formulation is to assume that $\mathbf{x}_i$ is a random vector and our formal assumption concerns the nature of the random process that produces $\mathbf{x}_i$. If $\mathbf{x}_i$ is taken to be a random vector, then Assumptions A1 through A4 become a statement about the joint distribution of y_i and $\mathbf{x}_i$. The precise nature of the regressor and how we view the sampling process will be a major determinant of our derivation of the statistical properties of our estimators and test statistics. In the end, the crucial assumption is A3, the uncorrelatedness of $\mathbf{X}$ and $\boldsymbol{\varepsilon}$. Now, we do note that this alternative is not completely satisfactory either, because $\mathbf{X}$ may well contain nonstochastic elements, including a constant, a time trend, and dummy variables that mark specific episodes in time. This makes for an ambiguous conclusion, but there is a straightforward and economically useful way out of it. We will allow $\mathbf{X}$ to be any mixture of constants and random variables, and the mean and variance of ε_i are both independent of all elements of $\mathbf{X}$.

ASSUMPTION A5: $\mathbf{X}$ may be fixed or random.	(2-10)
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2.3.6 NORMALITY

It is convenient to assume that the disturbances are **normally distributed**, with zero mean and constant variance. That is, we add normality of the distribution to Assumptions A3 and A4.

ASSUMPTION A6: $\varepsilon \mathbf{X} \sim N[\mathbf{0}, \sigma^2\mathbf{I}]$.	(2-11)
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In view of our description of the source of ε , the conditions of the central limit theorem will generally apply, at least approximately, and the normality assumption will be reasonable in most settings. A useful implication of Assumption A6 is that it implies that observations on ε_i are statistically independent as well as uncorrelated. [See the third point in Section B.9, (B-97) and (B-99).]

Normality is usually viewed as an unnecessary and possibly inappropriate addition to the regression model. Except in those cases in which some alternative distribution is explicitly assumed, as in the stochastic frontier model discussed in Chapter 19, the normality assumption may be quite reasonable. But the assumption is not necessary

to obtain most of the results we use in multiple regression analysis. It will prove useful as a starting point in constructing confidence intervals and test statistics, as shown in Section 4.7 and Chapter 5. But it will be possible to discard this assumption and retain for practical purposes the important statistical results we need for the investigation.

2.3.7 INDEPENDENCE AND EXOGENEITY

The term *independent* has been used several ways in this chapter.

In Section 2.2, the right-hand-side variables in the model are denoted the independent variables. Here, the notion of independence refers to the sources of variation. In the context of the model, the variation in the independent variables arises from sources that are outside of the process being described. Thus, in our health services versus income example in the introduction, we have suggested a theory for how variation in demand for services is associated with variation in income and, possibly, variation in insurance coverage. But, we have not suggested an explanation of the sample variation in income; income is assumed to vary for reasons that are outside the scope of the model. Nor have we suggested a behavioral model for insurance take up. This will be a convenient definition to use for **exogeneity** of a variable x .

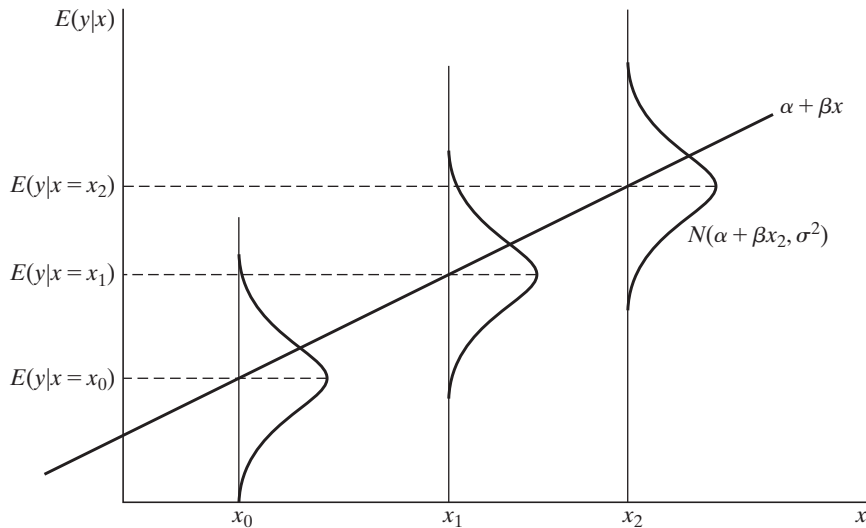
The assumption in (2-6), $E[\varepsilon_i | \mathbf{X}] = 0$, is mean independence. Its implication is that variation in the disturbances in our data is not explained by variation in the independent variables. Situations in which $E[\varepsilon_i | \mathbf{X}] \neq 0$ arise frequently, as we will explore in Chapter 8 and others. When $E[\varepsilon | x] \neq 0$, x is **endogenous** in the model. The most straightforward instance is a left-out variable. Consider the model in Example 2.2. In a simple model that contains only *Education* but which has inappropriately omitted *Age*, it would follow that *Age* implicitly appears in the disturbance:

$$\text{Income} = \gamma_1 + \gamma_2 \text{Education} + (\gamma_3 \text{Age} + u) = \gamma_1 + \gamma_2 \text{Education} + \varepsilon.$$

If *Education* and (the hidden variable) *Age* are correlated, then *Education* is endogenous in this equation, which is no longer a regression because $E[\varepsilon | \text{Education}] = \gamma_3 E[\text{Age} | \text{Education}] + E[u | \text{Education}] \neq 0$.

We have also assumed in Section 2.3.4 that the disturbances are uncorrelated with each other (Assumption A4 in Table 2.1). This implies that $E[\varepsilon_i | \varepsilon_j] = 0$ when $i \neq j$ —the disturbances are also mean independent of each other. Conditional normality of the disturbances assumed in Section 2.3.6 (Assumption A6) implies that they are statistically independent of each other, which is a stronger result than mean independence and stronger than we will need in most applications.

Finally, Section 2.3.2 discusses the **linear independence** of the columns of the data matrix, $\mathbf{X}$. The notion of independence here is an algebraic one relating to the column rank of $\mathbf{X}$. In this instance, the underlying interpretation is that it must be possible for the variables in the model to vary linearly independently of each other. Thus, in Example 2.6, we find that it is not possible for the logs of surface area, aspect ratio, and height of a painting all to vary independently of one another. The modeling implication is that, if the variables cannot vary independently of each other, then it is not possible to analyze them in a linear regression model that assumes the variables can each vary while holding the others constant. There is an ambiguity in this discussion of independence of the variables. We have both *age* and *age squared* in a model in Example 2.2. These cannot vary independently, but there is no obstacle to formulating a linear regression

FIGURE 2.3 The Normal Linear Regression Model.

model containing both *age* and *age squared*. The resolution is that *age* and *age squared*, though not *functionally* independent, are *linearly* independent in $\mathbf{X}$. That is the crucial assumption in the linear regression model.

2.4 SUMMARY AND CONCLUSIONS

This chapter has framed the linear regression model, the basic platform for model building in econometrics. The assumptions of the classical regression model are summarized in Figure 2.3, which shows the two-variable case.

Key Terms and Concepts

- Autocorrelation
- Central limit theorem
- Conditional mean
- Conditional median
- Conditional variance
- Conditional variation
- Constant elasticity
- Counterfactual
- Covariate
- Dependent variable
- Deterministic relationship
- Disturbance
- Endogeneity
- Exogeneity
- Explained variable
- Explanatory variable
- Flexible functional form
- Full rank
- Heteroscedasticity
- Homoscedasticity
- Identification condition
- Impact of treatment on the treated
- Independent variable
- Law of Iterated Expectations
- Linear independence
- Linear regression model
- Loglinear model
- Mean independence
- Multiple linear regression model
- Nonautocorrelation
- Nonstochastic regressors
- Normality
- Normally distributed
- Path diagram
- Population regression equation
- Random sample
- Regressand
- Regression function
- Regressor
- Semilog
- Translog model

LEAST SQUARES REGRESSION



3.1 INTRODUCTION

This chapter examines the computation of the least squares regression model. A useful understanding of what is being computed when one uses least squares to compute the coefficients of the model can be developed before we turn to the statistical aspects. Section 3.2 will detail the computations of least squares regression. We then examine two particular aspects of the fitted equation:

- The crucial feature of the multiple regression model is its ability to provide the analyst a device for “holding other things constant.” In an earlier example, we considered the “partial effect” of an additional year of education, holding age constant in

$$\text{Earnings} = \gamma_1 + \gamma_2 \text{Education} + \gamma_3 \text{Age} + \varepsilon.$$

The theoretical exercise is simple enough. How do we do this in practical terms? How does the actual computation of the linear model produce the interpretation of partial effects? An essential insight is provided by the notion of partial regression coefficients. Sections 3.3 and 3.4 use the **Frisch–Waugh theorem** to show how the regression model controls for (i.e., holds constant) the effects of intervening variables.

- The model is proposed to describe the movement of an explained variable. In broad terms, $y = \mu(\mathbf{x}) + \varepsilon$. How well does the model do this? How can we measure the success? Sections 3.5 and 3.6 examine fit measures for the linear regression.

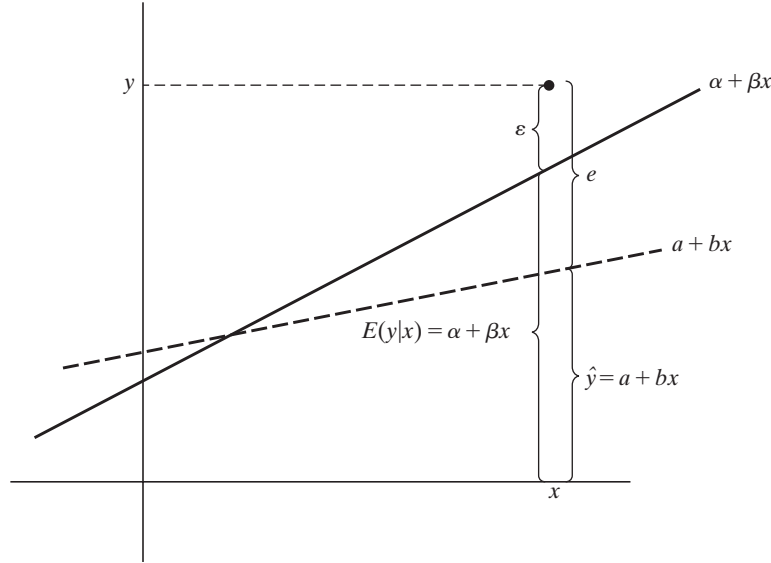
3.2 LEAST SQUARES REGRESSION

Consider a simple (the simplest) version of the model in the introduction,

$$\text{Earnings} = \alpha + \beta \text{Education} + \varepsilon.$$

The unknown parameters of the stochastic relationship, $y_i = \mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i$, are the objects of estimation. It is necessary to distinguish between unobserved population quantities, such as $\boldsymbol{\beta}$ and ε_i , and sample estimates of them, denoted $\mathbf{b}$ and e_i . The **population regression** is $E[y_i | \mathbf{x}_i] = \mathbf{x}_i' \boldsymbol{\beta}$, whereas our estimate of $E[y_i | \mathbf{x}_i]$ is denoted $\hat{y}_i = \mathbf{x}_i' \mathbf{b}$. The **disturbance** associated with the i th data point is $\varepsilon_i = y_i - \mathbf{x}_i' \boldsymbol{\beta}$. For any value of $\mathbf{b}$, we shall estimate ε_i with the **residual**

$$e_i = y_i - \mathbf{x}_i' \mathbf{b}.$$

FIGURE 3.1 Population and Sample Regression.

From the two definitions,

$$y_i = \mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i = \mathbf{x}_i' \mathbf{b} + e_i.$$

These results are summarized for a two-variable regression in Figure 3.1.

The **population quantity**, $\boldsymbol{\beta}$, is a vector of unknown parameters of the joint probability distribution of $(y, \mathbf{x})$ whose values we hope to estimate with our sample data, $(y_i, \mathbf{x}_i)$, $i = 1, \dots, n$. This is a problem of statistical inference that is discussed in Chapter 4 and much of the rest of the book. It is useful, however, to begin by considering the algebraic problem of choosing a vector $\mathbf{b}$ so that the fitted line $\mathbf{x}_i' \mathbf{b}$ is close to the data points. The measure of closeness constitutes a **fitting criterion**. The one used most frequently is **least squares**.¹

3.2.1 THE LEAST SQUARES COEFFICIENT VECTOR

The least squares coefficient vector minimizes the sum of squared residuals:

$$\sum_{i=1}^n e_{i0}^2 = \sum_{i=1}^n (y_i - \mathbf{x}_i' \mathbf{b}_0)^2, \quad (3-1)$$

where $\mathbf{b}_0$ denotes a choice for the coefficient vector. In matrix terms, minimizing the sum of squares in (3-1) requires us to choose $\mathbf{b}_0$ to

$$\text{Minimize}_{\mathbf{b}_0} S(\mathbf{b}_0) = \mathbf{e}_0' \mathbf{e}_0 = (\mathbf{y} - \mathbf{X} \mathbf{b}_0)' (\mathbf{y} - \mathbf{X} \mathbf{b}_0). \quad (3-2)$$

¹ We have yet to establish that the practical approach of fitting the line as closely as possible to the data by least squares leads to estimators with good statistical properties. This makes intuitive sense and is, indeed, the case. We shall return to the statistical issues in Chapter 4.

Expanding this gives

$$\mathbf{e}_0' \mathbf{e}_0 = \mathbf{y}' \mathbf{y} - \mathbf{b}_0' \mathbf{X}' \mathbf{y} - \mathbf{y}' \mathbf{X} \mathbf{b}_0 + \mathbf{b}_0' \mathbf{X}' \mathbf{X} \mathbf{b}_0 \quad (3-3)$$

or

$$S(\mathbf{b}_0) = \mathbf{y}' \mathbf{y} - 2\mathbf{y}' \mathbf{X} \mathbf{b}_0 + \mathbf{b}_0' \mathbf{X}' \mathbf{X} \mathbf{b}_0.$$

The necessary condition for a minimum is

$$\frac{\partial S(\mathbf{b}_0)}{\partial \mathbf{b}_0} = -2\mathbf{X}' \mathbf{y} + 2\mathbf{X}' \mathbf{X} \mathbf{b}_0 = \mathbf{0}.^2 \quad (3-4)$$

Let $\mathbf{b}$ be the solution (assuming it exists). Then, after manipulating (3-4), we find that $\mathbf{b}$ satisfies the **least squares normal equations**,

$$\mathbf{X}' \mathbf{X} \mathbf{b} = \mathbf{X}' \mathbf{y}. \quad (3-5)$$

If the inverse of $\mathbf{X}' \mathbf{X}$ exists, which follows from the full column rank assumption (Assumption A2 in Section 2.3), then the solution is

$$\mathbf{b} = (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' \mathbf{y}. \quad (3-6)$$

For this solution to minimize the sum of squares, the second derivatives matrix,

$$\frac{\partial^2 S(\mathbf{b}_0)}{\partial \mathbf{b}_0 \partial \mathbf{b}_0'} = 2\mathbf{X}' \mathbf{X},$$

must be a positive definite matrix. Let $q = \mathbf{c}' \mathbf{X}' \mathbf{X} \mathbf{c}$ for some arbitrary nonzero vector $\mathbf{c}$. (The multiplication by 2 is irrelevant.) Then

$$q = \mathbf{v}' \mathbf{v} = \sum_{i=1}^n v_i^2, \quad \text{where } \mathbf{v} = \mathbf{X} \mathbf{c}.$$

Unless every element of $\mathbf{v}$ is zero, q is positive. But if $\mathbf{v}$ could be zero, then $\mathbf{v}$ would be a linear combination of the columns of $\mathbf{X}$ that equals $\mathbf{0}$, which contradicts Assumption A2, that $\mathbf{X}$ has full column rank. Because $\mathbf{c}$ is arbitrary, q is positive for every nonzero $\mathbf{c}$, which establishes that $2\mathbf{X}' \mathbf{X}$ is positive definite. Therefore, if $\mathbf{X}$ has full column rank, then the least squares solution $\mathbf{b}$ is unique and minimizes the sum of squared residuals.

3.2.2 APPLICATION: AN INVESTMENT EQUATION

To illustrate the computations in a multiple regression, we consider an example based on the macroeconomic data in Appendix Table F3.1. To estimate an investment equation, we first convert the investment series in Table F3.1 to real terms by dividing them by the GDP deflator and then scale the series so that they are measured in trillions of dollars. The real GDP series is the quantity index reported in the Economic Report of the President (2016). The other variables in the regression are a time trend (1, 2, . . .), an interest rate (the prime rate), and the yearly rate of inflation in the Consumer Price Index. These produce the data matrices listed in Table 3.1. Consider first a regression of real investment on a constant, the time trend, and real GDP, which correspond to x_1 , x_2 ,

² See Appendix A.8 for discussion of calculus results involving matrices and vectors.

TABLE 3.1 Data Matrices

<i>Real Investment (Y)</i>	<i>Constant (I)</i>	<i>Trend (T)</i>	<i>Real GDP (G)</i>	<i>Interest Rate (R)</i>	<i>Inflation Rate (P)</i>
2.484	1	1	87.1	9.23	3.4
2.311	1	2	88.0	6.91	1.6
2.265	1	3	89.5	4.67	2.4
2.339	1	4	92.0	4.12	1.9
2.556	1	5	95.5	4.34	3.3
2.759	1	6	98.7	6.19	3.4
2.828	1	7	101.4	7.96	2.5
y = 2.717	X = 1	8	103.2	8.05	4.1
2.445	1	9	102.9	5.09	0.1
1.878	1	10	100.0	3.25	2.7
2.076	1	11	102.5	3.25	1.5
2.168	1	12	104.2	3.25	3.0
2.356	1	13	105.6	3.25	1.7
2.482	1	14	109.0	3.25	1.5
2.637	1	15	111.6	3.25	0.8

Notes:

1. Data from 2000–2014 obtained from Tables B-3, B-10, and B17 from Economic Report of the President: https://www.whitehouse.gov/sites/default/files/docs/2015_erp_appendix_b.pdf.
2. Results are based on the values shown. Slightly different results are obtained if the raw data on investment and the GNP deflator in Table F3.1 are input to the computer program and used to compute $\text{real investment} = \text{gross investment}/(0.01 * \text{GNP deflator})$ internally.

and x_3 . (For reasons to be discussed in Chapter 21, this is probably not a well-specified equation for these macroeconomic variables. It will suffice for a simple numerical example, however.) Inserting the specific variables of the example into (3-5), we have

$$\begin{aligned}
 b_1 n &+ b_2 \sum_i T_i &+ b_3 \sum_i G_i &= \sum_i Y_i, \\
 b_1 \sum_i T_i &+ b_2 \sum_i T_i^2 &+ b_3 \sum_i T_i G_i &= \sum_i T_i Y_i, \\
 b_1 \sum_i G_i &+ b_2 \sum_i T_i G_i &+ b_3 \sum_i G_i^2 &= \sum_i G_i Y_i.
 \end{aligned}$$

A solution for b_1 can be obtained by dividing the first equation by n and rearranging it to obtain

$$\begin{aligned}
 b_1 &= \bar{Y} - b_2 \bar{T} - b_3 \bar{G} \\
 &= 2.41882 - b_2 \times 8 - b_3 \times 99.4133.
 \end{aligned} \tag{3-7}$$

Insert this solution in the second and third equations, and rearrange terms again to yield a set of two equations:

$$\begin{aligned}
 b_2 \sum_i (T_i - \bar{T})^2 &+ b_3 \sum_i (T_i - \bar{T})(G_i - \bar{G}) &= \sum_i (T_i - \bar{T})(Y_i - \bar{Y}), \\
 b_2 \sum_i (G_i - \bar{G})(T_i - \bar{T}) &+ b_3 \sum_i (G_i - \bar{G})^2 &= \sum_i (G_i - \bar{G})(Y_i - \bar{Y}).
 \end{aligned}$$

This result shows the nature of the solution for the slopes, which can be computed from the sums of squares and cross products of the deviations of the variables from their

means. Letting lowercase letters indicate variables measured as deviations from the sample means, we find that the normal equations are

$$\begin{aligned} b_2 \sum_i t_i^2 + b_3 \sum_i t_i g_i &= \sum_i t_i y_i \\ b_2 \sum_i g_i t_i + b_3 \sum_i g_i^2 &= \sum_i g_i y_i \end{aligned}$$

and the least squares solutions for b_2 and b_3 are

$$\begin{aligned} b_2 &= \frac{\sum_i t_i y_i \sum_i g_i^2 - \sum_i g_i y_i \sum_i t_i g_i}{\sum_i t_i^2 \sum_i g_i^2 - (\sum_i g_i t_i)^2} = \frac{-1.6351(792.857) - 4.22255(451.9)}{280(792.857) - (451.9)^2} = -0.180169, \\ b_3 &= \frac{\sum_i g_i y_i \sum_i t_i^2 - \sum_i t_i y_i \sum_i t_i g_i}{\sum_i t_i^2 \sum_i g_i^2 - (\sum_i g_i t_i)^2} = \frac{4.22255(280) - (-1.6351)(451.9)}{280(792.857) - (451.9)^2} = 0.1080157. \end{aligned} \quad (3-8)$$

With these solutions in hand, b_1 can now be computed using (3-7); $b_1 = -6.8780284$.

Suppose that we just regressed investment on the constant and GDP, omitting the time trend. At least some of the correlation between real investment and real GDP that we observe in the data will be explainable because both variables have an obvious time trend. (The trend in investment clearly has two parts, before and after the crash of 2007–2008.) Consider how this shows up in the regression computation. Denoting by “ b_{yx} ” the slope in the simple, **bivariate regression** of variable y on a constant and the variable x , we find that the slope in this reduced regression would be

$$b_{YG} = \frac{\sum_i g_i y_i}{\sum_i g_i^2} = 0.00533. \quad (3-9)$$

By manipulating the earlier expression for b_3 and using the definition of the sample correlation between G and T , $r_{GT}^2 = (\sum_i g_i t_i)^2 / (\sum_i g_i^2 \sum_i t_i^2)$, we obtain

$$b_{YG|T} = \frac{b_{YG}}{1 - r_{GT}^2} - \frac{b_{YT} b_{TG}}{1 - r_{GT}^2} = b_{YG} - \left(\frac{b_{YT} b_{TG} - r_{GT}^2 b_{YG}}{1 - r_{GT}^2} \right) = 0.1080157. \quad (3-10)$$

(The notation “ $b_{YG|T}$ ” used on the left-hand side is interpreted to mean the slope in the regression of Y on G and a constant “in the presence of T .”) The slope in the **multiple regression** differs from that in the simple regression by a factor of 20, by including a correction that accounts for the influence of the additional variable T on both Y and G . For a striking example of this effect, in the simple regression of real investment on a time trend, $b_{YT} = -1.6351/280 = -0.00584$. But, in the multiple regression, after we account for the influence of GNP on real investment, the slope on the time trend is -0.180169 . The general result for a three-variable regression in which x_1 is a constant term is

$$b_{Y2|3} = \frac{b_{Y2} - b_{Y3} b_{32}}{1 - r_{23}^2}. \quad (3-11)$$

It is clear from this expression that the magnitudes of $b_{Y2|3}$ and b_{Y2} can be quite different. They need not even have the same sign. The result just seen is worth emphasizing; the coefficient on a variable in the simple regression [e.g., Y on $(1, G)$] will generally not be the same as the one on that variable in the multiple regression [e.g., Y on $(1, T, G)$] if the new variable and the old one are correlated. But, note that b_{YG} in (3-9) will be the same as $b_3 = b_{YG|T}$ in (3-8) if $\sum_i t_i g_i = 0$, that is, if T and G are not correlated.

In practice, you will never actually compute a multiple regression by hand or with a calculator. For a regression with more than three variables, the tools of matrix algebra are indispensable (as is a computer). Consider, for example, an enlarged model of investment that includes—in addition to the constant, time trend, and GDP—an interest rate and the rate of inflation. Least squares requires the simultaneous solution of five normal equations. Letting $\mathbf{X}$ and $\mathbf{y}$ denote the full data matrices shown previously, the normal equations in (3-5) are

$$\begin{bmatrix} 15.000 & 120.00 & 1491.2 & 76.05 & 33.90 \\ 120.000 & 1240.0 & 12381.5 & 522.06 & 244.10 \\ 1491.2 & 12381.5 & 149038 & 7453.03 & 3332.83 \\ 76.06 & 522.06 & 7453.03 & 446.323 & 186.656 \\ 33.90 & 244.10 & 3332.83 & 186.656 & 93.33 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \end{bmatrix} = \begin{bmatrix} 36.28230 \\ 288.624 \\ 3611.17 \\ 188.176 \\ 82.7731 \end{bmatrix}.$$

The solution is

$$\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = (-6.25441, -0.161342, 0.0994684, 0.0196656, -0.0107206)'.$$

3.2.3 ALGEBRAIC ASPECTS OF THE LEAST SQUARES SOLUTION

The normal equations are

$$\mathbf{X}'\mathbf{X}\mathbf{b} - \mathbf{X}'\mathbf{y} = -\mathbf{X}'(\mathbf{y} - \mathbf{X}\mathbf{b}) = -\mathbf{X}'\mathbf{e} = \mathbf{0}. \quad (3-12)$$

Hence, for every column $\mathbf{x}_k$ of $\mathbf{X}$, $\mathbf{x}_k'\mathbf{e} = 0$. If the first column of $\mathbf{X}$ is a column of 1s, which we denote $\mathbf{i}$, then there are three implications.

1. *The least squares residuals sum to zero.* This implication follows from $\mathbf{x}_1'\mathbf{e} = \mathbf{i}'\mathbf{e} = \sum_i e_i = 0$.
2. *The regression hyperplane passes through the point of means of the data.* The first normal equation implies that $\bar{y} = \bar{\mathbf{x}}'\mathbf{b}$. This follows from $\sum_i e_i = \sum_i (y_i - \mathbf{x}_i'\mathbf{b}) = 0$ by dividing by n .
3. *The mean of the fitted values from the regression equals the mean of the actual values.* This implication follows from point 2 because the fitted values are $\mathbf{x}_i'\mathbf{b}$.

It is important to note that none of these results need hold if the regression does not contain a constant term.

3.2.4 PROJECTION

The vector of least squares residuals is

$$\mathbf{e} = \mathbf{y} - \mathbf{X}\mathbf{b}. \quad (3-13)$$

Inserting the result in (3-6) for $\mathbf{b}$ gives

$$\mathbf{e} = \mathbf{y} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = (\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')\mathbf{y} = \mathbf{M}\mathbf{y}. \quad (3-14)$$

The $n \times n$ matrix $\mathbf{M}$ defined in (3-14) is fundamental in regression analysis. You can easily show that $\mathbf{M}$ is both symmetric ($\mathbf{M} = \mathbf{M}'$) and idempotent ($\mathbf{M} = \mathbf{M}^2$). In view of (3-13), we can interpret $\mathbf{M}$ as a matrix that produces the vector of least squares residuals

in the regression of $\mathbf{y}$ on $\mathbf{X}$ when it premultiplies any vector $\mathbf{y}$. It will be convenient later to refer to this matrix as a “**residual maker**.” Matrices of this form will appear repeatedly in our development to follow.

DEFINITION 3.1: Residual Maker

Let the $n \times K$ full column rank matrix, $\mathbf{X}$ be composed of columns $(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_K)$, and let $\mathbf{y}$ be an $n \times 1$ column vector. The matrix, $\mathbf{M} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is a “residual maker” in that when $\mathbf{M}$ premultiplies a vector, $\mathbf{y}$, the result, $\mathbf{M}\mathbf{y}$, is the column vector of residuals in the least squares regression of $\mathbf{y}$ on $\mathbf{X}$.

It follows from the definition that

$$\mathbf{MX} = \mathbf{0}, \quad (3-15)$$

because if a column of $\mathbf{X}$ is regressed on $\mathbf{X}$, a perfect fit will result and the residuals will be zero.

Result (3-13) implies that $\mathbf{y} = \mathbf{X}\mathbf{b} + \mathbf{e}$, which is the sample analog to Assumption A1, (2-3). (See Figure 3.1 as well.) The least squares results partition $\mathbf{y}$ into two parts, the fitted values $\hat{\mathbf{y}} = \mathbf{X}\mathbf{b}$ and the residuals, $\mathbf{e} = \mathbf{M}\mathbf{y}$. [See Section A.3.7, especially (A-54).] Because $\mathbf{MX} = \mathbf{0}$, these two parts are orthogonal. Now, given (3-13),

$$\hat{\mathbf{y}} = \mathbf{y} - \mathbf{e} = \mathbf{I}\mathbf{y} - \mathbf{M}\mathbf{y} = (\mathbf{I} - \mathbf{M})\mathbf{y} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{P}\mathbf{y}. \quad (3-16)$$

The matrix $\mathbf{P}$ is a **projection matrix**. It is the matrix formed from $\mathbf{X}$ such that when a vector $\mathbf{y}$ is premultiplied by $\mathbf{P}$, the result is the fitted values in the least squares regression of $\mathbf{y}$ on $\mathbf{X}$. This is also the **projection** of the vector $\mathbf{y}$ into the column space of $\mathbf{X}$. (See Sections A3.5 and A3.7.) By multiplying it out, you will find that, like $\mathbf{M}$, $\mathbf{P}$ is symmetric and idempotent. Given the earlier results, it also follows that $\mathbf{M}$ and $\mathbf{P}$ are orthogonal;

$$\mathbf{PM} = \mathbf{MP} = \mathbf{0}.$$

As might be expected from (3-15),

$$\mathbf{PX} = \mathbf{X}.$$

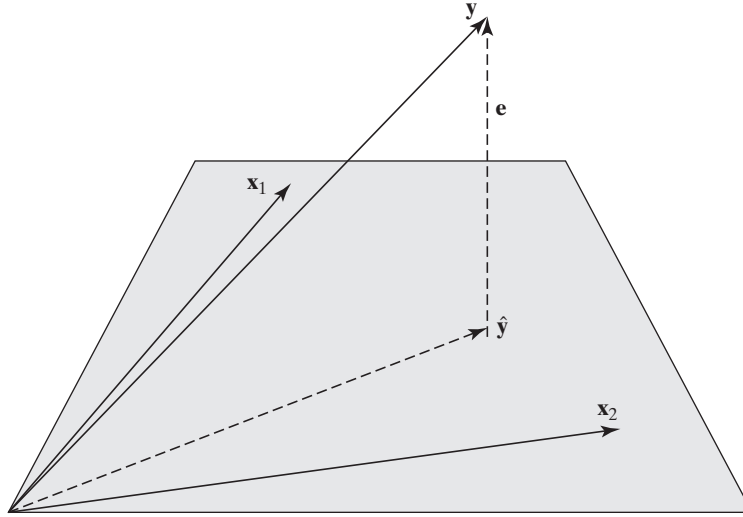
As a consequence of (3-14) and (3-16), we can see that least squares partitions the vector $\mathbf{y}$ into two orthogonal parts,

$$\mathbf{y} = \mathbf{P}\mathbf{y} + \mathbf{M}\mathbf{y} = \text{projection} + \text{residual}.$$

The result is illustrated in Figure 3.2 for the two-variable case. The gray-shaded plane is the column space of $\mathbf{X}$. The projection and residual are the orthogonal dashed rays. We can also see the Pythagorean theorem at work in the sums of squares,

$$\begin{aligned} \mathbf{y}'\mathbf{y} &= \mathbf{y}'\mathbf{P}'\mathbf{P}\mathbf{y} + \mathbf{y}'\mathbf{M}'\mathbf{M}\mathbf{y} \\ &= \hat{\mathbf{y}}'\hat{\mathbf{y}} + \mathbf{e}'\mathbf{e}. \end{aligned}$$

The sample linear projection of y on $\mathbf{x}$, $Proj(y|\mathbf{x})$, is an extremely useful device in empirical research. Linear least squares regression is often the starting point for model development. We will find in developing the regression model that if the population conditional mean function in Assumption A1, $E[y|\mathbf{x}]$, is linear in $\mathbf{x}$, then $E[y|\mathbf{x}]$ is also

FIGURE 3.2 Projection of $\mathbf{y}$ into the Column Space of $\mathbf{X}$.

the population counterpart to the projection of $\mathbf{y}$ on $\mathbf{x}$. We will be able to show that $\text{Proj}(\mathbf{y}|\mathbf{x})$ estimates $\mathbf{x}' \{E[\mathbf{x}\mathbf{x}']\}^{-1}E[\mathbf{x}\mathbf{y}]$, which appears implicitly in (3-16), is also $E[\mathbf{y}|\mathbf{x}]$. If the conditional mean function is not linear in $\mathbf{x}$, then the projection of $\mathbf{y}$ on $\mathbf{x}$ will still estimate a useful descriptor of the joint distribution of $\mathbf{y}$ and $\mathbf{x}$.

3.3 PARTITIONED REGRESSION AND PARTIAL REGRESSION

It is common to specify a multiple regression model when, in fact, interest centers on only one or a subset of the full set of variables—the remaining variables are often viewed as “controls.” Consider the earnings equation discussed in the Introduction. Although we are primarily interested in the effect of education on earnings, age is, of necessity, included in the model. The question we consider here is what computations are involved in obtaining, in isolation, the coefficients of a subset of the variables in a multiple regression (e.g., the coefficient of education in the aforementioned regression).

Suppose that the regression involves two sets of variables, $\mathbf{X}_1$ and $\mathbf{X}_2$. Thus,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}.$$

What is the algebraic solution for $\mathbf{b}_2$? The **normal equations** are

$$\begin{aligned} (1) \quad & \begin{bmatrix} \mathbf{X}_1'\mathbf{X}_1 & \mathbf{X}_1'\mathbf{X}_2 \end{bmatrix} \begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{X}_1'\mathbf{y} \\ \mathbf{X}_2'\mathbf{y} \end{bmatrix}. \\ (2) \quad & \end{aligned} \tag{3-17}$$

A solution can be obtained by using the partitioned inverse matrix of (A-74). Alternatively, (1) and (2) in (3-17) can be manipulated directly to solve for $\mathbf{b}_2$. We first solve (1) for $\mathbf{b}_1$:

$$\begin{aligned} \mathbf{X}_1'\mathbf{X}_1\mathbf{b}_1 + \mathbf{X}_1'\mathbf{X}_2\mathbf{b}_2 &= \mathbf{X}_1'\mathbf{y}, \\ \mathbf{b}_1 &= (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{y} - (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\mathbf{b}_2 = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'(\mathbf{y} - \mathbf{X}_2\mathbf{b}_2). \end{aligned} \tag{3-18}$$

This solution states that $\mathbf{b}_1$ is the set of coefficients in the regression of $\mathbf{y}$ on $\mathbf{X}_1$, minus a correction vector. We digress briefly to examine an important result embedded in (3-18). Suppose that $\mathbf{X}_1'\mathbf{X}_2 = \mathbf{0}$. Then, $\mathbf{b}_1 = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{y}$, which is simply the coefficient vector in the regression of $\mathbf{y}$ on $\mathbf{X}_1$. The general result is given in the following theorem.

THEOREM 3.1 Orthogonal Partitioned Regression

In the linear least squares multiple regression of $\mathbf{y}$ on two sets of variables $\mathbf{X}_1$ and $\mathbf{X}_2$, if the two sets of variables are orthogonal, then the separate coefficient vectors can be obtained by separate regressions of $\mathbf{y}$ on $\mathbf{X}_1$ alone and $\mathbf{y}$ on $\mathbf{X}_2$ alone.

Proof: The assumption of the theorem is that $\mathbf{X}_1'\mathbf{X}_2 = \mathbf{0}$ in the normal equations in (3-17). Inserting this assumption into (3-18) produces the immediate solution for $\mathbf{b}_1 = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{y}$ and likewise for $\mathbf{b}_2$.

If the two sets of variables $\mathbf{X}_1$ and $\mathbf{X}_2$ are not orthogonal, then the solutions for $\mathbf{b}_1$ and $\mathbf{b}_2$ found by (3-17) and (3-18) are more involved than just the simple regressions in Theorem 3.1. The more general solution is suggested by the following theorem:

THEOREM 3.2 Frisch–Waugh (1933)–Lovell (1963) Theorem³

In the linear least squares regression of vector $\mathbf{y}$ on two sets of variables, $\mathbf{X}_1$ and $\mathbf{X}_2$, the subvector $\mathbf{b}_2$ is the set of coefficients obtained when the residuals from a regression of $\mathbf{y}$ on $\mathbf{X}_1$ alone are regressed on the set of residuals obtained when each column of $\mathbf{X}_2$ is regressed on $\mathbf{X}_1$.

To prove Theorem 3.2, begin from equation (2) in (3-17), which is

$$\mathbf{X}_2'\mathbf{X}_1\mathbf{b}_1 + \mathbf{X}_2'\mathbf{X}_2\mathbf{b}_2 = \mathbf{X}_2'\mathbf{y}.$$

Now, insert the result for $\mathbf{b}_1$ that appears in (3-18) into this result. This produces

$$\mathbf{X}_2'\mathbf{X}_1(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{y} - \mathbf{X}_2'\mathbf{X}_1(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\mathbf{b}_2 + \mathbf{X}_2'\mathbf{X}_2\mathbf{b}_2 = \mathbf{X}_2'\mathbf{y}.$$

After collecting terms, the solution is

$$\begin{aligned}\mathbf{b}_2 &= [\mathbf{X}_2'(\mathbf{I} - \mathbf{X}_1(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1')\mathbf{X}_2]^{-1}[\mathbf{X}_2'(\mathbf{I} - \mathbf{X}_1(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1')\mathbf{y}] \\ &= (\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2)^{-1}(\mathbf{X}_2'\mathbf{M}_1\mathbf{y}).\end{aligned}\tag{3-19}$$

³The theorem, such as it was, appeared in the first volume of *Econometrica*, in the introduction to the paper: “The partial trend regression method can never, indeed, achieve anything which the individual trend method cannot, because the two methods lead by definition to identically the same results.” Thus, Frisch and Waugh were concerned with the (lack of) difference between a regression of a variable $\mathbf{y}$ on a time trend variable, $\mathbf{t}$, and another variable, $\mathbf{x}$, compared to the regression of a detrended $\mathbf{y}$ on a detrended $\mathbf{x}$, where detrending meant computing the residuals of the respective variable on a constant and the time trend, $\mathbf{t}$. A concise statement of the theorem and its matrix formulation were added later by Lovell (1963).

The $\mathbf{M}_1$ matrix appearing in the parentheses inside each set of parentheses is the “residual maker” defined in (3-14) and Definition 3.1, in this case defined for a regression on the columns of $\mathbf{X}_1$. Thus, $\mathbf{M}_1\mathbf{X}_2$ is a matrix of residuals; each column of $\mathbf{M}_1\mathbf{X}_2$ is a vector of residuals in the regression of the corresponding column of $\mathbf{X}_2$ on the variables in $\mathbf{X}_1$. By exploiting the fact that $\mathbf{M}_1$, like $\mathbf{M}$, is symmetric and idempotent, we can rewrite (3-19) as

$$\mathbf{b}_2 = (\mathbf{X}_2^*\mathbf{X}_2^*)^{-1}\mathbf{X}_2^*\mathbf{y}_*, \quad (3-20)$$

where $\mathbf{X}_2^* = \mathbf{M}_1\mathbf{X}_2$ and $\mathbf{y}_* = \mathbf{M}_1\mathbf{y}$. This result is fundamental in regression analysis.

This process is commonly called **partialing out** or **netting out** the effect of $\mathbf{X}_1$. For this reason, the coefficients in a multiple regression are often called the **partial regression coefficients**. The application of Theorem 3.2 to the computation of a single coefficient as suggested at the beginning of this section is detailed in the following: Consider the regression of $\mathbf{y}$ on a set of variables $\mathbf{X}$ and an additional variable $\mathbf{z}$. Denote the coefficients $\mathbf{b}$ and c , respectively.

COROLLARY 3.2.1 Individual Regression Coefficients

The coefficient on $\mathbf{z}$ in a multiple regression of $\mathbf{y}$ on $\mathbf{W} = [\mathbf{X}, \mathbf{z}]$ is computed as $c = (\mathbf{z}'\mathbf{M}_\mathbf{X}\mathbf{z})^{-1}(\mathbf{z}'\mathbf{M}_\mathbf{X}\mathbf{y}) = (\mathbf{z}_'\mathbf{z}_*)^{-1}\mathbf{z}_*'\mathbf{y}_*$ where $\mathbf{z}_*$ and $\mathbf{y}_*$ are the residual vectors from least squares regressions of $\mathbf{z}$ and $\mathbf{y}$ on $\mathbf{X}$; $\mathbf{z}_* = \mathbf{M}_\mathbf{X}\mathbf{z}$ and $\mathbf{y}_* = \mathbf{M}_\mathbf{X}\mathbf{y}$ where $\mathbf{M}_\mathbf{X}$ is defined in (3-14).*

Proof: This is an application of Theorem 3.2 in which $\mathbf{X}_1$ is $\mathbf{X}$ and $\mathbf{X}_2$ is $\mathbf{z}$.

In terms of Example 2.2, we could obtain the coefficient on education in the multiple regression by first regressing earnings and education on age (or age and age squared) and then using the residuals from these regressions in a simple regression. In the classic application of this latter observation, Frisch and Waugh (1933) noted that in a time-series setting, the same results were obtained whether a regression was fitted with a time-trend variable or the data were first “detrended” by netting out the effect of time, as noted earlier, and using just the detrended data in a simple regression.

Consider the case in which $\mathbf{X}_1$ is $\mathbf{i}$, a constant term that is a column of 1s in the first column of $\mathbf{X}$, and $\mathbf{X}_2$ is a set of variables. The solution for $\mathbf{b}_2$ in this case will then be the slopes in a regression that contains a constant term. Using Theorem 3.2 the vector of residuals for any variable, $\mathbf{x}$, in $\mathbf{X}_2$ will be

$$\begin{aligned} \mathbf{x}_* &= \mathbf{x} - \mathbf{i}(\mathbf{i}'\mathbf{i})^{-1}\mathbf{i}'\mathbf{x} \\ &= \mathbf{x} - \mathbf{i}(1/n)\mathbf{i}'\mathbf{x} \\ &= \mathbf{x} - \mathbf{i}\bar{x} \\ &= \mathbf{M}^0\mathbf{x}. \end{aligned} \quad (3-21)$$

(See Section A.5.4 where we have developed this result purely algebraically.) For this case, then, the residuals are deviations from the sample mean. Therefore, each column of $\mathbf{M}_1\mathbf{X}_2$ is the original variable, now in the form of deviations from the mean. This general result is summarized in the following corollary.

COROLLARY 3.2.2 Regression with a Constant Term

The slopes in a multiple regression that contains a constant term can be obtained by transforming the data to deviations from their means and then regressing the variable y in deviation form on the explanatory variables, also in deviation form.

[We used this result in (3-8).] Having obtained the coefficients on $\mathbf{X}_2$, how can we recover the coefficients on $\mathbf{X}_1$ (the constant term)? One way is to repeat the exercise while reversing the roles of $\mathbf{X}_1$ and $\mathbf{X}_2$. But there is an easier way. We have already solved for $\mathbf{b}_2$. Therefore, we can use (3-18) in a solution for $\mathbf{b}_1$. If $\mathbf{X}_1$ is just a column of 1s, then the first of these produces the familiar result

$$b_1 = \bar{y} - \bar{x}_2 b_2 - \cdots - \bar{x}_K b_K$$

[which is used in (3-7)].

Theorem 3.2 and Corollaries 3.2.1 and 3.2.2 produce a useful interpretation of the **partitioned regression** when the model contains a constant term. According to Theorem 3.1, if the columns of $\mathbf{X}$ are orthogonal, that is, $\mathbf{X}'_k \mathbf{x}_m = 0$ for columns k and m , then the separate regression coefficients in the regression of y on $\mathbf{X}$ when $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_K]$ are simply $\mathbf{x}'_k y / \mathbf{x}'_k \mathbf{x}_k$. When the regression contains a constant term, we can compute the multiple regression coefficients by regression of y in mean deviation form on the columns of $\mathbf{X}$, also in deviations from their means. In this instance, the *orthogonality* of the columns means that the sample covariances (and correlations) of the variables are zero. The result is another theorem:

THEOREM 3.3 Orthogonal Regression

If the multiple regression of y on $\mathbf{X}$ contains a constant term and the variables in the regression are uncorrelated, then the multiple regression slopes are the same as the slopes in the individual simple regressions of y on a constant and each variable in turn.

Proof: The result follows from Theorems 3.1 and 3.2.

3.4 PARTIAL REGRESSION AND PARTIAL CORRELATION COEFFICIENTS

The use of multiple regression involves a conceptual experiment that we might not be able to carry out in practice, the *ceteris paribus* analysis familiar in economics. To pursue the earlier example, a regression equation relating earnings to age and education enables us to do the experiment of comparing the earnings of two individuals of the same age with different education levels, *even if the sample contains no such pair of individuals*. It is this characteristic of the regression that is implied by the term partial regression coefficients. The way we obtain this result, as we have seen, is first to regress income and education on age and then to compute the residuals from this regression. By construction, age will not have any power in explaining variation in these residuals. Therefore, any

correlation between income and education after this “purging” is independent of (or after removing the effect of) age.

The same principle can be applied to the correlation between two variables. To continue our example, to what extent can we assert that this correlation reflects a direct relationship rather than that both income and education tend, on average, to rise as individuals become older? To find out, we would use a **partial correlation coefficient**, which is computed along the same lines as the partial regression coefficient. In the context of our example, the partial correlation coefficient between income and education, controlling for the effect of age, is obtained as follows:

1. y_* = the residuals in a regression of income on a constant and age.
2. z_* = the residuals in a regression of education on a constant and age.
3. The partial correlation r_{yz}^* is the simple correlation between y_* and z_* .

This calculation might seem to require a large amount of computation. Using Corollary 3.2.1, the two residual vectors in points 1 and 2 are $\mathbf{y}_* = \mathbf{M}\mathbf{y}$ and $\mathbf{z}_* = \mathbf{M}\mathbf{z}$ where $\mathbf{M} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is the residual maker defined in (3-14). We will assume that there is a constant term in $\mathbf{X}$ so that the vectors of residuals $\mathbf{y}_*$ and $\mathbf{z}_*$ have zero sample means. Then, the square of the partial correlation coefficient is

$$r_{yz}^{*2} = \frac{(\mathbf{z}_*' \mathbf{y}_*)^2}{(\mathbf{z}_*' \mathbf{z}_*)(\mathbf{y}_*' \mathbf{y}_*)}.$$

There is a convenient shortcut. Once the multiple regression is computed, the t ratio in (5-13) for testing the hypothesis that the coefficient equals zero (e.g., the last column of Table 4.6) can be used to compute

$$r_{yz}^{*2} = \frac{t_z^2}{t_z^2 + \text{degrees of freedom}}, \quad (3-22)$$

where the **degrees of freedom** is equal to $n - (K + 1)$; $K + 1$ is the number of variables in the regression plus the constant term. The proof of this less than perfectly intuitive result will be useful to illustrate some results on partitioned regression. We will rely on two useful theorems from least squares algebra. The first isolates a particular diagonal element of the inverse of a **moment matrix** such as $(\mathbf{X}'\mathbf{X})^{-1}$.

THEOREM 3.4 Diagonal Elements of the Inverse of a Moment Matrix

Let $\mathbf{W}$ denote the partitioned matrix $[\mathbf{X}, \mathbf{z}]$ —that is, the K columns of $\mathbf{X}$ plus an additional column labeled $\mathbf{z}$. The last diagonal element of $(\mathbf{W}'\mathbf{W})^{-1}$ is $(\mathbf{z}'\mathbf{M}_\mathbf{X}\mathbf{z})^{-1} = (\mathbf{z}_*' \mathbf{z}_*)^{-1}$ where $\mathbf{z}_* = \mathbf{M}_\mathbf{X}\mathbf{z}$ and $\mathbf{M}_\mathbf{X} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.

Proof: This is an application of the partitioned inverse formula in (A-74) where $A_{11} = \mathbf{X}'\mathbf{X}$, $A_{12} = \mathbf{X}'\mathbf{z}$, $A_{21} = \mathbf{z}'\mathbf{X}$ and $A_{22} = \mathbf{z}'\mathbf{z}$. Note that this theorem generalizes the development in Section A.2.8, where $\mathbf{X}$ contains only a constant term, $\mathbf{i}$.

We can use Theorem 3.4 to establish the result in (3-22). Let c and $\mathbf{u}$ denote the coefficient on $\mathbf{z}$ and the vector of residuals in the multiple regression of $\mathbf{y}$ on $\mathbf{W} = [\mathbf{X}, \mathbf{z}]$, respectively. Then, by definition, the squared t ratio that appears in (3-22) is

$$t_z^2 = \frac{c^2}{\left[\frac{\mathbf{u}'\mathbf{u}}{n - (K + 1)} \right] (\mathbf{W}'\mathbf{W})_{K+1, K+1}^{-1}},$$

where $(\mathbf{W}'\mathbf{W})_{K+1, K+1}^{-1}$ is the $(K + 1)$ (last) diagonal element of $(\mathbf{W}'\mathbf{W})^{-1}$. [The bracketed term appears in (4-17).] The theorem states that this element of the matrix equals $(\mathbf{z}'_*\mathbf{z}_*)^{-1}$. From Corollary 3.2.1, we also have that $c^2 = [(\mathbf{z}'_*\mathbf{y}_*)/(\mathbf{z}'_*\mathbf{z}_*)]^2$. For convenience, let $DF = n - (K + 1)$. Then, $t_z^2 = \frac{(\mathbf{z}'_*\mathbf{y}_*/\mathbf{z}'_*\mathbf{z}_*)^2}{(\mathbf{u}'\mathbf{u}/DF)(\mathbf{z}'_*\mathbf{z}_*)^{-1}} = \frac{(\mathbf{z}'_*\mathbf{y}_*)^2 DF}{(\mathbf{u}'\mathbf{u})(\mathbf{z}'_*\mathbf{z}_*)^{-1}}$. It follows that the result in (3-22) is equivalent to

$$\frac{t_z^2}{t_z^2 + DF} = \frac{\frac{(\mathbf{z}'_*\mathbf{y}_*)^2 DF}{(\mathbf{u}'\mathbf{u})(\mathbf{z}'_*\mathbf{z}_*)}}{\frac{(\mathbf{z}'_*\mathbf{y}_*)^2 DF}{(\mathbf{u}'\mathbf{u})(\mathbf{z}'_*\mathbf{z}_*)} + DF} = \frac{\frac{(\mathbf{z}'_*\mathbf{y}_*)^2}{(\mathbf{u}'\mathbf{u})(\mathbf{z}'_*\mathbf{z}_*)}}{\frac{(\mathbf{z}'_*\mathbf{y}_*)^2}{(\mathbf{u}'\mathbf{u})(\mathbf{z}'_*\mathbf{z}_*)} + 1} = \frac{(\mathbf{z}'_*\mathbf{y}_*)^2}{(\mathbf{z}'_*\mathbf{y}_*)^2 + (\mathbf{u}'\mathbf{u})(\mathbf{z}'_*\mathbf{z}_*)}.$$

Divide numerator and denominator by $(\mathbf{z}'_*\mathbf{z}_*)(\mathbf{y}'_*\mathbf{y}_*)$ to obtain

$$\frac{t_z^2}{t_z^2 + DF} = \frac{(\mathbf{z}'_*\mathbf{y}_*)^2/((\mathbf{z}'_*\mathbf{z}_*)(\mathbf{y}'_*\mathbf{y}_*))}{(\mathbf{z}'_*\mathbf{y}_*)^2/((\mathbf{z}'_*\mathbf{z}_*)(\mathbf{y}'_*\mathbf{y}_*)) + ((\mathbf{u}'\mathbf{u})(\mathbf{z}'_*\mathbf{z}_*)/((\mathbf{z}'_*\mathbf{z}_*)(\mathbf{y}'_*\mathbf{y}_*)))} = \frac{r_{yz}^{*2}}{r_{yz}^{*2} + (\mathbf{u}'\mathbf{u})/(\mathbf{y}'_*\mathbf{y}_*)}. \quad (3-23)$$

We will now use a second theorem to manipulate $\mathbf{u}'\mathbf{u}$ and complete the derivation. The result we need is given in Theorem 3.5.

Returning to the derivation, then, $\mathbf{e}'\mathbf{e} = \mathbf{y}'_*\mathbf{y}_*$ and $c^2(\mathbf{z}'_*\mathbf{z}_*) = (\mathbf{z}'_*\mathbf{y}_*)^2/(\mathbf{z}'_*\mathbf{z}_*)$. Therefore,

$$\frac{\mathbf{u}'\mathbf{u}}{\mathbf{y}'_*\mathbf{y}_*} = \frac{\mathbf{y}'_*\mathbf{y}_* - (\mathbf{z}'_*\mathbf{y}_*)^2/\mathbf{z}'_*\mathbf{z}_*}{\mathbf{y}'_*\mathbf{y}_*} = 1 - r_{yz}^{*2}.$$

Inserting this in the denominator of (3-23) produces the result we sought.

THEOREM 3.5 Change in the Sum of Squares When a Variable Is Added to a Regression

If $\mathbf{e}'\mathbf{e}$ is the sum of squared residuals when $\mathbf{y}$ is regressed on $\mathbf{X}$ and $\mathbf{u}'\mathbf{u}$ is the sum of squared residuals when $\mathbf{y}$ is regressed on $\mathbf{X}$ and $\mathbf{z}$, then

$$\mathbf{u}'\mathbf{u} = \mathbf{e}'\mathbf{e} - c^2(\mathbf{z}'_*\mathbf{z}_*) \leq \mathbf{e}'\mathbf{e}, \quad (3-24)$$

where c is the coefficient on $\mathbf{z}$ in the long regression of $\mathbf{y}$ on $[\mathbf{X}, \mathbf{z}]$ and $\mathbf{z}_* = \mathbf{Mz}$ is the vector of residuals when $\mathbf{z}$ is regressed on $\mathbf{X}$.

Proof: In the long regression of $\mathbf{y}$ on $\mathbf{X}$ and $\mathbf{z}$, the vector of residuals is $\mathbf{u} = \mathbf{y} - \mathbf{Xd} - \mathbf{zc}$. Note that unless $\mathbf{X}'\mathbf{z} = \mathbf{0}$, $\mathbf{d}$ will not equal $\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$. (See Section 4.3.2.) Moreover, unless $c = 0$, $\mathbf{u}$ will not equal $\mathbf{e} = \mathbf{y} - \mathbf{Xb}$. From Corollary 3.2.1, $c = (\mathbf{z}'_*\mathbf{z}_*)^{-1}(\mathbf{z}'_*\mathbf{y}_*)$. From (3-18), we also have that the coefficients on $\mathbf{X}$ in this long regression are

$$\mathbf{d} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{y} - \mathbf{zc}) = \mathbf{b} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{z}c.$$

Inserting this expression for $\mathbf{d}$ in that for $\mathbf{u}$ gives

$$\mathbf{u} = \mathbf{y} - \mathbf{X}\mathbf{b} + \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{z}\mathbf{c} - \mathbf{z}\mathbf{c} = \mathbf{e} - \mathbf{M}_{\mathbf{X}}\mathbf{z}\mathbf{c} = \mathbf{e} - \mathbf{z}_*\mathbf{c}.$$

Then,

$$\mathbf{u}'\mathbf{u} = \mathbf{e}'\mathbf{e} + \mathbf{c}'(\mathbf{z}'_*\mathbf{z}_*)\mathbf{c} - 2\mathbf{c}'(\mathbf{z}'_*\mathbf{e}).$$

But, $\mathbf{e} = \mathbf{M}_{\mathbf{X}}\mathbf{y} = \mathbf{y}_*$ and $\mathbf{z}'_*\mathbf{e} = \mathbf{z}'_*\mathbf{y}_* = \mathbf{c}'(\mathbf{z}'_*\mathbf{z}_*)$. Inserting this result in $\mathbf{u}'\mathbf{u}$ immediately above gives the result in the theorem.

Example 3.1 Partial Correlations

For the data in the application in Section 3.2.2, the simple correlations between investment and the regressors, r_{YK} , and the partial correlations, r_{YK}^* , between investment and the four regressors (given the other variables) are listed in Table 3.2. As is clear from the table, there is no necessary relation between the simple and partial correlation coefficients. One thing worth noting is that the signs of the partial correlations are the same as those of the coefficients, but not necessarily the same as the signs of the raw correlations. Note the difference in the coefficient on *Inflation*.

3.5 GOODNESS OF FIT AND THE ANALYSIS OF VARIANCE

The original fitting criterion, the sum of squared residuals, suggests a measure of the fit of the regression line to the data. However, as can easily be verified, the sum of squared residuals can be scaled arbitrarily just by multiplying all the values of y by the desired scale factor. Because the fitted values of the regression are based on the values of $\mathbf{x}$, we might ask instead whether *variation* in $\mathbf{x}$ is a good predictor of *variation* in y . Figure 3.3 shows three possible cases for a simple linear regression model, $y = \beta_1 + \beta_2x + \varepsilon$. The measure of fit described here embodies both the fitting criterion and the covariation of y and $\mathbf{x}$.

Variation of the dependent variable is defined in terms of deviations from its mean, $(y_i - \bar{y})$. The **total variation** in y is the sum of squared deviations:

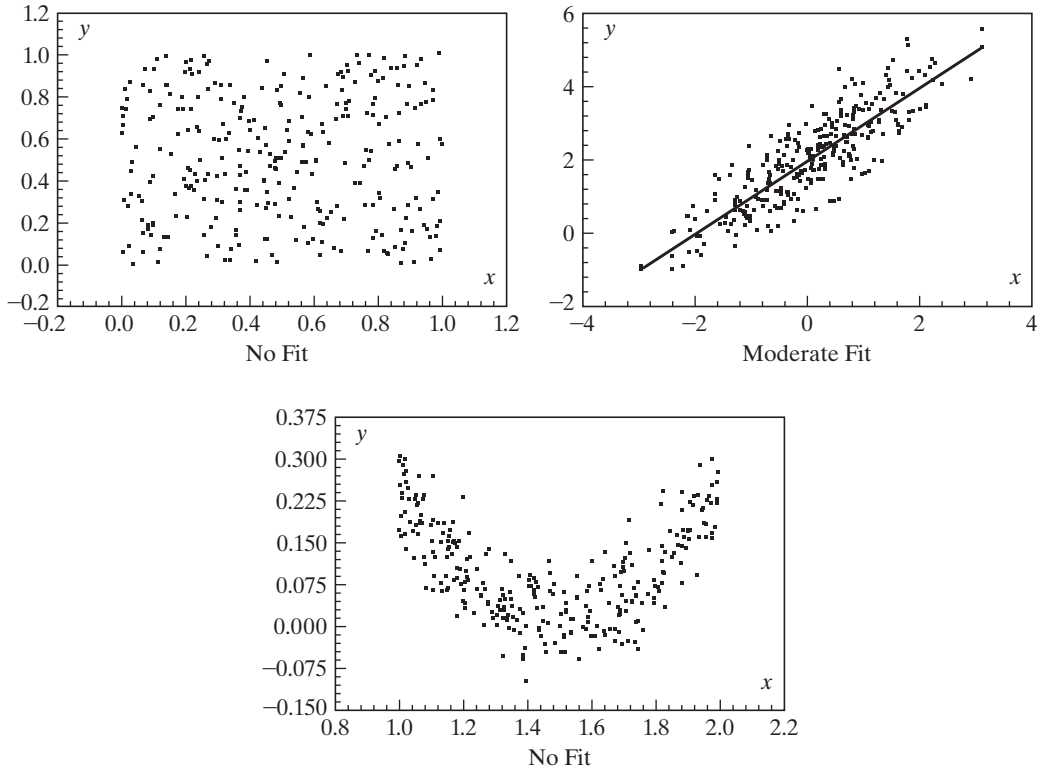
$$\text{SST} = \sum_{i=1}^n (y_i - \bar{y})^2.$$

In terms of the regression equation, we may write the full set of observations as

$$\mathbf{y} = \mathbf{X}\mathbf{b} + \mathbf{e} = \hat{\mathbf{y}} + \mathbf{e}.$$

TABLE 3.2 Correlations of Investment with Other Variables (DF = 10)

Variable	Coefficient	<i>t</i> Ratio	Simple Correlation	Partial Correlation
Trend	-0.16134	-3.42	-0.09965	-0.73423
RealGDP	0.09947	4.12	0.15293	0.79325
Interest	0.01967	0.58	0.55006	0.18040
Inflation	-0.01072	-0.27	0.19332	-0.08507

FIGURE 3.3 Sample Data.

For an individual observation, we have

$$y_i = \hat{y}_i + e_i = \mathbf{x}_i' \mathbf{b} + e_i.$$

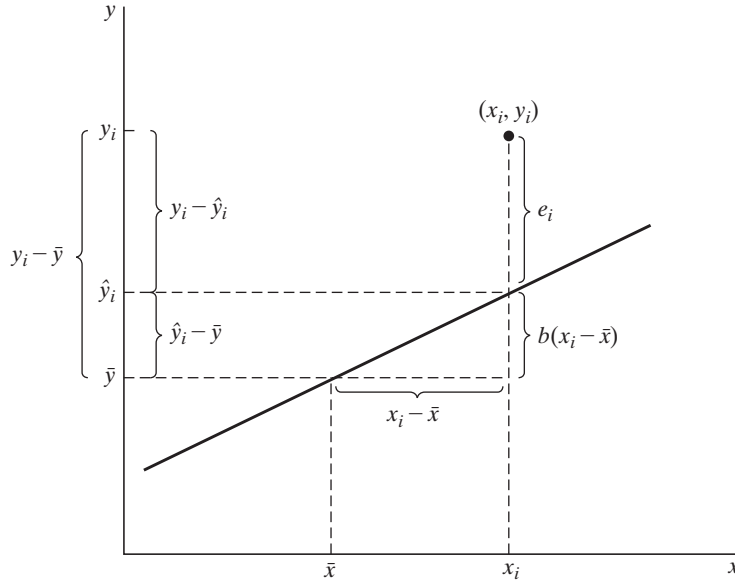
If the regression contains a constant term, then the residuals will sum to zero and the mean of the predicted values of y_i will equal the mean of the actual values. Subtracting $\bar{y}$ from both sides and using this result and result 2 in Section 3.2.3 gives

$$y_i - \bar{y} = \hat{y}_i - \bar{y} + e_i = (\mathbf{x}_i - \bar{\mathbf{x}})' \mathbf{b} + e_i.$$

Figure 3.4 illustrates the computation for the two-variable regression. Intuitively, the regression would appear to fit well if the deviations of y from its mean are more largely accounted for by deviations of x from its mean than by the residuals. Since both terms in this decomposition sum to zero, to quantify this fit, we use the sums of squares instead. For the full set of observations, we have

$$\mathbf{M}^0 \mathbf{y} = \mathbf{M}^0 \mathbf{X} \mathbf{b} + \mathbf{M}^0 \mathbf{e},$$

where $\mathbf{M}^0$ is the $n \times n$ idempotent matrix that transforms observations into deviations from sample means. [See (3-21) and Section A.2.8; $\mathbf{M}^0$ is a residual maker for $\mathbf{X} = \mathbf{i}$.] The column of $\mathbf{M}^0 \mathbf{X}$ corresponding to the constant term is zero, and, since the residuals

FIGURE 3.4 Decomposition of y_i .

already have mean zero, $\mathbf{M}^0 \mathbf{e} = \mathbf{e}$. Then, since $\mathbf{e}' \mathbf{M}^0 \mathbf{X} = \mathbf{e}' \mathbf{X} = \mathbf{0}$, the total sum of squares is

$$\mathbf{y}' \mathbf{M}^0 \mathbf{y} = \mathbf{b}' \mathbf{X}' \mathbf{M}^0 \mathbf{X} \mathbf{b} + \mathbf{e}' \mathbf{e}.$$

Write this as total sum of squares = regression sum of squares + error sum of squares, or

$$\text{SST} = \text{SSR} + \text{SSE}. \quad (3-25)$$

(Note that this is the same partitioning that appears at the end of Section 3.2.4.)

We can now obtain a measure of how well the regression line fits the data by using the

$$\text{coefficient of determination: } \frac{\text{SSR}}{\text{SST}} = \frac{\mathbf{b}' \mathbf{X}' \mathbf{M}^0 \mathbf{X} \mathbf{b}}{\mathbf{y}' \mathbf{M}^0 \mathbf{y}} = 1 - \frac{\mathbf{e}' \mathbf{e}}{\mathbf{y}' \mathbf{M}^0 \mathbf{y}} = 1 - \frac{\sum_{i=1}^n e_i^2}{\sum_{i=1}^n (y_i - \bar{y})^2}. \quad (3-26)$$

The coefficient of determination is denoted R^2 . As we have shown, it must be between 0 and 1, and it measures the proportion of the total variation in y that is accounted for by variation in the regressors. It equals zero if the regression is a horizontal line, that is, if all the elements of $\mathbf{b}$ except the constant term are zero. In this case, the predicted values of y are always $\bar{y}$, so deviations of $\mathbf{x}$ from its mean do not translate into different predictions for y . As such, $\mathbf{x}$ has no explanatory power. The other extreme, $R^2 = 1$, occurs if the values of $\mathbf{x}$ and y all lie in the same hyperplane (on a straight line for a two-variable regression) so that the residuals are all zero. If all the values of y_i lie on a vertical line, then R^2 has no meaning and cannot be computed.

Regression analysis is often used for forecasting. In this case, we are interested in how well the regression model predicts movements in the dependent variable. With this in mind, an equivalent way to compute R^2 is also useful. First, the sum of squares for the predicted values is

$$\sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = \hat{\mathbf{y}}' \mathbf{M}^0 \hat{\mathbf{y}} = \mathbf{b}' \mathbf{X}' \mathbf{M}^0 \mathbf{X} \mathbf{b},$$

but $\hat{\mathbf{y}} = \mathbf{X} \mathbf{b}$, $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{e}$, $\mathbf{M}^0 \mathbf{e} = \mathbf{e}$, and $\mathbf{X}' \mathbf{e} = \mathbf{0}$, so $\hat{\mathbf{y}}' \mathbf{M}^0 \hat{\mathbf{y}} = \hat{\mathbf{y}}' \mathbf{M}^0 \mathbf{y} = \sum_{i=1}^n (\hat{y}_i - \bar{y})(y_i - \bar{y})$. Multiply $R^2 = \hat{\mathbf{y}}' \mathbf{M}^0 \hat{\mathbf{y}} / \mathbf{y}' \mathbf{M}^0 \mathbf{y} = \hat{\mathbf{y}}' \mathbf{M}^0 \mathbf{y} / \mathbf{y}' \mathbf{M}^0 \mathbf{y}$ by 1 = $\hat{\mathbf{y}}' \mathbf{M}^0 \mathbf{y} / \hat{\mathbf{y}}' \mathbf{M}^0 \hat{\mathbf{y}}$ to obtain

$$R^2 = \frac{[\sum_i (y_i - \bar{y})(\hat{y}_i - \bar{y})]^2}{[\sum_i (y_i - \bar{y})^2][\sum_i (\hat{y}_i - \bar{y})^2]}, \quad (3-27)$$

which is the squared correlation between the observed values of y and the predictions produced by the estimated regression equation.

Example 3.2 Fit of a Consumption Function

The data plotted in Figure 2.1 are listed in Appendix Table F2.1. For these data, where y is C and x is X , we have $\bar{y} = 273.2727$, $\bar{x} = 323.2727$, $S_{yy} = 12,618.182$, $S_{xx} = 12,300.182$, and $S_{xy} = 8,423.182$, so $SST = 12,618.182$, $b = 8,423.182/12,300.182 = 0.6848014$, $SSR = b^2 S_{xx} = 5,768.2068$, and $SSE = SST - SSR = 6,849.975$. Then $R^2 = b^2 S_{xx} / SST = 0.457135$. As can be seen in Figure 2.1, this is a moderate fit, although it is not particularly good for aggregate time-series data. On the other hand, it is clear that not accounting for the anomalous wartime data has degraded the fit of the model. This value is the R^2 for the model indicated by the solid line in the figure. By simply omitting the years 1942–1945 from the sample and doing these computations with the remaining seven observations—the dashed line—we obtain an R^2 of 0.93379. Alternatively, by creating a variable WAR which equals 1 in the years 1942–1945 and zero otherwise and including this in the model, which produces the model shown by the two dashed lines, the R^2 rises to 0.94450.

We can summarize the calculation of R^2 in an **analysis of variance** table, which might appear as shown in Table 3.3.

Example 3.3 Analysis of Variance for the Investment Equation

The analysis of variance table for the investment equation of Section 3.2.2 is given in Table 3.4.

3.5.1 THE ADJUSTED R -SQUARED AND A MEASURE OF FIT

There are some problems with the use of R^2 in analyzing **goodness of fit**. The first concerns the number of degrees of freedom used up in estimating the parameters.

TABLE 3.3 Analysis of Variance Table

Source	Sum of Squares	Degrees of Freedom	Mean Square
Regression	$\mathbf{b}' \mathbf{X}' \mathbf{y} - n\bar{y}^2$	$K - 1$ (assuming a constant term)	
Residual	$\mathbf{e}' \mathbf{e}$	$n - K$ (including the constant term)	s^2
Total	$\mathbf{y}' \mathbf{y} - n\bar{y}^2$	$n - 1$	s_y^2
R^2	$1 - \mathbf{e}' \mathbf{e} / (\mathbf{y}' \mathbf{y} - n\bar{y}^2)$		

TABLE 3.4 Analysis of Variance for the Investment Equation

<i>Source</i>	<i>Sum of Squares</i>	<i>Degrees of Freedom</i>	<i>Mean Square</i>
Regression	0.75621	4	
Residual	0.20368	10	0.02037
Total	0.95989	14	0.06856
R^2	0.78781		

[See (3-22) and Table 3.3.] R^2 will never decrease when another variable is added to a regression equation. Equation (3-24) provides a convenient means for us to establish this result. Once again, we are comparing a regression of $\mathbf{y}$ on $\mathbf{X}$ with sum of squared residuals $\mathbf{e}'\mathbf{e}$ to a regression of $\mathbf{y}$ on $\mathbf{X}$ and an additional variable $\mathbf{z}$, which produces sum of squared residuals $\mathbf{u}'\mathbf{u}$. Recall the vectors of residuals $\mathbf{z}_* = \mathbf{M}\mathbf{z}$ and $\mathbf{y}_* = \mathbf{M}\mathbf{y} = \mathbf{e}$, which implies that $\mathbf{e}'\mathbf{e} = (\mathbf{y}'_*\mathbf{y}_*)$. Let c be the coefficient on $\mathbf{z}$ in the longer regression. Then $c = (\mathbf{z}'_*\mathbf{z}_*)^{-1}(\mathbf{z}'_*\mathbf{y}_*)$, and inserting this in (3-24) produces

$$\mathbf{u}'\mathbf{u} = \mathbf{e}'\mathbf{e} - \frac{(\mathbf{z}'_*\mathbf{y}_*)^2}{(\mathbf{z}'_*\mathbf{z}_*)} = \mathbf{e}'\mathbf{e}(1 - r_{yz}^{*2}), \quad (3-28)$$

where r_{yz}^* is the partial correlation between $\mathbf{y}$ and $\mathbf{z}$, controlling for $\mathbf{X}$. Now divide through both sides of the equality by $\mathbf{y}'\mathbf{M}^0\mathbf{y}$. From (3-26), $\mathbf{u}'\mathbf{u}/\mathbf{y}'\mathbf{M}^0\mathbf{y}$ is $(1 - R_{\mathbf{X}\mathbf{z}}^2)$ for the regression on $\mathbf{X}$ and $\mathbf{z}$ and $\mathbf{e}'\mathbf{e}/\mathbf{y}'\mathbf{M}^0\mathbf{y}$ is $(1 - R_{\mathbf{X}}^2)$. Rearranging the result produces the following:

THEOREM 3.6 Change in R^2 When a Variable Is Added to a Regression

Let $R_{\mathbf{X}\mathbf{z}}^2$ be the coefficient of determination in the regression of $\mathbf{y}$ on $\mathbf{X}$ and an additional variable $\mathbf{z}$, let $R_{\mathbf{X}}^2$ be the same for the regression of $\mathbf{y}$ on $\mathbf{X}$ alone, and let r_{yz}^* be the partial correlation between $\mathbf{y}$ and $\mathbf{z}$, controlling for $\mathbf{X}$. Then

$$R_{\mathbf{X}\mathbf{z}}^2 = R_{\mathbf{X}}^2 + (1 - R_{\mathbf{X}}^2) r_{yz}^{*2}. \quad (3-29)$$

Thus, the R^2 in the longer regression cannot be smaller. It is tempting to exploit this result by just adding variables to the model; R^2 will continue to rise to its limit of 1.⁴ The **adjusted R^2** (for degrees of freedom), which incorporates a penalty for these results, is computed as follows:

$$\bar{R}^2 = 1 - \frac{\mathbf{e}'\mathbf{e}/(n - K)}{\mathbf{y}'\mathbf{M}^0\mathbf{y}/(n - 1)}. \quad (3-30)$$

For computational purposes, the connection between R^2 and $\bar{R}^2$ is

$$\bar{R}^2 = 1 - \frac{n - 1}{n - K} (1 - R^2).$$

⁴This result comes at a cost, however. The parameter estimates become progressively less precise as we do so. We will pursue this result in Chapter 4.

The adjusted R^2 may decline when a variable is added to the set of independent variables. Indeed, $\bar{R}^2$ could even be negative. To consider an admittedly extreme case, suppose that $\mathbf{x}$ and $\mathbf{y}$ have a sample correlation of zero. Then the adjusted R^2 will equal $-1/(n - 2)$. Whether $\bar{R}^2$ rises or falls when a variable is added to the model depends on whether the contribution of the new variable to the fit of the regression more than offsets the correction for the loss of an additional degree of freedom. The general result (the proof of which is left as an exercise) is as follows.

THEOREM 3.7 Change in $\bar{R}^2$ When a Variable Is Added to a Regression

In a multiple regression, $\bar{R}^2$ will fall (rise) when the variable x is deleted from the regression if the square of the t ratio associated with this variable is greater (less) than 1.

We have shown that R^2 will never fall when a variable is added to the regression. We now consider this result more generally. The change in the residual sum of squares when a set of variables $\mathbf{X}_2$ is added to the regression is

$$\mathbf{e}'_1\mathbf{e}_1 - \mathbf{e}'_{1,2}\mathbf{e}_{1,2} = \mathbf{b}'_2\mathbf{X}'_2\mathbf{M}_1\mathbf{X}_2\mathbf{b}_2,$$

where $\mathbf{e}_1$ is the residuals when $\mathbf{y}$ is regressed on $\mathbf{X}_1$ alone and $\mathbf{e}_{1,2}$ indicates regression on *both* $\mathbf{X}_1$ and $\mathbf{X}_2$. The coefficient vector $\mathbf{b}_2$ is the coefficients on $\mathbf{X}_2$ in the multiple regression of $\mathbf{y}$ on $\mathbf{X}_1$ and $\mathbf{X}_2$. [See (3-19) and (3-20) for definitions of $\mathbf{b}_2$ and $\mathbf{M}_1$.] Therefore,

$$R^2_{1,2} = 1 - \frac{\mathbf{e}'_1\mathbf{e}_1 - \mathbf{b}'_2\mathbf{X}'_2\mathbf{M}_1\mathbf{X}_2\mathbf{b}_2}{\mathbf{y}'\mathbf{M}^0\mathbf{y}} = R^2_1 + \frac{\mathbf{b}'_2\mathbf{X}'_2\mathbf{M}_1\mathbf{X}_2\mathbf{b}_2}{\mathbf{y}'\mathbf{M}^0\mathbf{y}},$$

which is greater than R^2_1 unless $\mathbf{b}_2$ equals zero. ($\mathbf{M}_1\mathbf{X}_2$ could not be zero unless $\mathbf{X}_2$ is a linear function of $\mathbf{X}_1$, in which case the regression on $\mathbf{X}_1$ and $\mathbf{X}_2$ could not be computed.) This equation can be manipulated a bit further to obtain

$$R^2_{1,2} = R^2_1 + \frac{\mathbf{y}'\mathbf{M}_1\mathbf{y}}{\mathbf{y}'\mathbf{M}^0\mathbf{y}} \frac{\mathbf{b}'_2\mathbf{X}'_2\mathbf{M}_1\mathbf{X}_2\mathbf{b}_2}{\mathbf{y}'\mathbf{M}_1\mathbf{y}}.$$

But $\mathbf{y}'\mathbf{M}_1\mathbf{y} = \mathbf{e}'_1\mathbf{e}_1$, so the first term in the product is $1 - R^2_1$. The second is the **multiple correlation** in the regression of $\mathbf{M}_1\mathbf{y}$ on $\mathbf{M}_1\mathbf{X}_2$, or the partial correlation (after the effect of $\mathbf{X}_1$ is removed) in the regression of $\mathbf{y}$ on $\mathbf{X}_2$. Collecting terms, we have

$$R^2_{1,2} = R^2_1 + (1 - R^2_1)r^2_{y2.1}. \quad (3-31)$$

[This is the multivariate counterpart to (3-29).]

It is possible to push R^2 as high as desired (up to one) just by adding regressors to the model. This possibility motivates the use of the adjusted R^2 in (3-30), instead of R^2 as a method of choosing among alternative models. Since $\bar{R}^2$ incorporates a penalty for reducing the degrees of freedom while still revealing an improvement in fit, one possibility is to choose the specification that maximizes $\bar{R}^2$. It has been suggested that

the adjusted R^2 does not penalize the loss of degrees of freedom heavily enough.⁵ Some alternatives that have been proposed for comparing models (which we index by j) are a modification of the adjusted R squared, that minimizes Amemiya's (1985) **prediction criterion**,

$$PC_j = \frac{\mathbf{e}_j' \mathbf{e}_j}{n - K_j} \left(1 + \frac{K_j}{n} \right) = s_j^2 \left(1 + \frac{K_j}{n} \right),$$

$$\bar{R}_j^2 = 1 - \frac{n + K_j}{n - K_j} (1 - R_j^2).$$

Two other fitting criteria are the Akaike and Bayesian information criteria discussed in Section 5.10.1,

$$AIC_j = \ln \left(\frac{\mathbf{e}_j' \mathbf{e}_j}{n} \right) + \frac{2K}{n},$$

$$BIC_j = \ln \left(\frac{\mathbf{e}_j' \mathbf{e}_j}{n} \right) + \frac{K \ln n}{n}.$$

3.5.2 R -SQUARED AND THE CONSTANT TERM IN THE MODEL

A second difficulty with R^2 concerns the constant term in the model. The proof that $0 \leq R^2 \leq 1$ requires $\mathbf{X}$ to contain a column of 1s. If not, then (1) $\mathbf{M}^0 \mathbf{e} \neq \mathbf{e}$ and (2) $\mathbf{e}' \mathbf{M}^0 \mathbf{X} \neq \mathbf{0}$, and the term $2\mathbf{e}' \mathbf{M}^0 \mathbf{X} \mathbf{b}$ in $\mathbf{y}' \mathbf{M}^0 \mathbf{y} = (\mathbf{M}^0 \mathbf{X} \mathbf{b} + \mathbf{M}^0 \mathbf{e})' (\mathbf{M}^0 \mathbf{X} \mathbf{b} + \mathbf{M}^0 \mathbf{e})$ in the expansion preceding (3-25) will not drop out. Consequently, when we compute

$$R^2 = 1 - \frac{\sum_{i=1}^n e_i^2}{\sum_{i=1}^n (y_i - \bar{y})^2},$$

the result is unpredictable. It will never be higher and can be far lower than the same figure computed for the regression with a constant term included. It can even be negative. Computer packages differ in their computation of R^2 . An alternative computation,

$$R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \hat{\bar{y}})^2}{\sum_{i=1}^n (y_i - \bar{y})^2},$$

is equally problematic. Again, this calculation will differ from the one obtained with the constant term included; this time, R^2 may be larger than 1. Some computer packages bypass these difficulties by reporting a third " R^2 ," the squared sample correlation between the actual values of y and the fitted values from the regression. If the regression contains a constant term, then all three computations give the same answer. Even if not, this last one will always produce a value between zero and one. But it is not a proportion of variation explained. On the other hand, for the purpose of comparing models, this squared correlation might well be a useful descriptive device. It is important for users of computer packages to be aware of how the reported R^2 is computed.

⁵ See, for example, Amemiya (1985, pp. 50–51).

3.5.3 COMPARING MODELS

The value of R^2 of 0.94450 that we obtained for the consumption function in Example 3.2 seems high in an absolute sense. Is it? Unfortunately, there is no absolute basis for comparison. In fact, in using aggregate time-series data, coefficients of determination this high are routine. In terms of the values one normally encounters in cross sections, an R^2 of 0.5 is relatively high. Coefficients of determination in cross sections of individual data as high as 0.2 are sometimes noteworthy. The point of this discussion is that whether a regression line provides a good fit to a body of data depends on the setting.

Little can be said about the relative quality of fits of regression lines in different contexts or in different data sets even if they are supposedly generated by the same data-generating mechanism. One must be careful, however, even in a single context, to be sure to use the same basis for comparison for competing models. Usually, this concern is about how the dependent variable is computed. For example, a perennial question concerns whether a linear or loglinear model fits the data better. Unfortunately, the question cannot be answered with a direct comparison. An R^2 for the linear regression model is different from an R^2 for the loglinear model. Variation in y is different from variation in $\ln y$. The latter R^2 will typically be larger, but this does not imply that the loglinear model is a better fit in some absolute sense.

It is worth emphasizing that R^2 is a measure of *linear* association between x and y . For example, the third panel of Figure 3.3 shows data that might arise from the model

$$y_i = \alpha + \beta x_i + \gamma x_i^2 + \varepsilon_i.$$

The relationship between y and x in this model is nonlinear, and a linear regression of y on x would find no fit.

3.6 LINEARLY TRANSFORMED REGRESSION

As a final application of the tools developed in this chapter, we examine a purely algebraic result that is very useful for understanding the computation of linear regression models. In the regression of $\mathbf{y}$ on $\mathbf{X}$, suppose the columns of $\mathbf{X}$ are linearly transformed. Common applications would include changes in the units of measurement, say by changing units of currency, hours to minutes, or distances in miles to kilometers. Example 3.4 suggests a slightly more involved case. This is a useful practical, algebraic result. For example, it simplifies the analysis in the first application suggested, changing the units of measurement. If an independent variable is scaled by a constant, p , the regression coefficient will be scaled by $1/p$. There is no need to recompute the regression.

Example 3.4 Art Appreciation

Theory 1 of the determination of the auction prices of Monet paintings holds that the price is determined by the dimensions (width, W , and height, H) of the painting,

$$\begin{aligned}\ln \text{Price} &= \beta_1(1) + \beta_2 \ln W + \beta_3 \ln H + \varepsilon \\ &= \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon.\end{aligned}$$

Theory 2 claims, instead, that art buyers are interested specifically in surface area and aspect ratio,

$$\begin{aligned}\ln \text{Price} &= \gamma_1(1) + \gamma_2 \ln (WH) + \gamma_3 \ln (W/H) + \varepsilon \\ &= \gamma_1 z_1 + \gamma_2 z_2 + \gamma_3 z_3 + u.\end{aligned}$$

It is evident that $z_1 = x_1$, $z_2 = x_2 + x_3$, and $z_3 = x_2 - x_3$. In matrix terms, $\mathbf{Z} = \mathbf{XP}$ where

$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}, \mathbf{P}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 1/2 \\ 0 & 1/2 & -1/2 \end{bmatrix}.$$

The effect of a transformation on the linear regression of $\mathbf{y}$ on $\mathbf{X}$ compared to that of $\mathbf{y}$ on $\mathbf{Z}$ is given by Theorem 3.8. Thus, $\beta_1 = \gamma_1$, $\beta_2 = 1/2(\gamma_2 + \gamma_3)$, $\beta_3 = 1/2(\gamma_2 - \gamma_3)$.

THEOREM 3.8 Transformed Variables

In the linear regression of $\mathbf{y}$ on $\mathbf{Z} = \mathbf{XP}$ where $\mathbf{P}$ is a nonsingular matrix that transforms the columns of $\mathbf{X}$, the coefficients will equal $\mathbf{P}^{-1}\mathbf{b}$ where $\mathbf{b}$ is the vector of coefficients in the linear regression of $\mathbf{y}$ on $\mathbf{X}$, and the R^2 will be identical.

Proof: The coefficients are

$$\begin{aligned} \mathbf{d} &= (\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y} = [(\mathbf{XP})'(\mathbf{XP})]^{-1}(\mathbf{XP})'\mathbf{y} = (\mathbf{P}'\mathbf{X}'\mathbf{XP})^{-1}\mathbf{P}'\mathbf{X}'\mathbf{y} \\ &= \mathbf{P}^{-1}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{P}'\mathbf{P}^{-1}\mathbf{P}'\mathbf{X}'\mathbf{y} = \mathbf{P}^{-1}\mathbf{b}. \end{aligned}$$

The vector of residuals is $\mathbf{u} = \mathbf{y} - \mathbf{Z}(\mathbf{P}^{-1}\mathbf{b}) = \mathbf{y} - \mathbf{XPP}^{-1}\mathbf{b} = \mathbf{y} - \mathbf{Xb} = \mathbf{e}$. Since the residuals are identical, the numerator of $1 - R^2$ is the same, and the denominator is unchanged. This establishes the result.

3.7 SUMMARY AND CONCLUSIONS

This chapter has described the exercise of fitting a line (hyperplane) to a set of points using the method of least squares. We considered the primary problem first, using a data set of n observations on K variables. We then examined several aspects of the solution, including the nature of the projection and residual maker matrices and several useful algebraic results relating to the computation of the residuals and their sum of squares. We also examined the difference between gross or simple regression and correlation and multiple regression by defining partial regression coefficients and partial correlation coefficients. The Frisch–Waugh–Lovell Theorem (3.2) is a fundamentally useful tool in regression analysis that enables us to obtain the expression for a subvector of a vector of regression coefficients. We examined several aspects of the partitioned regression, including how the fit of the regression model changes when variables are added to it or removed from it. Finally, we took a closer look at the conventional measure of how well the fitted regression line predicts or “fits” the data.

Key Terms and Concepts

- Adjusted R^2
- Analysis of variance
- Bivariate regression
- Coefficient of determination
- Degrees of freedom
- Disturbance
- Fitting criterion
- Frisch–Waugh theorem
- Goodness of fit
- Least squares
- Least squares normal equations
- Moment matrix
- Multiple correlation
- Multiple regression
- Netting out
- Normal equations
- Orthogonal regression
- Partial correlation coefficient

- Partial regression coefficient
- Partialing out
- Partitioned regression
- Prediction criterion
- Population quantity
- Population regression
- Projection
- Projection matrix
- Residual
- Residual maker
- Total variation

Exercises

1. *The two-variable regression.* For the regression model $y = \alpha + \beta x + \varepsilon$,
 - a. Show that the least squares normal equations imply $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$.
 - b. Show that the solution for the constant term is $a = \bar{y} - b\bar{x}$.
 - c. Show that the solution for b is $b = [\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})] / [\sum_{i=1}^n (x_i - \bar{x})^2]$.
 - d. Prove that these two values uniquely minimize the sum of squares by showing that the diagonal elements of the second derivatives matrix of the sum of squares with respect to the parameters are both positive and that the determinant is $4n[(\sum_{i=1}^n x_i^2) - n\bar{x}^2] = 4n[\sum_{i=1}^n (x_i - \bar{x})^2]$, which is positive unless all values of x are the same.
2. *Change in the sum of squares.* Suppose that $\mathbf{b}$ is the least squares coefficient vector in the regression of $\mathbf{y}$ on $\mathbf{X}$ and that $\mathbf{c}$ is any other $K \times 1$ vector. Prove that the difference in the two sums of squared residuals is

$$(\mathbf{y} - \mathbf{Xc})'(\mathbf{y} - \mathbf{Xc}) - (\mathbf{y} - \mathbf{Xb})'(\mathbf{y} - \mathbf{Xb}) = (\mathbf{c} - \mathbf{b})'\mathbf{X}'\mathbf{X}(\mathbf{c} - \mathbf{b}).$$

Prove that this difference is positive.

3. *Partial Frisch and Waugh.* In the least squares regression of $\mathbf{y}$ on a constant and $\mathbf{X}$, to compute the regression coefficients on $\mathbf{X}$, we can first transform $\mathbf{y}$ to deviations from the mean $\bar{y}$ and, likewise, transform each column of $\mathbf{X}$ to deviations from the respective column mean; second, regress the transformed $\mathbf{y}$ on the transformed $\mathbf{X}$ without a constant. Do we get the same result if we only transform $\mathbf{y}$? What if we only transform $\mathbf{X}$?
4. *Residual makers.* What is the result of the matrix product $\mathbf{M}_1\mathbf{M}$ where $\mathbf{M}_1$ is defined in (3-19) and $\mathbf{M}$ is defined in (3-14)?
5. *Adding an observation.* A data set consists of n observations contained in $\mathbf{X}_n$ and $\mathbf{y}_n$. The least squares estimator based on these n observations is $\mathbf{b}_n = (\mathbf{X}_n'\mathbf{X}_n)^{-1}\mathbf{X}_n'\mathbf{y}_n$. Another observation, $\mathbf{x}_s$ and y_s , becomes available. Prove that the least squares estimator computed using this additional observation is

$$\mathbf{b}_{n,s} = \mathbf{b}_n + \frac{1}{1 + \mathbf{x}_s'(\mathbf{X}_n'\mathbf{X}_n)^{-1}\mathbf{x}_s} (\mathbf{X}_n'\mathbf{X}_n)^{-1} \mathbf{x}_s (y_s - \mathbf{x}_s'\mathbf{b}_n).$$

Note that the last term is e_s , the residual from the prediction of y_s using the coefficients based on $\mathbf{X}_n$ and $\mathbf{y}_n$. Conclude that the new data change the results of least squares only if the new observation on y cannot be perfectly predicted using the information already in hand.

6. *Deleting an observation.* A common strategy for handling a case in which an observation is missing data for one or more variables is to fill those missing variables with 0s and add a variable to the model that takes the value 1 for that one observation and 0 for all other observations. Show that this strategy is equivalent to discarding the observation as regards the computation of $\mathbf{b}$ but it does have an

effect on R^2 . Consider the special case in which $\mathbf{X}$ contains only a constant and one variable. Show that replacing missing values of x with the mean of the complete observations has the same effect as adding the new variable.

7. *Demand system estimation.* Let Y denote total expenditure on consumer durables, nondurables, and services and E_d , E_n , and E_s are the expenditures on the three categories. As defined, $Y = E_d + E_n + E_s$. Now, consider the expenditure system

$$\begin{aligned} E_d &= \alpha_d + \beta_d Y + \gamma_{dd} P_d + \gamma_{dn} P_n + \gamma_{ds} P_s + \varepsilon_d, \\ E_n &= \alpha_n + \beta_n Y + \gamma_{nd} P_d + \gamma_{nn} P_n + \gamma_{ns} P_s + \varepsilon_n, \\ E_s &= \alpha_s + \beta_s Y + \gamma_{sd} P_d + \gamma_{sn} P_n + \gamma_{ss} P_s + \varepsilon_s. \end{aligned}$$

Prove that if all equations are estimated by ordinary least squares, then the sum of the expenditure coefficients will be 1 and the four other column sums in the preceding model will be zero.

8. *Change in adjusted R^2 .* Prove that the adjusted R^2 in (3-30) rises (falls) when variable $\mathbf{x}_k$ is deleted from the regression if the square of the t ratio on $\mathbf{x}_k$ in the multiple regression is less (greater) than 1.
9. *Regression without a constant.* Suppose that you estimate a multiple regression first with, then without, a constant. Whether the R^2 is higher in the second case than the first will depend in part on how it is computed. Using the (relatively) standard method $R^2 = 1 - (\mathbf{e}'\mathbf{e}/\mathbf{y}'\mathbf{M}^0\mathbf{y})$, which regression will have a higher R^2 ?
10. Three variables, N , D , and Y , all have zero means and unit variances. A fourth variable is $C = N + D$. In the regression of C on Y , the slope is 0.8. In the regression of C on N , the slope is 0.5. In the regression of D on Y , the slope is 0.4. What is the sum of squared residuals in the regression of C on D ? There are 21 observations and all moments are computed using $1/(n - 1)$ as the divisor.
11. Using the matrices of sums of squares and cross products immediately preceding Section 3.2.3, compute the coefficients in the multiple regression of real investment on a constant, GNP, and the interest rate. Compute R^2 .
12. In the December 1969 *American Economic Review* (pp. 886–896), Nathaniel Leff reports the following least squares regression results for a cross section study of the effect of age composition on savings in 74 countries in 1964:

$$\begin{aligned} \ln S/Y &= 7.3439 + 0.1596 \ln Y/N + 0.0254 \ln G - 1.3520 \ln D_1 - 0.3990 \ln D_2, \\ \ln S/N &= 2.7851 + 1.1486 \ln Y/N + 0.0265 \ln G - 1.3438 \ln D_1 - 0.3966 \ln D_2, \end{aligned}$$

where S/Y = domestic savings ratio, S/N = per capita savings, Y/N = per capita income, D_1 = percentage of the population under 15, D_2 = percentage of the population over 64, and G = growth rate of per capita income. Are these results correct? Explain.⁶

13. *Is it possible to partition R^2 ?* The idea of “hierarchical partitioning” is to decompose R^2 into the contributions made by each variable in the multiple regression. That is, if $x_1, \dots, x_K$ are entered into a regression one at a time, then c_k is the incremental contribution of x_k such that given the order entered, $\sum_k c_k = R^2$ and the incremental

⁶ See Goldberger (1973) and Leff (1973) for discussion.

contribution of x_k is then c_k/R^2 . Of course, based on (3-31), we know that this is not a useful calculation.

- Argue based on (3-31) why it is not useful.
- Show using (3-31) that the computation is sensible if (and only if) all variables are orthogonal.
- For the investment example in Section 3.2.2, compute the incremental contribution of T if it is entered first in the regression. Now compute the incremental contribution of T if it is entered last.

Application

The data listed in Table 3.5 are extracted from Koop and Tobias's (2004) study of the relationship between wages and education, ability, and family characteristics. (See Appendix Table F3.2.) Their data set is a panel of 2,178 individuals with a total of 17,919 observations. Shown in the table are the first year and the time-invariant variables for the first 15 individuals in the sample. The variables are defined in the article.

Let $\mathbf{X}_1$ equal a constant, education, experience, and ability (the individual's own characteristics). Let $\mathbf{X}_2$ contain the mother's education, the father's education, and the number of siblings (the household characteristics). Let $\mathbf{y}$ be the log of the hourly wage.

- Compute the least squares regression coefficients in the regression of $\mathbf{y}$ on $\mathbf{X}_1$. Report the coefficients.
- Compute the least squares regression coefficients in the regression of $\mathbf{y}$ on $\mathbf{X}_1$ and $\mathbf{X}_2$. Report the coefficients.

TABLE 3.5 Subsample from Koop and Tobias Data

<i>Person</i>	<i>Education</i>	<i>In Wage</i>	<i>Experience</i>	<i>Ability</i>	<i>Mother's Education</i>	<i>Father's Education</i>	<i>Siblings</i>
1	13	1.82	1	1.00	12	12	1
2	15	2.14	4	1.50	12	12	1
3	10	1.56	1	-0.36	12	12	1
4	12	1.85	1	0.26	12	10	4
5	15	2.41	2	0.30	12	12	1
6	15	1.83	2	0.44	12	16	2
7	15	1.78	3	0.91	12	12	1
8	13	2.12	4	0.51	12	15	2
9	13	1.95	2	0.86	12	12	2
10	11	2.19	5	0.26	12	12	2
11	12	2.44	1	1.82	16	17	2
12	13	2.41	4	-1.30	13	12	5
13	12	2.07	3	-0.63	12	12	4
14	12	2.20	6	-0.36	10	12	2
15	12	2.12	3	0.28	10	12	3

- c. Regress each of the three variables in $\mathbf{X}_2$ on all the variables in $\mathbf{X}_1$ and compute the residuals from each regression. Arrange these new variables in the 15×3 matrix $\mathbf{X}_2^*$. What are the sample means of these three variables? Explain the finding.
- d. Using (3-26), compute the R^2 for the regression of $\mathbf{y}$ on $\mathbf{X}_1$ and $\mathbf{X}_2$. Repeat the computation for the case in which the constant term is omitted from $\mathbf{X}_1$. What happens to R^2 ?
- e. Compute the adjusted R^2 for the full regression including the constant term. Interpret your result.
- f. Referring to the result in part c, regress $\mathbf{y}$ on $\mathbf{X}_1$ and $\mathbf{X}_2^*$. How do your results compare to the results of the regression of $\mathbf{y}$ on $\mathbf{X}_1$ and $\mathbf{X}_2$? The comparison you are making is between the least squares coefficients when $\mathbf{y}$ is regressed on $\mathbf{X}_1$ and $\mathbf{M}_1\mathbf{X}_2$ and when $\mathbf{y}$ is regressed on $\mathbf{X}_1$ and $\mathbf{X}_2$. Derive the result theoretically. (Your numerical results should match the theory, of course.)

ESTIMATING THE REGRESSION MODEL BY LEAST SQUARES



4.1 INTRODUCTION

In this chapter, we will examine least squares in detail as an **estimator** of the parameters of the linear regression model (defined in Table 4.1). There are other candidates for estimating β . For example, we might use the coefficients that minimize the sum of absolute values of the residuals. We begin in Section 4.2 by considering the question “Why should we use least squares?” We will then analyze the estimator in detail. The question of which estimator to choose is based on the **statistical properties** of the candidates, such as unbiasedness, consistency, efficiency, and their **sampling distributions**. Section 4.3 considers **finite-sample properties** such as unbiasedness. The linear model is one of few settings in which the exact finite-sample properties of an estimator are known. In most cases, the only known properties are those that apply to large samples. We can approximate finite-sample behavior by using what we know about large-sample properties. In Section 4.4, we will examine the large-sample or **asymptotic properties** of the least squares estimator of the regression model.¹ Section 4.5 considers **robust inference**. The problem considered here is how to carry out inference when (real) data may not satisfy the assumptions of the basic linear model. Section 4.6 develops a method for inference based on functions of model parameters, rather than the estimates themselves.

Discussions of the properties of an estimator are largely concerned with **point estimation**—that is, in how to use the sample information as effectively as possible to produce the best single estimate of the model parameters. **Interval estimation**, considered in Section 4.7, is concerned with computing estimates that make explicit the uncertainty inherent in using randomly sampled data to estimate population quantities. We will consider some applications of interval estimation of parameters and some functions of parameters in Section 4.7. One of the most familiar applications of interval estimation is using the model to predict the dependent variable and to provide a plausible range of uncertainty for that prediction. Section 4.8 considers prediction and forecasting using the estimated regression model.

The analysis assumes that the data in hand correspond to the assumptions of the model. In Section 4.9, we consider several practical problems that arise in analyzing nonexperimental data. Assumption A2, full rank of $\mathbf{X}$, is taken as a given. As we noted in Section 2.3.2, when this assumption is not met, the model is not estimable, regardless of the sample size. **Multicollinearity**, the near failure of this assumption in real-world

¹This discussion will use results on asymptotic distributions. It may be helpful to review Appendix D before proceeding to Section 4.4.

TABLE 4.1 Assumptions of the Classical Linear Regression Model

- A1. Linearity:** $y_i = x_{i1}\beta_1 + x_{i2}\beta_2 + \cdots + x_{iK}\beta_K + \varepsilon_i = \mathbf{x}_i'\boldsymbol{\beta} + \varepsilon_i$. For the sample, $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$.
- A2. Full rank:** The $n \times K$ sample data matrix, $\mathbf{X}$, has full column rank for every $n \geq K$.
- A3. Exogeneity of the independent variables:** $E[\varepsilon_i | x_{j1}, x_{j2}, \dots, x_{jK}] = 0, i, j = 1, \dots, n$. There is no correlation between the disturbances and the independent variables. $E[\boldsymbol{\varepsilon} | \mathbf{X}] = \mathbf{0}$.
- A4. Homoscedasticity and nonautocorrelation:** Each disturbance, ε_i , has the same finite variance; $E[\varepsilon_i^2 | \mathbf{X}] = \sigma^2$. Every disturbance ε_i is uncorrelated with every other disturbance, ε_j , conditioned on $\mathbf{X}$; $E[\varepsilon_i \varepsilon_j | \mathbf{X}] = 0, i \neq j$. $E[\boldsymbol{\varepsilon} \boldsymbol{\varepsilon}' | \mathbf{X}] = \sigma^2 \mathbf{I}$.
- A5. Stochastic or nonstochastic data:** $(x_{i1}, x_{i2}, \dots, x_{iK}), i = 1, \dots, n$.
- A6. Normal distribution:** The disturbances, ε_i , are normally distributed. $\boldsymbol{\varepsilon} | \mathbf{X} \sim N[\mathbf{0}, \sigma^2 \mathbf{I}]$.

data, is examined in Sections 4.9.1 and 4.9.2. Missing data have the potential to derail the entire analysis. The benign case in which missing values are simply unexplainable random gaps in the data set is considered in Section 4.9.3. The more complicated case of nonrandomly missing data is discussed in Chapter 19. Finally, the problems of badly measured and outlying observations are examined in Section 4.9.4 and 4.9.5.

This chapter describes the properties of estimators. The assumptions in Table 4.1 will provide the framework for the analysis. (The assumptions are discussed in greater detail in Chapter 3.) For the present, it is useful to assume that the data are a cross section of independent, identically distributed random draws from the joint distribution of $(y_i, \mathbf{x}_i)$ with A1–A3 which defines $E[y_i | \mathbf{x}_i]$. Later in the text (and in Section 4.5), we will consider more general cases. The leading exceptions, which all bear some similarity, are stratified samples, cluster samples, **panel data**, and spatially correlated data. In these cases, groups of related individual observations constitute the observational units. The time-series case in Chapters 20 and 21 will deal with data sets in which potentially all observations are correlated. These cases will be treated later when they are developed in more detail. Under random (cross-section) sampling, with little loss of generality, we can easily obtain very general statistical results such as consistency and asymptotic normality. Later, such as in Chapter 11, we will be able to accommodate the more general cases fairly easily.

4.2 MOTIVATING LEAST SQUARES

Ease of computation is one reason that is occasionally offered to motivate least squares. But, with modern software, ease of computation is a minor (usually trivial) virtue. There are several theoretical justifications for this technique. First, least squares is a natural approach to estimation which makes explicit use of the structure of the model as laid out in the assumptions. Second, even if the true model is not a linear regression, the equation fit by least squares is an optimal linear predictor for the explained variable. Thus, it enjoys a sort of robustness that other estimators do not. Finally, under the specific assumptions of the classical model, by one reasonable criterion, least squares will be the most efficient use of the data.

4.2.1 POPULATION ORTHOGONALITY CONDITIONS

Let $\mathbf{x}$ denote the vector of independent variables in the population regression model. Assumption A3 states that $E[\varepsilon | \mathbf{x}] = 0$. Three useful results follow from this. First, by iterated expectations (Theorem B.1), $E_{\mathbf{x}} E[\varepsilon | \mathbf{x}] = E_{\mathbf{x}}[0] = E[\varepsilon] = 0$; ε has

zero mean, conditionally and unconditionally. Second, by Theorem B.2, $\text{Cov}[\mathbf{x}, \varepsilon] = \text{Cov}[\mathbf{x}, E[\varepsilon|\mathbf{x}]] = \text{Cov}[\mathbf{x}, 0] = \mathbf{0}$ so $\mathbf{x}$ and ε are uncorrelated. Finally, combining the earlier results, $E[\mathbf{x}\varepsilon] = \text{Cov}[\mathbf{x}, \varepsilon] + E[\varepsilon]E[\mathbf{x}] = \mathbf{0}$. We write the third of these as $E[\mathbf{x}\varepsilon] = E[\mathbf{x}(y - \mathbf{x}'\beta)] = \mathbf{0}$ or

$$E[\mathbf{x}y] = E[\mathbf{x}\mathbf{x}']\beta. \quad (4-1)$$

Now, recall the least squares normal equations (3-5) based on the sample of n observations, $\mathbf{X}'\mathbf{y} = \mathbf{X}'\mathbf{X}\mathbf{b}$. Divide this by n and write it as a summation to obtain

$$\left(\frac{1}{n}\sum_{i=1}^n \mathbf{x}_i y_i\right) = \left(\frac{1}{n}\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i'\right)\mathbf{b}. \quad (4-2)$$

Equation (4-1) is a population relationship. Equation (4-2) is a sample analog. Assuming the conditions underlying the laws of large numbers presented in Appendix D are met, the means in (4-2) are estimators of their counterparts in (4-1). Thus, by using least squares, we are mimicking in the sample the relationship that holds in the population.

4.2.2 MINIMUM MEAN SQUARED ERROR PREDICTOR

Consider the problem of finding an **optimal linear predictor** for y . Once again, ignore Assumption A6 and, in addition, drop Assumption A1. The conditional mean function, $E[y|\mathbf{x}]$, might be nonlinear. For the criterion, we will use the **mean squared error** rule, so we seek the minimum mean squared error *linear* predictor of y , which we'll denote $\mathbf{x}'\gamma$. (The minimum mean squared error predictor would be the conditional mean function in all cases. Here, we consider only a linear predictor.) The expected squared error of the linear predictor is

$$\text{MSE} = E[y - \mathbf{x}'\gamma]^2.$$

This can be written as

$$\text{MSE} = E[y - E[y|\mathbf{x}]]^2 + E\{E[y|\mathbf{x}] - \mathbf{x}'\gamma\}^2.$$

We seek the γ that minimizes this expectation. The first term is not a function of γ , so only the second term needs to be minimized. The necessary condition is

$$\begin{aligned} \frac{\partial E\{E(y|\mathbf{x}) - \mathbf{x}'\gamma\}^2}{\partial \gamma} &= E\left\{\frac{\partial \{E(y|\mathbf{x}) - \mathbf{x}'\gamma\}^2}{\partial \gamma}\right\} \\ &= -2E\{\mathbf{x}[E(y|\mathbf{x}) - \mathbf{x}'\gamma]\} = \mathbf{0}. \end{aligned}$$

We arrive at the equivalent condition

$$E[\mathbf{x}E(y|\mathbf{x})] = E[\mathbf{x}\mathbf{x}']\gamma.$$

The left-hand side of this result is $E[\mathbf{x}E(y|\mathbf{x})] = \text{Cov}[\mathbf{x}, E(y|\mathbf{x})] + E[\mathbf{x}]E[E(y|\mathbf{x})] = \text{Cov}[\mathbf{x}, y] + E[\mathbf{x}]E[y] = E[\mathbf{x}y]$. (We have used Theorem B.2.) Therefore, the necessary condition for finding the minimum MSE predictor is

$$E[\mathbf{x}y] = E[\mathbf{x}\mathbf{x}']\gamma. \quad (4-3)$$

This is the same as (4-1), which takes us back to the least squares condition. Assuming that these expectations exist, they would be estimated by the sums in (4-2), which means

THEOREM 4.1 Minimum Mean Squared Error Predictor

If the mechanism generating the data $(\mathbf{x}_i, y_i)$, $i = 1, \dots, n$, is such that the law of large numbers applies to the estimators in (4-2) of the matrices in (4-1), then the slopes of the minimum expected squared error linear predictor of y are estimated by the least squares coefficient vector.

that regardless of the form of the conditional mean, least squares is an estimator of the coefficients of the minimum expected squared error linear predictor of $y|\mathbf{x}$.

4.2.3 MINIMUM VARIANCE LINEAR UNBIASED ESTIMATION

Finally, consider the problem of finding a **linear unbiased estimator**. If we seek the one that has smallest variance, we will be led once again to least squares. This proposition will be proved in Section 4.3.5.

4.3 STATISTICAL PROPERTIES OF THE LEAST SQUARES ESTIMATOR

An *estimator* is a strategy, or formula, for using the sample data that are drawn from a population. The *properties* of that estimator are a description of how it can be expected to behave when it is applied to a sample of data. To consider an example, the concept of unbiasedness implies that on average an estimator (strategy) will correctly estimate the parameter in question; it will not be systematically too high or too low. It is not obvious how one could know this if they were only going to analyze a single sample of data from the population. The argument adopted in econometrics is provided by the **sampling properties** of the estimation strategy. A conceptual experiment lies behind the description. One imagines repeated sampling from the population and characterizes the behavior of the sample of samples. The underlying statistical theory of the estimator provides the basis of the description. Example 4.1 illustrates.

The development of the properties of least squares as an estimator can be viewed in three stages. The **finite sample properties** based on Assumptions A1–A6 are precise, and are independent of the sample size. They establish the essential characteristics of the estimator, such as unbiasedness and the broad approach to be used to estimate the sampling variance. Finite sample results have two limiting aspects. First, they can only be obtained for a small number of statistics—essentially only for the basic least squares estimator. Second, the sharpness of the finite sample results is obtained by making assumptions about the data-generating process that we would prefer not to impose, such as normality of the disturbances (Assumption A6 in Table 4.1). Asymptotic properties of the estimator are obtained by deriving reliable results that will provide good approximations in moderate sized or large samples. For example, the large sample property of consistency of the least squares estimator is looser than unbiasedness in one respect, but at the same time, is more informative about how the estimator improves as more sample data are used. Finally, robust inference methods are a refinement of the asymptotic results. The essential asymptotic theory for least squares modifies the finite sample results after relaxing certain assumptions, mainly A5 (data-generating process

for $\mathbf{X}$) and A6 (normality). Assumption A4 (homoscedasticity and nonautocorrelation) remains a limitation on the generality of the model assumptions. Real-world data are likely to be heteroscedastic in ways that cannot be precisely quantified. They may also be autocorrelated as a consequence of the sample design, such as the within household correlation of panel data observations. These possibilities may taint the inferences that use standard errors that are based on A4. Robust methods are used to accommodate possible violations of Assumption A4 without redesigning the estimation strategy. That is, we continue to use least squares, but employ inference procedures that will be appropriate whether A4 is reasonable or not.

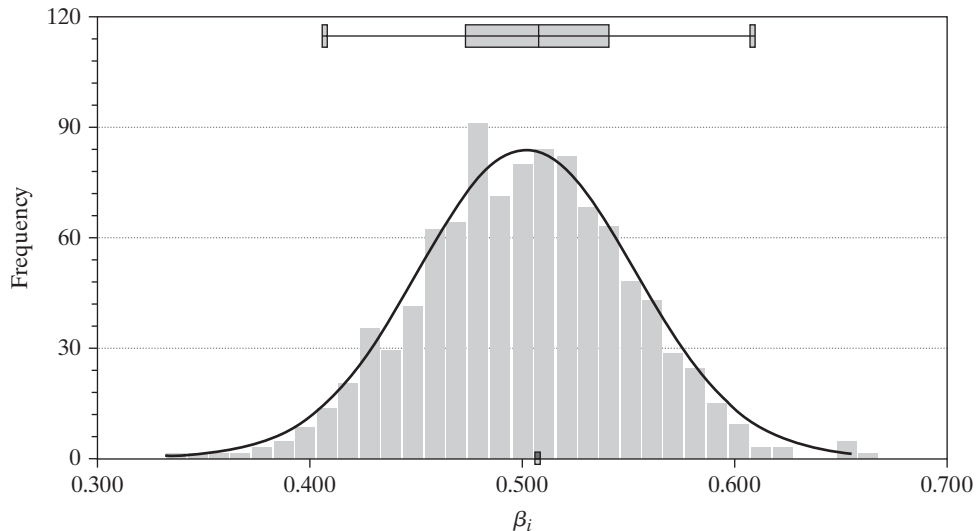
Example 4.1 The Sampling Distribution of a Least Squares Estimator

The following sampling experiment shows the nature of a sampling distribution and the implication of unbiasedness. We drew two samples of 10,000 random draws on variables w_i and x_i from the standard normal population (mean 0, variance 1). We generated a set of ε_i 's equal to $0.5w_i$ and then $y_i = 0.5 + 0.5x_i + \varepsilon_i$. We take this to be our population. We then drew 1,000 random samples of 100 observations on (y_i, x_i) from this population (without replacement), and with each one, computed the least squares slope, using at replication r ,

$$b_r = \left[\sum_{i=1}^{100} (x_{ir} - \bar{x}_r) y_{ir} \right] / \left[\sum_{i=1}^{100} (x_{ir} - \bar{x}_r)^2 \right].$$

The histogram in Figure 4.1 shows the result of the experiment. Note that the distribution of slopes has mean and median roughly equal to the *true value* of 0.5, and it has a substantial variance, reflecting the fact that the regression slope, like any other statistic computed from the sample, is a random variable. The concept of unbiasedness relates to the central tendency of this distribution of values obtained in repeated sampling from the population. The shape of the histogram also suggests the normal distribution of the estimator that we will show theoretically in Section 4.3.6.

FIGURE 4.1 Histogram for Sampled Least Squares Regression Slopes.



4.3.1 UNBIASED ESTIMATION

The least squares estimator is unbiased in every sample. To show this, write

$$\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}. \quad (4-4)$$

Now, take expectations, iterating over $\mathbf{X}$:

$$E[\mathbf{b}|\mathbf{X}] = \boldsymbol{\beta} + E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}|\mathbf{X}].$$

By Assumption A3, the expected value of the second term is $(\mathbf{X}'\mathbf{X}/n)^{-1}E[\sum_i \mathbf{x}_i \varepsilon_i/n|\mathbf{X}]$. Each term in the sum has expectation zero, which produces the result we need:

$$E[\mathbf{b}|\mathbf{X}] = \boldsymbol{\beta}. \quad (4-5)$$

Therefore,

$$E[\mathbf{b}] = E_{\mathbf{x}}\{E[\mathbf{b}|\mathbf{x}]\} = E_{\mathbf{x}}[\boldsymbol{\beta}] = \boldsymbol{\beta}. \quad (4-6)$$

The interpretation of this result is that for any sample of observations, $\mathbf{X}$, the least squares estimator has expectation $\boldsymbol{\beta}$. When we average this over the possible values of $\mathbf{X}$, we find the unconditional mean is $\boldsymbol{\beta}$ as well.

4.3.2 OMITTED VARIABLE BIAS

Suppose that a correctly specified regression model would be

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{z}\gamma + \boldsymbol{\varepsilon}, \quad (4-7)$$

where the two parts have K and 1 columns, respectively. If we regress $\mathbf{y}$ on $\mathbf{X}$ without including the relevant variable, $\mathbf{z}$, then the estimator is

$$\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{z}\gamma + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}. \quad (4-8)$$

(Note, “relevant” means $\gamma \neq 0$.) Taking the expectation, we see that unless $\mathbf{X}'\mathbf{z} = \mathbf{0}$, $\mathbf{b}$ is biased. The well-known result is the **omitted variable formula**:

$$E[\mathbf{b}|\mathbf{X},\mathbf{z}] = \boldsymbol{\beta} + \mathbf{p}_{\mathbf{X},\mathbf{z}}\gamma, \quad (4-9)$$

where

$$\mathbf{p}_{\mathbf{X},\mathbf{z}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{z}. \quad (4-10)$$

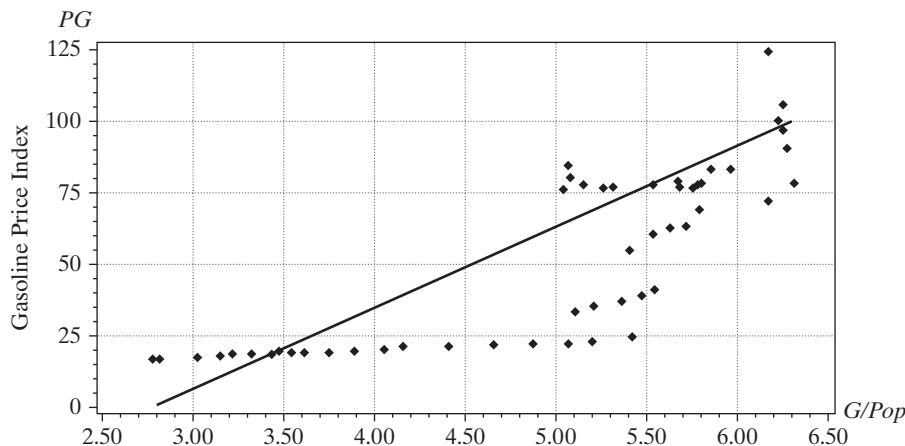
The vector $\mathbf{p}_{\mathbf{X},\mathbf{z}}$ is the column of slopes in the least squares regression of $\mathbf{z}$ on $\mathbf{X}$. Theorem 3.2 (Frisch-Waugh) and Corollary 3.2.1 provide some insight for this result. For each coefficient in (4-9), we have

$$E[b_k|\mathbf{X},\mathbf{z}] = \beta_k + \gamma \left(\frac{\text{Cov}(z, x_k | \text{all other } x\text{'s})}{\text{Var}(x_k | \text{all other } x\text{'s})} \right) \quad (4-11)$$

Example 4.2 Omitted Variable in a Demand Equation

If a demand equation is estimated without the relevant income variable, then (4-11) shows how the estimated price elasticity will be biased. The gasoline market data we have examined in Example 2.3 provides a clear example. The base demand model is

$$\text{Quantity} = \alpha + \beta \text{Price} + \gamma \text{Income} + \varepsilon.$$

FIGURE 4.2 Per Capita Gasoline Consumption Versus Price, 1953–2004.

Letting b be the slope coefficient in the regression of *Quantity* on *Price*, we obtain

$$E[b | \text{Price}, \text{Income}] = \beta + \gamma \frac{\text{Cov}[\text{Price}, \text{Income}]}{\text{Var}[\text{Price}]}.$$

In aggregate data, it is unclear whether the missing covariance would be positive or negative. The sign of the bias in b would be the same as this covariance, however, because $\text{Var}[\text{Price}]$ and γ would both be positive for a normal good such as gasoline. Figure 4.2 shows a simple plot of per capita gasoline consumption, G/Pop , against the price index PG (in inverted Marshallian form). The plot disagrees with what one might expect. But a look at the data in Appendix Table F2.2 shows clearly what is at work. In these aggregate data, the simple correlations for $(G/\text{Pop}, \text{Income}/\text{Pop})$ and for $(PG, \text{Income}/\text{Pop})$ are 0.938 and 0.934, respectively. To see if the expected relationship between price and consumption shows up, we will have to purge our price and quantity data of the intervening effect of income. To do so, we rely on the Frisch–Waugh result in Theorem 3.2. In the simple regression of the log of per capita gasoline consumption on a constant and the log of the price index, the coefficient is 0.29904, which, as expected, has the *wrong* sign. In the multiple regression of the log of per capita gasoline consumption on a constant, the log of the price index and the log of per capita income, the estimated price elasticity, $\hat{\beta}$, is -0.16949 and the estimated income elasticity, $\hat{\gamma}$, is 0.96595. This agrees with expectations.

In this development, it is straightforward to deduce the directions of bias when there is a single included variable and one omitted variable, as in Example 4.2. It is important to note, however, that if more than one variable is included in $\mathbf{X}$, then the terms in the omitted variable formula, (4-9) and (4-10), involve *multiple* regression coefficients, which have the signs of partial, not simple correlations. For example, in the demand model of the previous example, if the price of a closely related product, say new cars, had been included as well, then the simple correlation between gasoline price and income would be insufficient to determine the direction of the bias in the price elasticity. What would be required is the sign of the correlation between price and income net of the effect of the other price:

$$E[b_{\text{Gasoline Price}}|\mathbf{X}, \mathbf{z}] = \beta_{\text{Gasoline Price}} + \left(\frac{\text{Cov}(\text{Income, Gasoline Price} | \text{New Cars Price})}{\text{Var}(\text{Gasoline Price} | \text{New Cars Price})} \right) \gamma. \quad (4-12)$$

This sign might not be obvious, and it would become even less so as more regressors are added to the equation. However, (4-12) does suggest what would be needed for an argument that the least squares estimator remains unbiased, at least for coefficients that correspond to zero partial correlations.

4.3.3 INCLUSION OF IRRELEVANT VARIABLES

We can view the omission of a set of relevant variables as equivalent to imposing an incorrect restriction on (4-7). In particular, omitting $\mathbf{z}$ is equivalent to *incorrectly* estimating (4-7) subject to the restriction $\gamma = 0$. Incorrectly imposing a restriction produces a biased estimator. Suppose, however, that our error is a failure to use some information that is *correct*. If the regression model is correctly given by $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ and we estimate it as if (4-7) were correct [i.e., we include an (or some) extra variable(s)], then the inclusion of the irrelevant variable $\mathbf{z}$ in the regression is equivalent to failing to impose $\gamma = 0$ on (4-7) in estimation. But (4-7) is not incorrect; it simply fails to incorporate $\gamma = 0$. The least squares estimator of $(\boldsymbol{\beta}, \gamma)$ in (4-7) is still unbiased *even given* the restriction:

$$E\left[\begin{pmatrix} \mathbf{b} \\ c \end{pmatrix} | \mathbf{X}, \mathbf{z}\right] = \begin{pmatrix} \boldsymbol{\beta} \\ \gamma \end{pmatrix} = \begin{pmatrix} \boldsymbol{\beta} \\ 0 \end{pmatrix}. \quad (4-13)$$

The broad result is that *including irrelevant variables in the estimation equation does not lead to bias in the estimation of the nonzero coefficients*. Then where is the problem? It would seem that to be conservative, one might generally want to overfit the model. As we will show in Section 4.9.1, the covariance matrix in the regression that properly omits the irrelevant $\mathbf{z}$ is generally smaller than the covariance matrix for the estimator obtained in the presence of the superfluous variables. *The cost of overspecifying the model is larger variances (less precision) of the estimators.*

4.3.4 VARIANCE OF THE LEAST SQUARES ESTIMATOR

The least squares coefficient vector is

$$\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \boldsymbol{\beta} + \mathbf{A}\boldsymbol{\varepsilon}, \quad (4-14)$$

where $\mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$. By Assumption A4, $E[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' | \mathbf{X}] = \text{Var}[\boldsymbol{\varepsilon} | \mathbf{X}] = \sigma^2\mathbf{I}$. The conditional covariance matrix of the least squares slope estimator is

$$\begin{aligned} \text{Var}[\mathbf{b} | \mathbf{X}] &= E[(\mathbf{b} - \boldsymbol{\beta})(\mathbf{b} - \boldsymbol{\beta})' | \mathbf{X}] \\ &= E[\mathbf{A}\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'\mathbf{A}' | \mathbf{X}] \\ &= \mathbf{A}E[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' | \mathbf{X}]\mathbf{A}' \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}. \end{aligned} \quad (4-15)$$

If we wish to use $\mathbf{b}$ to test hypotheses about $\boldsymbol{\beta}$ or to form confidence intervals, then we will require a sample estimate of this matrix. The population parameter σ^2 remains to be estimated. Because σ^2 is the expected value of ε_i^2 and e_i is an estimate of ε_i , $\hat{\sigma}^2 = (1/n) \sum_{i=1}^n e_i^2$ would seem to be the natural estimator. But the least squares

residuals are imperfect estimates of their population counterparts; $e_i = y_i - \mathbf{x}_i'\mathbf{b} = \varepsilon_i - \mathbf{x}_i'(\mathbf{b} - \boldsymbol{\beta})$. The estimator $\hat{\sigma}^2$ is distorted because $\boldsymbol{\beta}$ must be estimated.

The least squares residuals are $\mathbf{e} = \mathbf{M}\mathbf{y} = \mathbf{M}[\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}] = \mathbf{M}\boldsymbol{\varepsilon}$, as $\mathbf{M}\mathbf{X} = \mathbf{0}$. [See Definition 3.1 and (3-15).] An estimator of σ^2 will be based on the sum of squared residuals:

$$\mathbf{e}'\mathbf{e} = \boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}.$$

The expected value of this quadratic form is $E[\mathbf{e}'\mathbf{e}|\mathbf{X}] = E[\boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}|\mathbf{X}]$. The scalar $\boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}$ is a 1×1 matrix, so it is equal to its trace. By using (A-94), $E[\text{tr}(\boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon})|\mathbf{X}] = E[\text{tr}(\mathbf{M}\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}')|\mathbf{X}]$. Because $\mathbf{M}$ is a function of $\mathbf{X}$, the result is $\text{tr}(\mathbf{M}E[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'|\mathbf{X}]) = \text{tr}(\mathbf{M}\sigma^2\mathbf{I}) = \sigma^2\text{tr}(\mathbf{M})$. The trace of $\mathbf{M}$ is $\text{tr}[\mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'] = \text{tr}(\mathbf{I}_n) - \text{tr}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}] = \text{tr}(\mathbf{I}_n) - \text{tr}(\mathbf{I}_K) = n - K$. Therefore,

$$E[\mathbf{e}'\mathbf{e}|\mathbf{X}] = (n - K)\sigma^2. \quad (4-16)$$

The natural estimator is biased toward zero, but the bias becomes smaller as the sample size increases. An unbiased estimator of σ^2 is

$$s^2 = \frac{\mathbf{e}'\mathbf{e}}{n - K}. \quad (4-17)$$

Like $\mathbf{b}$, s^2 is unbiased unconditionally, because $E[s^2] = E_{\mathbf{x}}\{E[s^2|\mathbf{X}]\} = E_{\mathbf{x}}[\sigma^2] = \sigma^2$. The **standard error of the regression** is s , the square root of s^2 . We can then compute

$$\text{Est. Var}[\mathbf{b}|\mathbf{X}] = s^2(\mathbf{X}'\mathbf{X})^{-1}. \quad (4-18)$$

Henceforth, we shall use the notation $\text{Est. Var}[\text{Est. Var}[\cdot]]$ to indicate a sample estimate of the **sampling variance** of an estimator. The square root of the k th diagonal element of this matrix, $\{[s^2(\mathbf{X}'\mathbf{X})^{-1}]_{kk}\}^{1/2}$, is the **standard error** of the estimator b_k , which is often denoted simply the standard error of b_k .

4.3.5 THE GAUSS-MARKOV THEOREM

We will now obtain a general result for the class of linear unbiased estimators of $\boldsymbol{\beta}$. Because $\mathbf{b}|\mathbf{X} = \mathbf{A}\mathbf{y}$, where $\mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, is a linear function of $\boldsymbol{\varepsilon}$, by the definition we will use here, it is a **linear estimator** of $\boldsymbol{\beta}$. Because $E[\mathbf{A}\boldsymbol{\varepsilon}|\mathbf{X}] = \mathbf{0}$, regardless of the distribution of $\boldsymbol{\varepsilon}$, under our other assumptions, $\mathbf{b}$ is a linear, unbiased estimator of $\boldsymbol{\beta}$.

THEOREM 4.2 Gauss–Markov Theorem

In the linear regression model with given regressor matrix $\mathbf{X}$, (1) the least squares estimator, $\mathbf{b}$, is the minimum variance linear unbiased estimator of $\boldsymbol{\beta}$ and (2) for any vector of constants $\mathbf{w}$, the minimum variance linear unbiased estimator of $\mathbf{w}'\boldsymbol{\beta}$ is $\mathbf{w}'\mathbf{b}$.

Note that the theorem makes no use of Assumption A6, normality of the distribution of the disturbances. Only A1 to A4 are necessary. Let $\mathbf{b}_0 = \mathbf{C}\mathbf{y}$ be a different linear unbiased estimator of $\boldsymbol{\beta}$, where $\mathbf{C}$ is a $K \times n$ matrix. If $\mathbf{b}_0$ is unbiased, then $E[\mathbf{C}\mathbf{y}|\mathbf{X}] = E[(\mathbf{C}\mathbf{X}\boldsymbol{\beta} + \mathbf{C}\boldsymbol{\varepsilon})|\mathbf{X}] = \boldsymbol{\beta}$, which implies that $\mathbf{C}\mathbf{X} = \mathbf{I}$ and $\mathbf{b}_0 = \boldsymbol{\beta} + \mathbf{C}\boldsymbol{\varepsilon}$, so $\text{Var}[\mathbf{b}_0|\mathbf{X}] = \sigma^2\mathbf{C}\mathbf{C}'$. Now, let $\mathbf{D} = \mathbf{C} - \mathbf{A}$ so $\mathbf{D}\mathbf{y} = \mathbf{b}_0 - \mathbf{b}$. Because $\mathbf{C}\mathbf{X} = \mathbf{I}$ and

$\mathbf{AX} = \mathbf{I}$, $\mathbf{DX} = \mathbf{0}$ and $\mathbf{DA}' = \mathbf{0}$. Then, $\text{Var}[\mathbf{b}_0|\mathbf{X}] = \sigma^2[(\mathbf{D} + \mathbf{A})(\mathbf{D} + \mathbf{A})']$. By multiplying the terms, we find

$$\text{Var}[\mathbf{b}_0|\mathbf{X}] = \sigma^2(\mathbf{X}'\mathbf{X})^{-1} + \sigma^2\mathbf{DD}' = \text{Var}[\mathbf{b}|\mathbf{X}] + \sigma^2\mathbf{DD}'.$$

The quadratic form in $\mathbf{DD}'$ is $\mathbf{q}'\mathbf{DD}'\mathbf{q} = \mathbf{v}'\mathbf{v} \geq 0$. The conditional covariance matrix of $\mathbf{b}_0$ equals that of $\mathbf{b}$ plus a nonnegative definite matrix. Every quadratic form in $\text{Var}[\mathbf{b}_0|\mathbf{X}]$ is larger than the corresponding quadratic form in $\text{Var}[\mathbf{b}|\mathbf{X}]$, which establishes result (1).

The proof of result (2) of the theorem follows from the previous derivation, because the variance of $\mathbf{w}'\mathbf{b}$ is a quadratic form in $\text{Var}[\mathbf{b}|\mathbf{X}]$, and likewise for any $\mathbf{b}_0$, and implies that each individual slope estimator b_k is the best linear unbiased estimator of β_k . (Let $\mathbf{w}$ be all zeros except for a one in the k th position.) The result applies to every linear combination of the elements of $\boldsymbol{\beta}$. *The implication is that under Assumptions A1–A5, $\mathbf{b}$ is the most efficient (linear unbiased) estimator of $\boldsymbol{\beta}$.*

4.3.6 THE NORMALITY ASSUMPTION

To this point, the specification and analysis of the regression model are **semiparametric** (see Section 12.3). We have not used Assumption A6, normality of $\boldsymbol{\varepsilon}$, in any of the results. In (4-4), $\mathbf{b}$ is a linear function of the disturbance vector, $\boldsymbol{\varepsilon}$. If $\boldsymbol{\varepsilon}$ has a multivariate normal distribution, then we may use the results of Section B.10.2 and the mean vector and covariance matrix derived earlier to state that

$$\mathbf{b}|\mathbf{X} \sim N[\boldsymbol{\beta}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1}].$$

Each element of $\mathbf{b}|\mathbf{X}$ is normally distributed:

$$b_k|\mathbf{X} \sim N[\beta_k, \sigma^2(\mathbf{X}'\mathbf{X})_{kk}^{-1}].$$

We found evidence of this result in Figure 4.1 in Example 4.1.

The exact distribution of $\mathbf{b}$ is conditioned on $\mathbf{X}$. The normal distribution of $\mathbf{b}$ in a finite sample is a consequence of the specific assumption of normally distributed disturbances. The normality assumption is useful for constructing test statistics and for forming **confidence intervals**. But we will ultimately find that we will be able to establish the results we need for inference about $\boldsymbol{\beta}$ based only on the sampling behavior of the statistics without tying the analysis to a narrow assumption of normality of $\boldsymbol{\varepsilon}$.

4.4 ASYMPTOTIC PROPERTIES OF THE LEAST SQUARES ESTIMATOR

The finite sample properties of the least squares estimator are helpful in suggesting the range of results that can be obtained from a sample of data. But the list of settings in which exact finite sample results can be obtained is extremely small. The assumption of normality likewise narrows the range of the applications. Estimation and inference can be based on approximate results that will be reliable guides in even moderately sized data sets, and require fewer assumptions.

4.4.1 CONSISTENCY OF THE LEAST SQUARES ESTIMATOR OF $\boldsymbol{\beta}$

Unbiasedness is a useful starting point for assessing the virtues of an estimator. It assures the analyst that their estimator will not persistently miss its target, either systematically too high or too low. However, as a guide to estimation strategy, unbiasedness has

two shortcomings. First, save for the least squares slope estimator we are discussing in this chapter, it is rare for an econometric estimator to be unbiased. In nearly all cases beyond the multiple linear regression model, the best one can hope for is that the estimator improves in the sense suggested by unbiasedness as more information (data) is brought to bear on the study. As such, we will need a broader set of tools to guide the econometric inquiry. Second, the property of unbiasedness does not, in fact, imply that more information is better than less in terms of estimation of parameters. The sample means of random samples of two, 20 and 20,000 are all unbiased estimators of a population mean—by this criterion all are equally desirable. Logically, one would hope that a larger sample is better than a smaller one in some sense that we are about to define. The property of **consistency** improves on unbiasedness in both of these directions.

To begin, we leave the data-generating mechanism for $\mathbf{X}$ unspecified— $\mathbf{X}$ may be any mixture of constants and random variables generated independently of the process that generates $\boldsymbol{\varepsilon}$. We do make two crucial assumptions. The first is a modification of Assumption A5; **A5a.** $(\mathbf{x}_i, \varepsilon_i)$, $i = 1, \dots, n$ is a sequence of independent, identically distributed *observations*.

The second concerns the behavior of the data in large samples:

$$\text{plim}_{n \rightarrow \infty} \frac{\mathbf{X}'\mathbf{X}}{n} = \mathbf{Q}, \text{ a positive definite matrix.} \quad (4-19)$$

Note how this extends A2. If every $\mathbf{X}$ has full column rank, then $\mathbf{X}'\mathbf{X}/n$ is a positive definite matrix in a specific sample of $n \geq K$ observations. Assumption (4-19) extends that to all samples with at least K observations. A straightforward way to reach (4-19) based on A5a is to assume

$$E[\mathbf{x}_i \mathbf{x}_i'] = \mathbf{Q},$$

so that by the law of large numbers, $(1/n)\sum_i \mathbf{x}_i \mathbf{x}_i'$ converges in probability to its expectation, $\mathbf{Q}$, and via Theorem D.14, $(\mathbf{X}'\mathbf{X}/n)^{-1}$ converges in probability to $\mathbf{Q}^{-1}$.

Time-series settings that involve trends, polynomial time series, and trending variables often pose cases in which the preceding assumptions are too restrictive. A somewhat weaker set of assumptions about $\mathbf{X}$ that is broad enough to include most of these is the **Grenander Conditions** listed in Table 4.2.² The conditions ensure that the data matrix is “well behaved” in large samples. The assumptions are very weak and likely to be satisfied by almost any data set encountered in practice.

At many points from here forward, we will make an assumption that the data are well behaved so that an estimator or statistic will converge to a result. Without repeating them in each instance, we will broadly rely on conditions such as those in Table 4.2.

The least squares estimator may be written

$$\mathbf{b} = \boldsymbol{\beta} + \left(\frac{\mathbf{X}'\mathbf{X}}{n} \right)^{-1} \left(\frac{\mathbf{X}'\boldsymbol{\varepsilon}}{n} \right). \quad (4-20)$$

Then,

$$\text{plim } \mathbf{b} = \boldsymbol{\beta} + \mathbf{Q}^{-1} \text{plim} \left(\frac{\mathbf{X}'\boldsymbol{\varepsilon}}{n} \right).$$

²See Grenander (1956), Palma (2016, p. 373) and Judge et al. (1985, p. 162).

TABLE 4.2 Grenander Conditions for Well-Behaved Data

- G1.** For each column of $\mathbf{X}$, $\mathbf{x}_k$, if $d_{nk}^2 = \mathbf{x}_k' \mathbf{x}_k$, then $\lim_{n \rightarrow \infty} d_{nk}^2 = +\infty$. Hence, $\mathbf{x}_k$ does not degenerate to a sequence of zeros. *Sums of squares will continue to grow as the sample size increases.*
- G2.** $\lim_{n \rightarrow \infty} x_{ik}^2/d_{nk}^2 = 0$ for all $i = 1, \dots, n$. No single observation will ever dominate $\mathbf{x}_k' \mathbf{x}_k$. *As $n \rightarrow \infty$, individual observations will become less important.*
- G3.** Let $\mathbf{C}_n$ be the sample correlation matrix of the columns of $\mathbf{X}$, excluding the constant term if there is one. Then $\lim_{n \rightarrow \infty} \mathbf{C}_n = \mathbf{C}$, a positive definite matrix. This condition implies that the full rank condition will always be met. We have already assumed that $\mathbf{X}$ has full rank in a finite sample. *This rank condition will not be violated as the sample size increases.*

We require the probability limit of the last term. In Section 4.2.1, we found that $E[\varepsilon|\mathbf{x}] = 0$ implies $E[\mathbf{x}\varepsilon] = \mathbf{0}$. Based on this result, again invoking D.4., we find $\mathbf{X}'\varepsilon/n = (1/n)\sum_i \mathbf{x}_i \varepsilon_i$ converges in probability to its expectation of zero, so

$$\text{plim}\left(\frac{\mathbf{X}'\varepsilon}{n}\right) = \mathbf{0}. \quad (4-21)$$

It follows that

$$\text{plim } \mathbf{b} = \boldsymbol{\beta} + \mathbf{Q}^{-1} \cdot \mathbf{0} = \boldsymbol{\beta}. \quad (4-22)$$

This result establishes that under Assumptions A1–A4 and the additional assumption (4-19), $\mathbf{b}$ is a **consistent estimator** of $\boldsymbol{\beta}$ in the linear regression model. Note how consistency improves on unbiasedness. The asymptotic result does not insist that $\mathbf{b}$ be unbiased. But, by the definition of consistency (see Definition D.6), it will follow that $\lim_{n \rightarrow \infty} \text{Prob}[|b_k - \beta_k| > \delta] = 0$ for any positive δ . This means that with increasing sample size, the estimator will be ever closer to the target. This is sometimes (loosely) labeled “asymptotic unbiasedness.”

4.4.2 THE ESTIMATOR OF Asy. Var[b]

To complete the derivation of the asymptotic properties of $\mathbf{b}$, we will require an estimator of $\text{Asy. Var}[\mathbf{b}] = (\sigma^2/n)\mathbf{Q}^{-1}$. With (4-19), it is sufficient to restrict attention to s^2 , so the purpose here is to assess the consistency of s^2 as an estimator of σ^2 . Expanding $s^2 = \varepsilon' \mathbf{M} \varepsilon / (n - K)$ produces

$$s^2 = \frac{1}{n - K} [\varepsilon' \varepsilon - \varepsilon' \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \varepsilon] = \frac{n}{n - K} \left[\frac{\varepsilon' \varepsilon}{n} - \left(\frac{\varepsilon' \mathbf{X}}{n} \right) \left(\frac{\mathbf{X}' \mathbf{X}}{n} \right)^{-1} \left(\frac{\mathbf{X}' \varepsilon}{n} \right) \right].$$

The leading constant clearly converges to 1. We can apply (4-19), (4-21) (twice), and the product rule for **probability limits** (Theorem D.14) to assert that the second term in the brackets converges to 0. That leaves $\frac{1}{n} \sum_{i=1}^n \varepsilon_i^2$. This is a narrow case in which the random variables ε_i^2 are independent with the same finite mean σ^2 , so not much is required to get the mean to converge almost surely to $\sigma^2 = E[\varepsilon_i^2]$. By the Markov theorem (D.8), what is needed is for $E[(\varepsilon_i^2)^{1+\delta}]$ to be finite, so the minimal assumption thus far is that ε_i have finite moments up to slightly greater than 2. Indeed, if we further assume that every ε_i has the same distribution, then by the Khinchine theorem (D.5) or the corollary to D8, finite moments (of ε_i) up to 2 is sufficient. So, under

fairly weak conditions, the first term in brackets converges in probability to σ^2 , which gives our result,

$$\text{plim } s^2 = \sigma^2,$$

and, by the product rule,

$$\text{plim } s^2(\mathbf{X}'\mathbf{X}/n)^{-1} = \sigma^2\mathbf{Q}^{-1}. \quad (4-23)$$

The appropriate *estimator* of the **asymptotic covariance matrix** of $\mathbf{b}$ is the familiar one,

$$\text{Est.Asy.Var}[\mathbf{b}] = s^2 (\mathbf{X}'\mathbf{X})^{-1}. \quad (4-24)$$

4.4.3 ASYMPTOTIC NORMALITY OF THE LEAST SQUARES ESTIMATOR

By relaxing assumption A6, we will lose the exact normal distribution of the estimator that will enable us to form confidence intervals in Section 4.7. However, normality of the disturbances is not necessary for establishing the distributional results we need to allow statistical inference, including confidence intervals and testing hypotheses. Under generally reasonable assumptions about the process that generates the sample data, large sample distributions will provide a reliable foundation for statistical inference in the regression model (and more generally, as we develop more elaborate estimators later in the book).

To derive the **asymptotic distribution** of the least squares estimator, we shall use the results of Section D.3. We will make use of some basic central limit theorems, so in addition to Assumption A3 (uncorrelatedness), we will assume that observations are *independent*. It follows from (4-20) that

$$\sqrt{n}(\mathbf{b} - \boldsymbol{\beta}) = \left(\frac{\mathbf{X}'\mathbf{X}}{n} \right)^{-1} \left(\frac{1}{\sqrt{n}} \right) \mathbf{X}'\boldsymbol{\varepsilon}. \quad (4-25)$$

If the limiting distribution of the random vector in (4-25) exists, then that limiting distribution is the same as that of

$$\left[\text{plim} \left(\frac{\mathbf{X}'\mathbf{X}}{n} \right)^{-1} \right] \left(\frac{1}{\sqrt{n}} \right) \mathbf{X}'\boldsymbol{\varepsilon} = \mathbf{Q}^{-1} \left(\frac{1}{\sqrt{n}} \right) \mathbf{X}'\boldsymbol{\varepsilon}. \quad (4-26)$$

Thus, we must establish the limiting distribution of

$$\left(\frac{1}{\sqrt{n}} \right) \mathbf{X}'\boldsymbol{\varepsilon} = \sqrt{n}(\bar{\mathbf{w}} - E[\bar{\mathbf{w}}]), \quad (4-27)$$

where $\mathbf{w}_i = \mathbf{x}_i\boldsymbol{\varepsilon}_i$ and $E[\mathbf{w}_i] = E[\bar{\mathbf{w}}] = \mathbf{0}$. The mean vector $\bar{\mathbf{w}}$ is the average of n independent identically distributed random vectors with means $\mathbf{0}$ and variances

$$\text{Var}[\mathbf{x}_i\boldsymbol{\varepsilon}_i] = \sigma^2 E[\mathbf{x}_i\mathbf{x}_i'] = \sigma^2\mathbf{Q}. \quad (4-28)$$

The variance of $\sqrt{n}\bar{\mathbf{w}} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{x}_i\boldsymbol{\varepsilon}_i$ is

$$\sigma^2 \left(\frac{1}{n} \right) [\mathbf{Q} + \mathbf{Q} + \cdots + \mathbf{Q}] = \sigma^2\mathbf{Q}. \quad (4-29)$$

We may apply the Lindeberg–Levy central limit theorem (D.18) to the vector $\sqrt{n} \bar{\mathbf{w}}$, as we did in Section D.3 for the univariate case $\sqrt{n} \bar{x}$. If $[\mathbf{x}_i, \varepsilon_i]$, $i = 1, \dots, n$ are independent vectors, each distributed with mean $\mathbf{0}$ and variance $\sigma^2 \mathbf{Q} < \infty$, and if (4-19) holds, then

$$\left(\frac{1}{\sqrt{n}} \right) \mathbf{X}' \boldsymbol{\varepsilon} \xrightarrow{d} N[\mathbf{0}, \sigma^2 \mathbf{Q}]. \quad (4-30)$$

It then follows that

$$\mathbf{Q}^{-1} \left(\frac{1}{\sqrt{n}} \right) \mathbf{X}' \boldsymbol{\varepsilon} \xrightarrow{d} N[\mathbf{Q}^{-1} \mathbf{0}, \mathbf{Q}^{-1} (\sigma^2 \mathbf{Q}) \mathbf{Q}^{-1}]. \quad (4-31)$$

Combining terms,

$$\sqrt{n}(\mathbf{b} - \boldsymbol{\beta}) \xrightarrow{d} N[\mathbf{0}, \sigma^2 \mathbf{Q}^{-1}]. \quad (4-32)$$

Using the technique of Section D.3, we then obtain the **asymptotic distribution** of $\mathbf{b}$:

THEOREM 4.3 Asymptotic Distribution of $\mathbf{b}$ with IID Observations

If $\{\varepsilon_i\}$ are independently distributed with mean zero and finite variance σ^2 and x_{ik} is such that the Grenander conditions are met, then

$$\mathbf{b} \stackrel{a}{\sim} N \left[\boldsymbol{\beta}, \frac{\sigma^2}{n} \mathbf{Q}^{-1} \right]. \quad (4-33)$$

*The development here has relied on random sampling from $(\mathbf{x}_i, \varepsilon_i)$. If observations are not identically distributed, for example, if $E[\mathbf{x}_i \mathbf{x}_i'] = \mathbf{Q}_i$, then under suitable, more general assumptions, an argument could be built around the **Lindeberg–Feller Central Limit Theorem** (D.19A). The essential results would be the same.*

In practice, it is necessary to estimate $(1/n)\mathbf{Q}^{-1}$ with $(\mathbf{X}'\mathbf{X})^{-1}$ and σ^2 with $\mathbf{e}'\mathbf{e}/(n - K)$.

If $\boldsymbol{\varepsilon}$ is normally distributed, then normality of $\mathbf{b}|\mathbf{X}$ holds in *every* sample, so it holds asymptotically as well. The important implication of this derivation is that *if the regressors are well behaved and observations are independent, then the **asymptotic normality** of the least squares estimator does not depend on normality of the disturbances; it is a consequence of the Central Limit Theorem.*

4.4.4 ASYMPTOTIC EFFICIENCY

It remains to establish whether the large-sample properties of the least squares estimator are optimal by any measure. The Gauss–Markov theorem establishes finite sample conditions under which least squares is optimal. The requirements that the estimator be linear and unbiased limit the theorem’s generality, however. One of the main purposes of the analysis in this chapter is to broaden the class of estimators in the linear regression model to those which might be biased, but which are consistent. Ultimately, we will be interested in nonlinear estimators as well. These cases extend beyond the reach of the Gauss–Markov theorem. To make any progress in this direction, we will require an alternative estimation criterion.

DEFINITION 4.1 Asymptotic Efficiency

An estimator is asymptotically efficient if it is consistent, asymptotically normally distributed, and has an asymptotic covariance matrix that is not larger than the asymptotic covariance matrix of any other consistent, asymptotically normally distributed estimator.

We can compare estimators based on their asymptotic variances. The complication in comparing two consistent estimators is that both converge to the true parameter as the sample size increases. Moreover, it usually happens (as in our Example 4.3), that they converge at the same rate—that is, in both cases, the asymptotic variances of the two estimators are of the same order, such as $O(1/n)$. In such a situation, we can sometimes compare the asymptotic variances for the same n to resolve the ranking. The least absolute deviations estimator as an alternative to least squares provides a leading example.

Example 4.3 Least Squares Vs. Least Absolute Deviations—A Monte Carlo Study

Least absolute deviations (LAD) is an alternative to least squares. (The LAD estimator is considered in more detail in Section 7.3.1.) The LAD estimator is obtained as

$$\mathbf{b}_{\text{LAD}} = \text{the minimizer of } \sum_{i=1}^n |y_i - \mathbf{x}_i' \mathbf{b}_0|,$$

in contrast to the linear least squares estimator, which is

$$\mathbf{b}_{\text{LS}} = \text{the minimizer of } \sum_{i=1}^n (y_i - \mathbf{x}_i' \mathbf{b}_0)^2.$$

Suppose the regression model is defined by

$$y_i = \mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i,$$

where the distribution of ε_i has conditional mean zero, constant variance σ^2 , and conditional median zero as well—the distribution is symmetric—and $\text{plim}(1/n)\mathbf{X}'\boldsymbol{\varepsilon} = \mathbf{0}$. That is, all the usual regression assumptions, but with the normality assumption replaced by symmetry of the distribution. Then, under our assumptions, $\mathbf{b}_{\text{LS}}$ is a consistent and asymptotically normally distributed estimator with asymptotic covariance matrix given in Theorem 4.3, which we will call $\sigma^2 \mathbf{A}$. As Koenker and Bassett (1978, 1982), Huber (1987), Rogers (1993), and Koenker (2005) have discussed, under these assumptions, $\mathbf{b}_{\text{LAD}}$ is also consistent. A good estimator of the asymptotic variance of $\mathbf{b}_{\text{LAD}}$ would be $(1/2)^2 [1/f(0)]^2 \mathbf{A}$ where $f(0)$ is the density of ε at its median, zero. This means that we can compare these two estimators based on their asymptotic variances. The ratio of the asymptotic variance of the k th element of $\mathbf{b}_{\text{LAD}}$ to the corresponding element of $\mathbf{b}_{\text{LS}}$ would be

$$q_k = \text{Var}(b_{k,\text{LAD}})/\text{Var}(b_{k,\text{LS}}) = (1/2)^2 (1/\sigma^2) [1/f(0)]^2.$$

If ε did actually have a normal distribution with mean (and median) zero, then $f(\varepsilon) = (2\pi\sigma^2)^{-1/2} \exp(-\varepsilon^2/(2\sigma^2))$ so $f(0) = (2\pi\sigma^2)^{-1/2}$ and for this special case $q_k = \pi/2$. If the disturbances are normally distributed, then LAD will be asymptotically less efficient by a factor of $\pi/2 = 1.573$.

The usefulness of the LAD estimator arises precisely in cases in which we cannot assume normally distributed disturbances. Then it becomes unclear which is the better estimator. It has been found in a long body of research that the advantage of the LAD estimator is most likely to appear in small samples and when the distribution of ε has thicker tails than the

normal—that is, when outlying values of y_i are more likely. As the sample size grows larger, one can expect the LS estimator to regain its superiority. We will explore this aspect of the estimator in a small **Monte Carlo study**.

Examples 2.6 and 3.4 note an intriguing feature of the fine art market. At least in some settings, large paintings sell for more at auction than small ones. Appendix Table F4.1 contains the sale prices, widths, and heights of 430 Monet paintings. These paintings sold at auction for prices ranging from \$10,000 to \$33 million. A linear regression of the log of the price on a constant term, the log of the surface area, and the aspect ratio produces the results in the top line of Table 4.3. This is the focal point of our analysis. In order to study the different behaviors of the LS and LAD estimators, we will do the following Monte Carlo study: We will draw without replacement 100 samples of R observations from the 430. For each of the 100 samples, we will compute $\mathbf{b}_{LS,r}$ and $\mathbf{b}_{LAD,r}$. We then compute the average of the 100 vectors and the sample variance of the 100 observations.³ The sampling variability of the 100 sets of results corresponds to the notion of “variation in repeated samples.” For this experiment, we will do this for $R = 10, 50$, and 100. The overall sample size is fairly large, so it is reasonable to take the full sample results as at least approximately the “true parameters.” The standard errors reported for the full sample LAD estimator are computed using **bootstrapping**. Briefly, the procedure is carried out by drawing B —we used $B = 100$ —samples of n (430) observations *with replacement*, from the full sample of n observations. The estimated variance of the LAD estimator is then obtained by computing the mean squared deviation of these B estimates around the mean of the B estimates. This procedure is discussed in detail in Section 15.4.

TABLE 4.3 Estimated Equations for Art Prices

<i>Full Sample</i>	<i>Constant</i>		<i>Log Area</i>		<i>Aspect Ratio</i>	
	<i>Mean</i>	<i>Standard Error*</i>	<i>Mean</i>	<i>Standard Error</i>	<i>Mean</i>	<i>Standard Error</i>
LS	−8.34327	0.67820	1.31638	0.09205	−0.09623	0.15784
LAD	−8.22726	0.82480	1.25904	0.13718	0.04195	0.22762
R = 10						
LS	−10.6218	8.39355	1.65525	1.21002	−0.07655	1.55330
LAD	−12.0635	11.1734	1.81531	1.53662	0.18269	2.11369
R = 50						
LS	−8.57755	1.94898	1.35026	0.27509	−0.08521	0.46600
LAD	−8.33638	2.18488	1.31408	0.36047	−0.06011	0.60910
R = 100						
LS	−8.38235	1.38332	1.32946	0.19682	−0.09378	0.33765
LAD	−8.37291	1.52613	1.31028	0.24277	−0.07908	0.47906

* For the full sample, standard errors for LS use (4-18). Standard errors for LAD are based on 100 bootstrap replications. For the $R = 10, 50$, and 100 experiments, standard errors are the sample standard deviations of the 100 sets of results from the runs of the experiments.

³The sample size R is not a negligible fraction of the population size, 430 for each replication. However, this does not call for a finite population correction of the variances in Table 4.3. We are not computing the variance of a sample of R observations drawn from a population of 430 paintings. We are computing the variance of a sample of R statistics, each computed from a different subsample of the full population. There about 10^{20} different samples of 10 observations we can draw. The number of different samples of 50 or 100 is essentially infinite.

If the assumptions underlying the regression model are correct, we should observe the following:

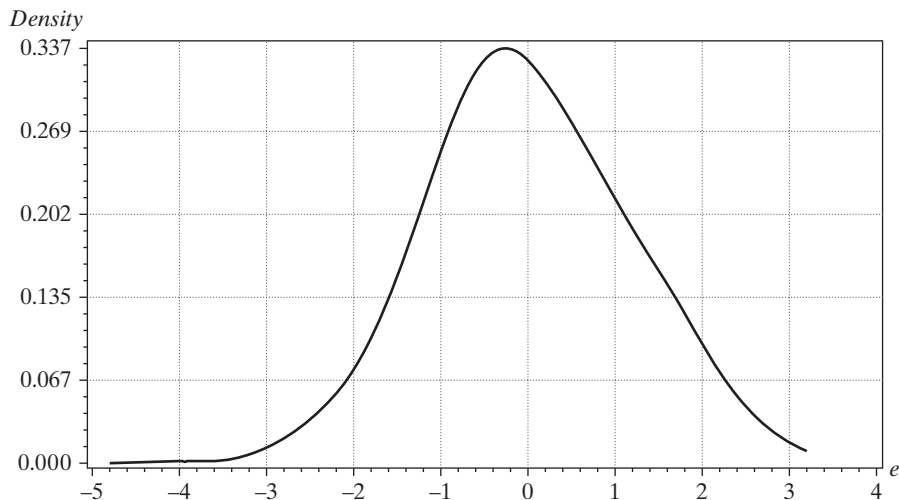
1. Because both estimators are consistent, the averages should resemble the full sample results, the more so as R increases.
2. As R increases, the sampling variance of the estimators should decline.
3. We should observe generally that the standard deviations of the LAD estimates are larger than the corresponding values for the LS estimator.
4. When R is small, the LAD estimator should compare more favorably to the LS estimator, but as R gets larger, the advantage of the LS estimator should become apparent.

A kernel density estimate for the distribution of the least squares residuals appears in Figure 4.3. There is a bit of skewness in the distribution, so a main assumption underlying our experiment may be violated to some degree. Results of the experiments are shown in Table 4.3. The force of the asymptotic results can be seen most clearly in the column for the coefficient on log Area. The decline of the standard deviation as R increases is evidence of the consistency of both estimators. In each pair of results (LS, LAD), we can also see that the estimated standard deviation of the LAD estimator is greater by a factor of about 1.2 to 1.4, which is also to be expected. Based on the normal distribution, we would have expected this ratio to be $\sqrt{\pi/2} = 1.253$.

4.4.5 LINEAR PROJECTIONS

Assumptions A1–A6 define the conditional mean function (CMF) in the joint distribution of $(y_i, \mathbf{x}_i)$, $E[y|\mathbf{x}] = \mathbf{x}'\boldsymbol{\beta}$, and the conditional distribution of $y|\mathbf{x}$ (normal). Based on Assumptions A1–A6, we find that least squares is a consistent estimator of the slopes of the linear conditional mean under quite general conditions. A useful question for modeling is “What is estimated by linear least squares if the conditional mean function is not linear?” To consider this, we begin with a more

FIGURE 4.3 Kernel Density Estimator for Least Squares Residuals.



general statement of the structural model—this is sometimes labeled the “error form” of the model—in which

$$y = E[y|\mathbf{x}] + \varepsilon = \mu(\mathbf{x}) + \varepsilon.$$

We have shown earlier using the law of iterated expectations that $E[\varepsilon|\mathbf{x}] = E[\varepsilon] = 0$ regardless of whether $\mu(\mathbf{x})$ is linear or not. As a side result to modeling a conditional mean function without the linearity assumption, the modeler might use the results of linear least squares as an easily estimable, interesting feature of the population.

To examine the idea, we retain only the assumption of well-behaved data on $\mathbf{x}$, A2, and A5, and assume, as well, that $(y_i, \mathbf{x}_i)$, $i = 1, \dots, n$ are a random sample from the joint population of $(y, \mathbf{x})$. We leave the marginal distribution of $\mathbf{x}$ and the conditional distribution of $y|\mathbf{x}$ both unspecified, but assume that all variables in $(y_i, \mathbf{x}_i)$ have finite means, variances, and covariances. The **linear projection** of y on $\mathbf{x}$, $Proj[y|\mathbf{x}]$, is defined by

$$y = \gamma_0 + \mathbf{x}'\boldsymbol{\gamma} + w = Proj[y|\mathbf{x}] + w,$$

where

$$\gamma_0 = E[y] - E[\mathbf{x}']\boldsymbol{\gamma}$$

and

$$\boldsymbol{\gamma} = (\text{Var}[\mathbf{x}])^{-1}\text{Cov}[\mathbf{x}, y]. \quad (4-34)$$

As noted earlier, if $E[w|\mathbf{x}] = 0$, then this would define the CMF, but we have not assumed that. It does follow by inserting the expression for γ_0 in $E[y] = \gamma_0 + E[\mathbf{x}']\boldsymbol{\gamma} + E[w]$ that $E[w] = 0$, and by expanding $\text{Cov}[\mathbf{x}, y]$ that $\text{Cov}[\mathbf{x}, w] = \mathbf{0}$. The linear projection is a characteristic of the joint distribution of $(y_i, \mathbf{x}_i)$. As we have seen, if the CMF in the joint distribution is linear, then the projection will be the conditional mean. But, in the more general case, the linear projection will simply be a feature of the joint distribution. Some aspects of the linear projection function follow from the specification of the model:

1. Because the linear projection is generally not a structural model—that would usually be the CMF—the coefficients in the linear projection will generally not have a *causal* interpretation; indeed, the elements of $\boldsymbol{\gamma}$ will usually not have any direct economic interpretation other than as approximations (of uncertain quality) to the slopes of the CMF.
2. As we saw in Section 4.2.1, linear least squares regression of y on $\mathbf{X}$ (under the assumed sampling conditions) always estimates the γ_0 and $\boldsymbol{\gamma}$ of the projection regardless of the form of the conditional mean.
3. The CMF is the **minimum mean squared error** predictor of y in the joint distribution of $(y, \mathbf{x})$. We showed in Section 4.2.2 that the linear projection would be the minimum mean squared error *linear* predictor of y . Because both functions are predicting the same thing, it is tempting to infer that the linear projection is a linear approximation to the conditional mean function—and the approximation is exact if the conditional mean is linear. This approximation aspect of the projection function is a common motivation for its use. How effective it is likely to be is obviously dependent on the CMF—a linear function is only going to be able to approximate a nonlinear function locally, and how accurate that is will depend generally on how much curvature there is in the CMF. No generality seems possible; this would be application specific.
4. The interesting features in a structural model are often the partial effects or derivatives of the CMF—in the context of a structural model these are generally the objects of a search for causal effects. A widely observed empirical regularity that

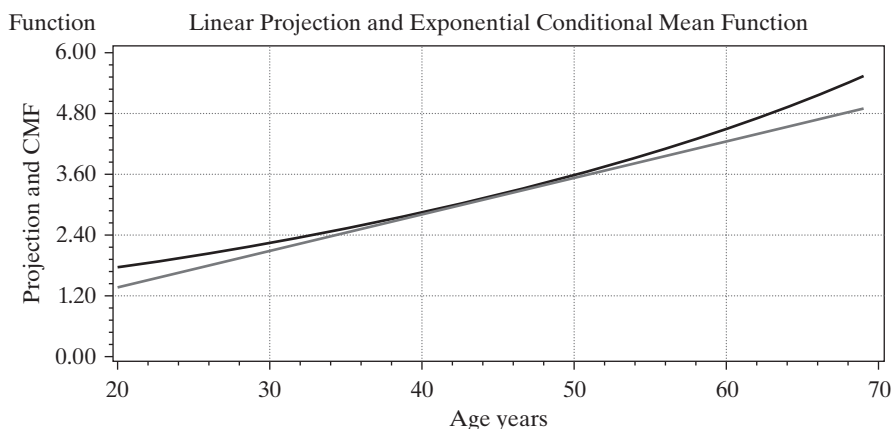
remains to be established with a firm theory is that γ in the linear projection often produces a good approximation to the average partial effects based on the CMF.

Example 4.4 Linear Projection: A Sampling Experiment

Table F7.1 describes panel data on 7,293 German households observed from 1 to 7 times for a total of 27,326 household-year observations. Looking ahead to Section 18.4, we examine a model with a nonlinear conditional mean function, a Poisson regression for the number of doctor visits by the household head, conditioned on the age of the survey respondent. We carried out the following experiment: Using all 27,326 observations, we fit a pooled Poisson regression by maximum likelihood in which the conditional mean function is $\lambda_i = \exp(\beta_0 + \beta_1 \text{Age}_i)$. The estimated values of (β_0, β_1) are $[0.11384, 0.02332]$. We take this to be the population; $f(y_i | x_i) = \text{Poisson}(\lambda_i)$. We then used the observed data on age to (1) compute this true λ_i for each of the 27,326 observations. (2) We used a random number generator to draw 27,326 observations on y_i from the Poisson population with mean equal to this constructed λ_i . Note that the generated data conform exactly to the model with nonlinear conditional mean. The true value of the average partial effect is computed from $\partial E[y_i | x_i] / \partial x_i = \beta_1 \lambda_i$. We computed this for the full sample. The true APE is $(1/27,326) \sum_i \beta_1 \lambda_i = 0.07384$. For the last step, we randomly sampled 1,000 observations from the population and fit the Poisson regression. The estimated coefficient was $b_1 = 0.02334$. The estimated average partial effect based on the MLEs is 0.07141. Finally, we linearly regressed the random draws y_i on Age_i using the 1,000 values. The estimated slope is 0.07163—nearly identical to the estimated average partial effect from the CMF. The estimated CMF and the linear projection are shown in Figure 4.4. The closest correspondence of the two functions occurs in the center of the data—the average age is 43 years. Several runs of the experiment (samples of 1,000 observations) produced the same result (not surprisingly).

As noted earlier, no firm theoretical result links the CMF to the linear projection save for the case when they are equal. As suggested by Figure 4.4, how good an approximation it provides will depend on the curvature of the CMF, and is an empirical question. For the present example, the fit is excellent in the middle of the data. Likewise, it is not possible to tie the slopes of the CMF at any particular point to the coefficients of the linear projection. The widely observed empirical regularity is that the linear projection can deliver good approximations to average partial effects in models with nonlinear CMFs. This is the underlying motivation

FIGURE 4.4 Nonlinear Conditional Mean Function and Linear Projection.



for recent applications of “linear probability models”—that is, for using linear least squares to fit a familiar nonlinear model. See Angrist and Pischke (2010) and Section 17.3 for further examination.

4.5 ROBUST ESTIMATION AND INFERENCE

Table 4.1 lists six assumptions that define the “Classical Linear Regression Model.” A1–A3 define the linear regression framework. A5 suggests a degree of flexibility—the model is broad enough to encompass a wide variety of data generating processes. Assumptions A4 and A6, however, specifically narrow the situations in which the model applies. In particular, A4 seems to preclude the approach developed so far if the disturbances are heteroscedastic or autocorrelated, while A6 limits the stochastic specification to normally distributed disturbances. In fact, we have established all of the finite sample properties save for normality of $\mathbf{b}|\mathbf{X}$, and all of the asymptotic properties without actually using Assumption A6. As such, by these results, the least squares estimator is “robust” to violations of the normality assumption. In particular, it appears to be possible to establish the properties we need for least squares without any specific assumption about the distribution of ε (again, so long as the other assumptions are met).

An estimator of a model is said to be “robust” if it is insensitive to departures from the base assumptions of the model. In practical econometric terms, **robust estimators** retain their desirable properties in spite of violations of some of the assumptions of the model that motivate the estimator. We have seen, for example, that the unbiased least squares estimator is robust to a departure from the normality assumption, A6. In fact, the unbiasedness of least squares is also robust to violations of assumption A4. But, as regards unbiasedness, it is certainly not robust to violations of A3. Also, whether consistency for least squares can be established without A4 remains to be seen. Robustness is usually defined with respect to specific violations of the model assumptions. Estimators are not globally “robust.” Robustness is not necessarily a precisely defined feature of an estimator, however. For example, the LAD estimator examined in Example 4.4 is often viewed as a more robust estimator than least squares, at least in small samples, because of its numerical insensitivity to the presence of outlying observations in the data.

For our practical purposes, we will take robustness to be a broad characterization of the asymptotic properties of certain estimators and procedures. We will specifically focus on and distinguish between **robust estimation** and **robust inference**. A robust estimator, in most settings, will be a consistent estimator that remains consistent in spite of violations of assumptions used to motivate it. To continue the example, with some fairly innocuous assumptions about the alternative specification, the least squares estimator will be robust to violations of the homoscedasticity assumption $\text{Var}[\varepsilon_i|\mathbf{x}_i] = \sigma^2$. In most applications, inference procedures are robust when they are based on estimators of asymptotic variances that are appropriate even when assumptions are violated.

Applications of econometrics rely heavily on robust estimation and inference. The development of robust methods has greatly simplified the development of models, as we shall see, by obviating assumptions that would otherwise limit their generality. We will develop a variety of robust estimators and procedures as we proceed.

4.5.1 CONSISTENCY OF THE LEAST SQUARES ESTIMATOR

In the context of A1–A6, we established consistency of $\mathbf{b}$ by invoking two results. Assumption A2 is an assumption about existence. Without A2, discussion of consistency is moot, because if $\mathbf{X}'\mathbf{X}/n$ does not have full rank, $\mathbf{b}$ does not exist. We also relied on A4. The central result is $\text{plim } \mathbf{X}'\boldsymbol{\varepsilon}/n = \mathbf{0}$, which we could establish if $E[\mathbf{x}_i\boldsymbol{\varepsilon}_i] = \mathbf{0}$. The remaining element would be a law of large numbers by which the sample mean would converge to its population counterpart. Collecting terms, it turns out that normality, homoscedasticity and nonautocorrelation are not needed for consistency of $\mathbf{b}$, so, in turn, consistency of the least squares estimator is robust to violations of these three assumptions. Broadly, random sampling is sufficient.

4.5.2 A HETEROSCEDASTICITY ROBUST COVARIANCE MATRIX FOR LEAST SQUARES

The derivations in Sections 4.4.2 of $\text{Asy.Var}[\mathbf{b}] = (\sigma^2/n)\mathbf{Q}^{-1}$ relied specifically on Assumption A4. In the analysis of a cross section, in which observations are uncorrelated, the issue will be the implications of violations of the homoscedasticity assumption. (We will consider the heteroscedasticity case here. Autocorrelation in time-series data is examined in Section 20.5.2.) For the most general case, suppose $\text{Var}[\boldsymbol{\varepsilon}_i|\mathbf{x}_i] = \sigma_i^2$, with variation assumed to be over $\mathbf{x}_i$. In this case,

$$\mathbf{b} = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1} \sum_i \mathbf{x}_i \boldsymbol{\varepsilon}_i.$$

Then,

$$\text{Var}[\mathbf{b}|\mathbf{X}] = (\mathbf{X}'\mathbf{X})^{-1} \left[\sum_i \sigma_i^2 \mathbf{x}_i \mathbf{x}_i' \right] (\mathbf{X}'\mathbf{X})^{-1}. \quad (4-35)$$

Based on this finite sample result, the asymptotic variance will be

$$\text{Asy.Var}[\mathbf{b}] = \frac{1}{n} \mathbf{Q}^{-1} \left[\text{plim} \frac{1}{n} \sum_i \sigma_i^2 \mathbf{x}_i \mathbf{x}_i' \right] \mathbf{Q}^{-1} = \frac{1}{n} \mathbf{Q}^{-1} \mathbf{Q}^* \mathbf{Q}^{-1}. \quad (4-36)$$

Two points to consider are (1) is $s^2(\mathbf{X}'\mathbf{X})^{-1}$ likely to be a valid estimator of $\text{Asy.Var}[\mathbf{b}]$ in this case? and, if not, (2) is there a strategy available that is “robust” to unspecified heteroscedasticity? The first point is pursued in detail in Section 9.3. The answer to the second is yes. What is required is a feasible estimator of $\mathbf{Q}^*$. White’s (1980) heteroscedasticity robust estimator of $\mathbf{Q}^*$ is

$$\mathbf{W}_{het} = \frac{1}{n} \sum_i e_i^2 \mathbf{x}_i \mathbf{x}_i',$$

where e_i is the least squares residual, $y_i - \mathbf{x}_i'\mathbf{b}$. With $\mathbf{W}_{het}$ in hand, an estimator of $\text{Asy.Var}[\mathbf{b}]$ that is robust to unspecified heteroscedasticity is

$$\text{Est.Asy.Var}[\mathbf{b}] = n(\mathbf{X}'\mathbf{X})^{-1} \mathbf{W}_{het} (\mathbf{X}'\mathbf{X})^{-1}. \quad (4-37)$$

The implication to this point will be that we can discard the homoscedasticity assumption in A4 and recover appropriate standard errors by using (4-37) to estimate the asymptotic standard errors for the coefficients.

4.5.3 ROBUSTNESS TO CLUSTERING

Settings in which the sample data consist of groups of related observations are increasingly common. Panel data applications such as that in Example 4.5 and in Chapter 11 are an obvious case. Samples of firms grouped by industries, students in schools, home prices in neighborhoods, and so on are other examples. In this application, we suppose that the sample consists of C groups, or “clusters” of observations, labeled $c = 1, \dots, C$. There are N_c observations in cluster c where N_c is one or more. The n observations in the entire sample therefore comprise $n = \sum_c N_c$ observations. The regression model is

$$y_{i,c} = \mathbf{x}'_{i,c} \boldsymbol{\beta} + \varepsilon_{i,c}.$$

The observations within a cluster are grouped by the correlation across observations within the group. Consider, for example, student test scores where students are grouped by their class. The common teacher will induce a cross-student correlation of $\varepsilon_{i,c}$. An intuitively appealing formulation of such teacher effects would be the “random effects” formulation,

$$y_{i,c} = \mathbf{x}'_{i,c} \boldsymbol{\beta} + w_c + u_{i,c}. \quad (4-38)$$

By this formulation, the common within cluster effect (e.g., the common teacher effect) would induce the same correlation across all members of the group. This random effects specification is considered in detail in Chapter 11. For present purposes, the assumption is stronger than necessary—note that in (4-38), assuming $u_{i,c}$ is independent across observations, $\text{Cov}(\varepsilon_{i,c}, \varepsilon_{j,c}) = \sigma_w^2$. At this point, we prefer to allow the correlation to be unspecified, and possibly vary for different pairs of observations.

The least squares estimator is

$$\mathbf{b} = \boldsymbol{\beta} + \left(\sum_{c=1}^C \mathbf{X}'_c \mathbf{X}_c \right)^{-1} \left[\sum_{c=1}^C \left(\sum_{i=1}^{N_c} \mathbf{x}_{i,c} \varepsilon_{i,c} \right) \right] = \boldsymbol{\beta} + \left(\mathbf{X}' \mathbf{X} \right)^{-1} \left[\sum_{c=1}^C \left(\mathbf{X}'_c \boldsymbol{\varepsilon}_c \right) \right],$$

where $\mathbf{X}_c$ is the $N_c \times K$ matrix of exogenous variables for cluster c and $\boldsymbol{\varepsilon}_c$ is the N_c disturbances for the group. Assuming that the clusters are independent,

$$\text{Var}[\mathbf{b} | \mathbf{X}] = \left(\mathbf{X}' \mathbf{X} \right)^{-1} \left[\sum_{c=1}^C \mathbf{X}_c \boldsymbol{\Omega}_c \mathbf{X}'_c \right] \left(\mathbf{X}' \mathbf{X} \right)^{-1}. \quad (4-39)$$

Like σ_i^2 before, $\boldsymbol{\Omega}_c$ is not meant to suggest a particular set of population parameters. Rather, $\boldsymbol{\Omega}_c$ represents the possibly unstructured correlations allowed among the N_c disturbances in cluster c . The construction is essentially the same as the White estimator, though $\boldsymbol{\Omega}_c$ is the matrix of variances and covariances for the full vector $\boldsymbol{\varepsilon}_c$. (It would be identical to the White estimator if each cluster contained one observation.) Taking the same approach as before, we obtain the asymptotic variance

$$\text{Asy. Var}[\mathbf{b}] = \frac{1}{C} \mathbf{Q}^{-1} \left[\text{plim} \frac{1}{C} \sum_{c=1}^C \mathbf{X}_c \boldsymbol{\Omega}_c \mathbf{X}'_c \right] \mathbf{Q}^{-1}.^4 \quad (4-40)$$

⁴Since the observations in a cluster are not assumed to be independent, the number of observations in the sample is no longer n . Logically, the sample would now consist of C multivariate observations. In order to employ the asymptotic theory used to obtain $\text{Asy. Var}[\mathbf{b}]$, we are implicitly assuming that C is large while N_c is relatively small, and asymptotic results would relate to increasing C , not n . In practical applications, the number of clusters is often rather small, and the group sizes relatively large. We will revisit these complications in Section 11.3.3.

A feasible estimator of the bracketed matrix based on the least squares residuals is

$$\mathbf{W}_{cluster} = \frac{1}{C} \sum_{c=1}^C \left(\mathbf{X}'_c \mathbf{e}_c \right) \left(\mathbf{e}'_c \mathbf{X}_c \right) = \frac{1}{C} \sum_{c=1}^C \left(\sum_{i=1}^{N_c} \mathbf{x}_{ic} e_{ic} \right) \left(\sum_{i=1}^{N_c} \mathbf{x}_{ic} e_{ic} \right)'. \quad (4-41)$$

Then,

$$\text{Est.Asy.Var}[\mathbf{b}] = C(\mathbf{X}'\mathbf{X})^{-1} \mathbf{W}_{cluster} (\mathbf{X}'\mathbf{X})^{-1}. \quad (4-42)$$

[A refinement intended to accommodate a possible downward bias induced by a small number of clusters is to multiply $\mathbf{W}_{cluster}$ by $C/(C-1)$ (*SAS*) or by $[C/(C-1)] \times [(n-1)/(n-K)]$ (*Stata*, *NLOGIT*).]

Example 4.5 Robust Inference About the Art Market

The Monet paintings examined in Example 4.3 were sold at auction over 1989–2006. Our model thus far is

$$\ln \text{Price}_{it} = \beta_1 + \beta_2 \ln \text{Area}_{it} + \beta_3 \text{AspectRatio}_{it} + \varepsilon_{it}$$

The subscript “*it*” uniquely identifies the painting and when it was sold. Prices in open outcry auctions reflect (at least) three elements, the common (public), observable features of the item, the public unobserved (by the econometrician) elements of the asset, and the private unobservable preferences of the winning bidder. For example, it will turn out (in a later example) that whether the painting is signed or not has a large and significant influence on the price. For now, we assume (for sake of the example), that we do not observe whether the painting is signed or not, though, of course, the winning bidders do observe this. It does seem reasonable to suggest that the presence of a signature is uncorrelated with the two attributes we do observe, area and aspect ratio. We respecify the regression as

$$\ln \text{Price}_{it} = \beta_1 + \beta_2 \ln \text{Area}_{it} + \beta_3 \text{AspectRatio}_{it} + w_{it} + u_{it},$$

where w_{it} represents the intrinsic, unobserved features of the painting and u_{it} represents the unobserved preferences of the buyer. In fact, the sample of 430 sales involves 376 unique paintings. Several of the sales are repeat sales of the same painting. The numbers of sales per painting were one, 333; two, 34; three, 7; and four, 2. Figure 4.5 shows the configuration of the sample. For those paintings that sold more than once, the terms w_{it} do relate to the same i , and, moreover, would naturally be correlated. [They needn’t be identical as in (4-38), however. The valuation of attributes of paintings or other assets sold at auction could vary over time.]

FIGURE 4.5 Repeat Sales of Monet Paintings.

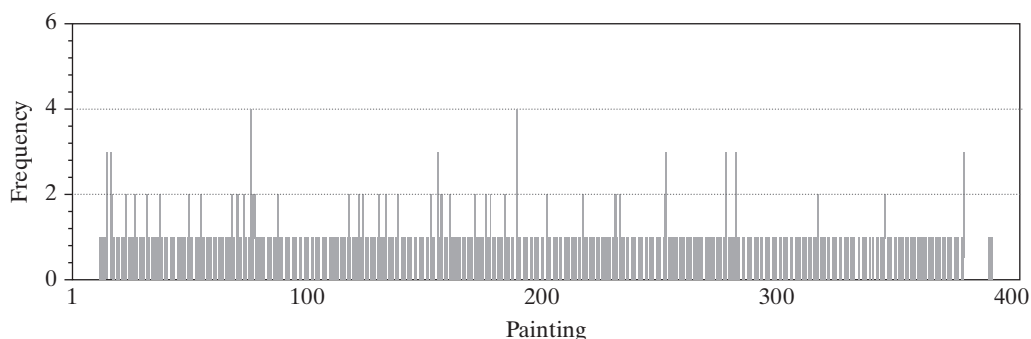


TABLE 4.4 Robust Standard Errors

	<i>Estimated</i>	<i>LS Standard</i>	<i>Heteroscedasticity</i>	<i>Cluster Robust</i>
<i>Variable</i>	<i>Coefficient</i>	<i>Error</i>	<i>Robust Std. Error</i>	<i>Std. Error</i>
<i>Constant</i>	−8.34237	0.67820	0.73342	0.75873
<i>ln Area</i>	1.31638	0.09205	0.10598	0.10932
<i>Aspect Ratio</i>	−0.09623	0.15784	0.16706	0.17776

The least squares estimates and three sets of estimated standard errors are shown in Table 4.4. Even with only a small amount of clustering, the correction produces a tangible adjustment of the standard errors. Perhaps surprisingly, accommodating possible heteroscedasticity produces a more pronounced effect than the cluster correction. Note, finally, in contrast to common expectations, the robust covariance matrix does not always have larger standard errors. The standard errors do increase slightly in this example, however.

4.5.4 BOOTSTRAPPED STANDARD ERRORS WITH CLUSTERED DATA

The sampling framework that underlies the treatment of clustering in the preceding section assumes that the sample consists of a reasonably large number of clusters, drawn randomly from a very large population of clusters. Within each cluster reside a number of observations generated by the linear regression model. Thus,

$$y_{i,c} = \mathbf{x}'_{i,c} \boldsymbol{\beta} + \varepsilon_{i,c},$$

where within each cluster, $E[\varepsilon_{i,c}, \varepsilon_{j,c}]$ may be nonzero—observations may be freely correlated. Clusters are assumed to be independent. Each cluster consists of N_c observations, $(\mathbf{y}_c, \mathbf{X}_c, \boldsymbol{\varepsilon}_c)$ and the cluster is the unit of observation. For example, we might be examining student test scores in a state where students are grouped by classroom, and there are potentially thousands of classrooms in the state. The sample consists of a sample of classrooms. (Higher levels of grouping, such as classrooms in a school, and schools in districts, would require some extensions. We will consider this possibility later in Chapter 11.) The essential feature of the data is the likely correlation across observations in the group. Another natural candidate for this type of process would be a panel data set such as the labor market data examined in Example 4.6, where a sample of 595 individuals is each observed in 7 consecutive years. The common feature is the large number of relatively small or moderately sized clusters in the sample.

The method of estimating a robust asymptotic covariance matrix for the least squares estimator that was introduced in the preceding section involves a method of using the data and the least squares residuals to build a covariance matrix. **Bootstrapping** is another method that is likely to be effective under these assumed sampling conditions. (We emphasize, if the number of clusters is quite small and/or group sizes are very large relative to the number of clusters, then bootstrapping, like the previous method, is likely not to be effective.⁵ Bootstrapping was introduced in Example 4.3 where we used the

⁵See, for example, Wooldridge (2010, Chapter 20).

method to estimate an asymptotic covariance matrix for the LAD estimator. The basic steps in the methodology are:

1. For R repetitions, draw a random sample of N_c observations from the full sample of N_c observations *with replacement*. Estimate the parameters of the regression model with each of the R constructed samples.
2. The estimator of the asymptotic covariance matrix is the sample variance of the R sets of estimated coefficients.

Keeping in mind that in the current case, the cluster is the unit of observation, we use a **block bootstrap**. In the example below, the block is the 7 observations for individual i , so each observation in the bootstrap replication is a block of 7 observations. Example 4.6 below illustrates the use of block bootstrap.

Example 4.6 Clustering and Block Bootstrapping

Cornwell and Rupert (1988) examined the returns to schooling in a panel data set of 595 heads of households observed in seven years, 1976–1982. The sample data (Appendix Table F8.1) are drawn from years 1976 to 1982 from the *Non-Survey of Economic Opportunity* from the Panel Study of Income Dynamics. A slightly modified version of their regression model is

$$\ln Wage_{it} = \beta_1 + \beta_2 Exp_{it} + \beta_3 Exp_{it}^2 + \beta_4 Wks_{it} + \beta_5 Occ_{it} + \beta_6 Ind_{it} + \beta_7 South_{it} \\ + \beta_8 SMSA_{it} + \beta_9 MS_{it} + \beta_{10} Union_{it} + \beta_{11} Ed_i + \beta_{12} Fem_i + \beta_{13} Blk_i + \varepsilon_{it}.$$

The variables in the model are as follows:

<i>Exp</i>	= years of full time work experience,
<i>Wks</i>	= weeks worked,
<i>Occ</i>	= 1 if blue-collar occupation, 0 if not,
<i>Ind</i>	= 1 if the individual works in a manufacturing industry, 0 if not,
<i>South</i>	= 1 if the individual resides in the south, 0 if not,
<i>SMSA</i>	= 1 if the individual resides in an SMSA, 0 if not,
<i>MS</i>	= 1 if the individual is married, 0 if not,
<i>Union</i>	= 1 if the individual wage is set by a union contract, 0 if not,
<i>Ed</i>	= years of education as of 1976,
<i>Fem</i>	= 1 if the individual is female, 0 if not,
<i>Blk</i>	= 1 if the individual is black.

See Appendix Table F8.1 for the data source.

Table 4.5 presents the least squares and three sets of asymptotic standard errors. The first is the conventional results based on $s^2(\mathbf{X}'\mathbf{X})^{-1}$. Compared to the other estimates, it appears that the uncorrected standard errors substantially understate the variability of the least squares estimator. The clustered standard errors are computed using (4-42). The values are 50%–100% larger. The bootstrapped standard errors are quite similar to the robust estimates, as would be expected.

4.6 ASYMPTOTIC DISTRIBUTION OF A FUNCTION OF $\mathbf{b}$: THE DELTA METHOD

We can extend Theorem D.22 to functions of the least squares estimator. Let $\mathbf{f}(\mathbf{b})$ be a set of J continuous, linear, or nonlinear and continuously differentiable functions of the least squares estimator, and let

$$\mathbf{C}(\mathbf{b}) = \frac{\partial \mathbf{f}(\mathbf{b})}{\partial \mathbf{b}'},$$

TABLE 4.5 Clustered, Robust, and Bootstrapped Standard Errors

<i>Variable</i>	<i>Least Squares Estimate</i>	<i>Standard Error</i>	<i>Clustered Std. Error</i>	<i>Bootstrapped Std. Error</i>	<i>White Hetero. Robust Std. Error</i>
<i>Constant</i>	5.25112	0.07129	0.12355	0.11171	0.07435
<i>Exp</i>	0.00401	0.00216	0.00408	0.00434	0.00216
<i>ExpSq</i>	−0.00067	0.00005	0.00009	0.00010	0.00005
<i>Wks</i>	0.00422	0.00108	0.00154	0.00164	0.00114
<i>Occ</i>	−0.14001	0.01466	0.02724	0.02555	0.01494
<i>Ind</i>	0.04679	0.01179	0.02366	0.02153	0.01199
<i>South</i>	−0.05564	0.01253	0.02616	0.02414	0.01274
<i>SMSA</i>	0.15167	0.01207	0.02410	0.02323	0.01208
<i>MS</i>	0.04845	0.02057	0.04094	0.03749	0.02049
<i>Union</i>	0.09263	0.01280	0.02367	0.02553	0.01233
<i>Ed</i>	0.05670	0.00261	0.00556	0.00483	0.00273
<i>Fem</i>	−0.36779	0.02510	0.04557	0.04460	0.02310
<i>Blk</i>	−0.16694	0.02204	0.04433	0.05221	0.02075

where $\mathbf{C}$ is the $J \times K$ matrix whose j th row is the vector of derivatives of the j th function with respect to $\mathbf{b}'$. By the Slutsky theorem (D.12),

$$\text{plim } \mathbf{f}(\mathbf{b}) = \mathbf{f}(\boldsymbol{\beta})$$

and

$$\text{plim } \mathbf{C}(\mathbf{b}) = \frac{\partial \mathbf{f}(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}'} = \boldsymbol{\Gamma}.$$

Using a linear Taylor series approach, we expand this set of functions in the approximation

$$\mathbf{f}(\mathbf{b}) = \mathbf{f}(\boldsymbol{\beta}) + \boldsymbol{\Gamma} \times (\mathbf{b} - \boldsymbol{\beta}) + \text{higher-order terms.}$$

The higher-order terms become negligible in large samples if $\text{plim } \mathbf{b} = \boldsymbol{\beta}$. Then, the asymptotic distribution of the function on the left-hand side is the same as that on the right. The mean of the asymptotic distribution is $\text{plim } \mathbf{f}(\mathbf{b}) = \mathbf{f}(\boldsymbol{\beta})$, and the asymptotic covariance matrix is $\{\boldsymbol{\Gamma}[\text{Asy. Var}(\mathbf{b} - \boldsymbol{\beta})]\boldsymbol{\Gamma}'\}$, which gives us the following theorem:

THEOREM 4.4 Asymptotic Distribution of a Function of $\mathbf{b}$

If $\mathbf{f}(\mathbf{b})$ is a set of continuous and continuously differentiable functions of $\mathbf{b}$ such that $\mathbf{f}(\text{plim } \mathbf{b})$ exists and $\boldsymbol{\Gamma} = \partial \mathbf{f}(\boldsymbol{\beta})/\partial \boldsymbol{\beta}'$ and if Theorem 4.4 holds, then

$$\mathbf{f}(\mathbf{b}) \stackrel{a}{\sim} N\left[\mathbf{f}(\boldsymbol{\beta}), \boldsymbol{\Gamma}\left(\text{Asy. Var}[\mathbf{b}]\right)\boldsymbol{\Gamma}'\right]. \quad (4-43)$$

In practice, the estimator of the asymptotic covariance matrix would be

$$\text{Est. Asy. Var}[\mathbf{f}(\mathbf{b})] = \mathbf{C}\{\text{Est. Asy. Var}[\mathbf{b}]\}\mathbf{C}'.$$

If any of the functions are nonlinear, then the property of unbiasedness that holds for $\mathbf{b}$ may not carry over to $\mathbf{f}(\mathbf{b})$. Nonetheless, $\mathbf{f}(\mathbf{b})$ is a consistent estimator of $\mathbf{f}(\boldsymbol{\beta})$, and the asymptotic covariance matrix is readily available.

Example 4.7 Nonlinear Functions of Parameters: The Delta Method

A dynamic version of the demand for gasoline model in Example 2.3 would be used to separate the short- and long-term impacts of changes in income and prices. The model would be

$$\begin{aligned}\ln(G/Pop)_t = & \beta_1 + \beta_2 \ln P_{G,t} + \beta_3 \ln(Income/Pop)_t + \beta_4 \ln P_{nc,t} \\ & + \beta_5 \ln P_{uc,t} + \gamma \ln(G/Pop)_{t-1} + \varepsilon_t,\end{aligned}$$

where P_{nc} and P_{uc} are price indexes for new and used cars. In this model, the short-run price and income elasticities are β_2 and β_3 . The long-run elasticities are $\phi_2 = \beta_2/(1 - \gamma)$ and $\phi_3 = \beta_3/(1 - \gamma)$, respectively. To estimate the long-run elasticities, we will estimate the parameters by least squares and then compute these two nonlinear functions of the estimates. We can use the delta method to estimate the standard errors.

Least squares estimates of the model parameters with standard errors and t ratios are given in Table 4.6. (Because these are aggregate time-series data, we have not computed a robust covariance matrix.) The estimated short-run elasticities are the estimates given in the table. The two estimated long-run elasticities are $f_2 = b_2/(1 - c) = -0.069532/(1 - 0.830971) = -0.411358$ and $f_3 = 0.164047/(1 - 0.830971) = 0.970522$. To compute the estimates of the standard errors, we need the estimated partial derivatives of these functions with respect to the six parameters in the model:

$$\begin{aligned}\hat{\Gamma}'_2 = \partial\phi_2(\hat{\boldsymbol{\beta}})/\partial\hat{\boldsymbol{\beta}}' &= [0, 1/(1 - \hat{\gamma}), 0, 0, 0, \hat{\boldsymbol{\beta}}_2/(1 - \hat{\gamma})^2] = [0, 5.91613, 0, 0, 0, -2.43365], \\ \hat{\Gamma}'_3 = \partial\phi_3(\hat{\boldsymbol{\beta}})/\partial\hat{\boldsymbol{\beta}}' &= [0, 0, 1/(1 - \hat{\gamma}), 0, 0, \hat{\boldsymbol{\beta}}_3/(1 - \hat{\gamma})^2] = [0, 0, 5.91613, 0, 0, 5.74174].\end{aligned}$$

Using (4-43), we can now compute the estimates of the asymptotic variances for the two estimated long-run elasticities by computing $\mathbf{g}'_2[s^2(\mathbf{X}'\mathbf{X})^{-1}]\mathbf{g}_2$ and $\mathbf{g}'_3[s^2(\mathbf{X}'\mathbf{X})^{-1}]\mathbf{g}_3$. The results are 0.023194 and 0.0263692, respectively. The two asymptotic standard errors are the square roots, 0.152296 and 0.162386.

TABLE 4.6 Regression Results for a Demand Equation

Sum of squared residuals:	0.0127352		
Standard error of the regression:	0.0168227		
R^2 based on 51 observations	0.9951081		
<i>Variable</i>	<i>Coefficient</i>	<i>Standard Error</i>	<i>t Ratio</i>
<i>Constant</i>	-3.123195	0.99583	-3.136
$\ln P_G$	-0.069532	0.01473	-4.720
$\ln Income / Pop$	0.164047	0.05503	2.981
$\ln P_{nc}$	-0.178395	0.05517	-3.233
$\ln P_{uc}$	0.127009	0.03577	3.551
last period $\ln G / Pop$	0.830971	0.04576	18.158

Estimated Covariance Matrix for $\mathbf{b}$ ($e - n = \text{times } 10^{-n}$)					
<i>Constant</i>	$\ln P_G$	$\ln (\text{Income/Pop})$	$\ln P_{nc}$	$\ln P_{uc}$	$\ln (G/Pop)_{t-1}$
0.99168					
-0.0012088	0.00021705				
-0.052602	1.62165e-5	0.0030279			
0.0051016	-0.00021705	-0.00024708	0.0030440		
0.0091672	-4.0551e-5	-0.00060624	-0.0016782	0.0012795	
0.043915	-0.0001109	-0.0021881	0.00068116	8.57001e-5	0.0020943

4.7 INTERVAL ESTIMATION

The objective of interval estimation is to present the best estimate of a parameter with an explicit expression of the uncertainty attached to that estimate. A general approach for estimation of a parameter θ would be

$$\hat{\theta} \pm \text{sampling variability.} \quad (4-44)$$

(We are assuming that the interval of interest would be symmetric around $\hat{\theta}$.) Following the logic that the range of the sampling variability should convey the degree of (un)certainty, we consider the logical extremes. We can be absolutely (100%) certain that the true value of the parameter we are estimating lies in the range $\hat{\theta} \pm \infty$. Of course, this is not particularly informative. At the other extreme, we should place no certainty (0.0%) on the range $\hat{\theta} \pm 0$. The probability that our estimate precisely hits the true parameter value should be considered zero. The point is to choose a value of α —0.05 or 0.01 is conventional—such that we can attach the desired confidence (probability), $100(1 - \alpha)\%$, to the interval in (4-44). We consider how to find that range and then apply the procedure to three familiar problems, calculating an interval for one of the regression parameters, estimating a function of the parameters, and predicting the value of the dependent variable in the regression using a specific setting of the independent variables. For this latter purpose, we will rely on the asymptotic normality of the estimator.

4.7.1 FORMING A CONFIDENCE INTERVAL FOR A COEFFICIENT

If the disturbances are normally distributed, then for any particular element of $\mathbf{b}$,

$$b_k \sim N[\beta_k, \sigma^2 S^{kk}],$$

where S^{kk} denotes the k th diagonal element of $(\mathbf{X}'\mathbf{X})^{-1}$. By standardizing the variable, we find

$$z_k = \frac{b_k - \beta_k}{\sqrt{\sigma^2 S^{kk}}} \quad (4-45)$$

has a standard normal distribution. Note that z_k , which is a function of b_k , β_k , σ^2 , and S^{kk} , nonetheless has a distribution that involves none of the model parameters or the data. Using

the conventional 95% confidence level, we know that $\text{Prob}[-1.96 \leq z_k \leq 1.96] = 0.95$. By a simple manipulation, we find that

$$\text{Prob}[b_k - 1.96\sqrt{\sigma^2 S^{kk}} \leq \beta_k \leq b_k + 1.96\sqrt{\sigma^2 S^{kk}}] = 0.95. \quad (4-46)$$

This states the probability that the random interval, $[b_k \pm \text{the sampling variability}]$, contains β_k , not the probability that β_k lies in the specified interval. If we wish to use some other level of confidence, not 95%, then the 1.96 in (4-46) is replaced by the appropriate $z_{(1-\alpha/2)}$. (We are using the notation $z_{(1-\alpha/2)}$ to denote the value of z such that for the standard normal variable z , $\text{Prob}[z \leq z_{(1-\alpha/2)}] = 1 - \alpha/2$. Thus, $z_{0.975} = 1.96$, which corresponds to $\alpha = 0.05$.)

We would have the desired confidence interval in (4-46), save for the complication that σ^2 is not known, so the interval is not operational. Using s^2 from the regression instead, the ratio

$$t_k = \frac{b_k - \beta_k}{\sqrt{s^2 S^{kk}}} \quad (4-47)$$

has a t distribution with $(n - K)$ degrees of freedom.⁶ We can use t_k to test hypotheses or form confidence intervals about the individual elements of β . A confidence interval for β_k would be formed using

$$\text{Prob}\left[b_k - t_{(1-\alpha/2), [n-K]} \sqrt{s^2 S^{kk}} \leq \beta_k \leq b_k + t_{(1-\alpha/2), [n-K]} \sqrt{s^2 S^{kk}}\right] = 1 - \alpha, \quad (4-48)$$

where $t_{(1-\alpha/2), [n-K]}$ is the appropriate critical value from the t distribution. The distribution of the pivotal statistic depends on the sample size through $(n - K)$, but, once again, not on the parameters or the data.

If the disturbances are not normally distributed, then the theory for the t distribution in (4-48) does not apply. But, the large sample results in Section 4.4 provide an alternative approach. Based on the development that we used to obtain Theorem 4.3 and (4-33), the limiting distribution of the statistic

$$z_k = \frac{\sqrt{n}(b_k - \beta_k)}{\sqrt{\sigma^2 Q^{kk}}}$$

is standard normal, where $\mathbf{Q} = [\text{plim}(\mathbf{X}'\mathbf{X}/n)]^{-1}$ and Q^{kk} is the k th diagonal element of $\mathbf{Q}$. Based on the Slutsky theorem (D.16), we may replace σ^2 with a consistent estimator, s^2 , and obtain a statistic with the same limiting distribution. We estimate $\mathbf{Q}$ with $(\mathbf{X}'\mathbf{X}/n)^{-1}$. This gives us precisely (4-47), which states that under the assumptions in Section 4.4, the “ t ” statistic in (4-47) converges to standard normal even if the disturbances are not normally distributed. The implication would be that to employ the asymptotic distribution of $\mathbf{b}$, we should use (4-48) to compute the confidence interval but use the critical values from the standard normal table (e.g., 1.96) rather than from the t distribution. In practical terms, if the degrees of freedom in (4-48) are moderately large, say greater than 100, then the t distribution will be indistinguishable from the standard normal, and this large sample result would apply in any event. For smaller sample sizes, however, in the interest of conservatism, one might be advised to use the critical

⁶See (B-36) in Section B.4.2. It is the ratio of a standard normal variable to the square root of a chi-squared variable divided by its degrees of freedom.

values from the t table rather than the standard normal, even in the absence of the normality assumption. In the application in Example 4.8, based on a sample of 52 observations, we form a confidence interval for the income elasticity of demand using the critical value of 2.012 from the t table with 47 degrees of freedom. If we chose to base the interval on the asymptotic normal distribution, rather than the standard normal, we would use the 95% critical value of 1.96. One might think this is a bit optimistic, however, and retain the value 2.012, again, in the interest of conservatism.

The preceding analysis starts from Assumption A6, normally distributed disturbance, then shows how the procedure is adjusted to rely on the asymptotic properties of the estimator rather than the narrow possibly unwarranted assumption of normally distributed disturbances. It continues to rely on the homoscedasticity assumption in A4. (For the present, we are assuming away possible autocorrelation.) Section 4.5 showed how the estimator of the asymptotic covariance matrix can be refined to allow for unspecified heteroscedasticity or cluster effects. The final adjustment of the confidence intervals would be to replace (4-48) with

$$\text{Prob}[b_k - z_{(1-\alpha/2)} \sqrt{\text{Est.Asy.Var}[b_k]} \leq \beta_k \leq b_k + z_{(1-\alpha/2)} \sqrt{\text{Est.Asy.Var}[b_k]}] = 1 - \alpha, \quad (4-49)$$

Example 4.8 *Confidence Interval for the Income Elasticity of Demand for Gasoline*

Using the gasoline market data discussed in Examples 4.2 and 4.4, we estimated the following demand equation using the 52 observations:

$$\ln(G/\text{Pop}) = \beta_1 + \beta_2 \ln P_G + \beta_3 \ln(\text{Income}/\text{Pop}) + \beta_4 \ln P_{nc} + \beta_5 \ln P_{uc} + \varepsilon.$$

Least squares estimates of the model parameters with standard errors and t ratios are given in Table 4.7. To form a confidence interval for the income elasticity, we need the critical value from the t distribution with $n - K = 52 - 5 = 47$ degrees of freedom. The 95% critical value is 2.012. Therefore a 95% confidence interval for β_3 is $1.095874 \pm 2.012 (0.07771) = [0.9395, 1.2522]$.

4.7.2 **CONFIDENCE INTERVAL FOR A LINEAR COMBINATION OF COEFFICIENTS: THE OAXACA DECOMPOSITION**

In Example 4.8, we showed how to form a confidence interval for one of the elements of β . By extending those results, we can show how to form a confidence interval for a

TABLE 4.7 Regression Results for a Demand Equation

Sum of squared residuals:	0.120871		
Standard error of the regression:	0.050712		
R^2 based on 52 observations	0.958443		
<i>Variable</i>	<i>Coefficient</i>	<i>Standard Error</i>	<i>t Ratio</i>
<i>Constant</i>	-21.21109	0.75322	-28.160
$\ln P_G$	-0.02121	0.04377	-0.485
$\ln \text{Income}/\text{Pop}$	1.09587	0.07771	14.102
$\ln P_{nc}$	-0.37361	0.15707	-2.379
$\ln P_{uc}$	0.02003	0.10330	0.194

linear function of the parameters. **Oaxaca's (1973) and Blinder's (1973) decomposition** provides a frequently used application.⁷

Let $\mathbf{w}$ denote a $K \times 1$ vector of known constants. Then, the linear combination $c = \mathbf{w}'\mathbf{b}$ is asymptotically normally distributed with mean $\gamma = \mathbf{w}'\boldsymbol{\beta}$ and variance $\sigma_c^2 = \mathbf{w}'[Asy.Var[\mathbf{b}]]\mathbf{w}$, which we estimate with $s_c^2 = \mathbf{w}'[Est.Asy.Var[\mathbf{b}]]\mathbf{w}$. With these in hand, we can use the earlier results to form a confidence interval for γ :

$$\text{Prob}[c - z_{(1-\alpha/2)}s_c \leq \gamma \leq c + z_{(1-\alpha/2)}s_c] = 1 - \alpha. \quad (4-50)$$

This general result can be used, for example, for the sum of the coefficients or for a difference.

Consider, then, Oaxaca's (1973) application. In a study of labor supply, separate wage regressions are fit for samples of n_m men and n_f women. The underlying regression models are

$$\ln \text{wage}_{m,i} = \mathbf{x}'_{m,i}\boldsymbol{\beta}_m + \varepsilon_{m,i}, \quad i = 1, \dots, n_m$$

and

$$\ln \text{wage}_{f,j} = \mathbf{x}'_{f,j}\boldsymbol{\beta}_f + \varepsilon_{f,j}, \quad j = 1, \dots, n_f.$$

The regressor vectors include sociodemographic variables, such as age, and human capital variables, such as education and experience. We are interested in comparing these two regressions, particularly to see if they suggest wage discrimination. Oaxaca suggested a comparison of the regression functions. For any two vectors of characteristics,

$$\begin{aligned} E[\ln \text{wage}_{m,i} | \mathbf{x}_{m,i}] - E[\ln \text{wage}_{f,j} | \mathbf{x}_{f,i}] &= \mathbf{x}'_{m,i}\boldsymbol{\beta}_m - \mathbf{x}'_{f,i}\boldsymbol{\beta}_f \\ &= \mathbf{x}'_{m,i}\boldsymbol{\beta}_m - \mathbf{x}'_{m,i}\boldsymbol{\beta}_f + \mathbf{x}'_{m,i}\boldsymbol{\beta}_f - \mathbf{x}'_{f,i}\boldsymbol{\beta}_f \\ &= \mathbf{x}'_{m,i}(\boldsymbol{\beta}_m - \boldsymbol{\beta}_f) + (\mathbf{x}_{m,i} - \mathbf{x}_{f,i})'\boldsymbol{\beta}_f. \end{aligned}$$

The second term in this decomposition is identified with differences in human capital that would explain wage differences naturally, assuming that labor markets respond to these differences in ways that we would expect. The first term shows the differential in log wages that is attributable to differences unexplainable by human capital; holding these factors constant at $\mathbf{x}_m$ makes the first term attributable to other factors. Oaxaca suggested that this decomposition be computed at the means of the two regressor vectors, $\bar{\mathbf{x}}_m$ and $\bar{\mathbf{x}}_f$, and the least squares coefficient vectors, $\mathbf{b}_m$ and $\mathbf{b}_f$. If the regressions contain constant terms, then this process will be equivalent to analyzing $\ln y_m - \ln y_f$.

We are interested in forming a confidence interval for the first term, which will require two applications of our result. We will treat the two vectors of sample means as known vectors. Assuming that we have two independent sets of observations, our two estimators, $\mathbf{b}_m$ and $\mathbf{b}_f$, are independent with means $\boldsymbol{\beta}_m$ and $\boldsymbol{\beta}_f$ and estimated asymptotic covariance matrices $Est.Asy.Var[\mathbf{b}_m]$ and $Est.Asy.Var[\mathbf{b}_f]$. The covariance matrix of the difference is the sum of these two matrices. We are forming a confidence interval for $\bar{\mathbf{x}}'_m \mathbf{d}$ where $\mathbf{d} = \mathbf{b}_m - \mathbf{b}_f$. The estimated covariance matrix is

$$Est.Asy.Var[\mathbf{d}] = Est.Asy.Var[\mathbf{b}_m] + Est.Asy.Var[\mathbf{b}_f]. \quad (4-51)$$

Now we can apply the result above. We can also form a confidence interval for the second term; just define $\mathbf{w} = \bar{\mathbf{x}}_m - \bar{\mathbf{x}}_f$ and apply the earlier result to $\mathbf{w}'\mathbf{b}_f$.

⁷See Bourguignon et al. (2002) for an extensive application.

Example 4.9 Oaxaca Decomposition of Home Sale Prices

The town of Shaker Heights, Ohio, a suburb of Cleveland, developed in the twentieth century as a patchwork of neighborhoods associated with neighborhood-based school districts. Responding to changes in the demographic composition of the city, in 1987, Shaker Heights redistricted the neighborhoods. Some houses in some neighborhoods remained in the same school district while others in the same neighborhood were removed to other school districts. Bogart and Cromwell (2000) examined how this abrupt policy change affected home values in Shaker Heights by studying sale prices of houses before and after the change. Several econometric approaches were used.

- **Difference in Differences Regression:** Houses that did not change districts constituted a control group while those that did change constitute a treatment group. Sales take place both before and after the treatment date, 1987. A hedonic regression of home sale prices on attributes and the treatment and policy dummy variables reveals the causal effect of the policy change. (We will examine this method in Chapter 6.)
- **Repeat Sales:** Some homes were sold more than once. For those that sold both before and after the redistricting, a regression of the form

$$\ln \text{Price}_{i1} - \ln \text{Price}_{i0} = \text{time effects} + \text{school effects} + \Delta \text{redistricted}.$$

The advantage of the first difference regression is that it effectively controls for and eliminates the characteristics of the house, and leaves only the persistent school effects and the effect of the policy change.

- **Oaxaca Decomposition:** Two hedonic regressions based on house characteristics are fit for different parts of neighborhoods where there are both houses that are in the neighborhood school areas and houses that are districted to other schools. The decomposition approach described above is applied to the two groups. The differences in the means of the sale prices are decomposed into a component that can be explained by differences in the house attributes and a residual effect that is suggested to be related to the benefit of having a neighborhood school. Figure 4.6 below shows the authors' main results for this part of the analysis.⁸

FIGURE 4.6 Results of Oaxaca Decomposition.

TABLE 6	
Within Neighborhood Estimates of Neighborhood Schools Effect, Lomond Neighborhood (1987–1994)	
Difference in mean house value	\$6,545
Percent of difference due to district change	52.9%–59.1%
Effect of district change on mean house value (decrease)	\$3462–\$3868 \$3779
Dummy variable estimate of effect of district change	476—same district 186—change district
Number of observations (662 total sales)	
<i>Note:</i> Percent of difference due to district change equals 100% minus the percent explained by differences in observable characteristics. Included characteristics are <i>heavy traffic</i>, <i>ln(frontage)</i>, <i>ln(living area)</i>, <i>ln(lot size)</i>, <i>ln(age of house)</i>, <i>average room size</i>, <i>plumbing fixtures</i>, <i>attached garage</i>, <i>finished attic</i>, <i>construction grade AA/A+</i>, <i>construction grade A</i>, <i>construction grade B or C or D</i>, <i>bad or fair condition</i>, <i>excellent condition</i>, and a set of year dummies. Regressions estimated using data from 1987 to 1994. Complete regression results available on request.	

⁸Bogart and Cromwell (2000, p. 298).

4.8 PREDICTION AND FORECASTING

After the estimation of the model parameters, a common use of regression modeling is for prediction of the dependent variable. We make a distinction between *prediction* and *forecasting* most easily based on the difference between cross section and time-series modeling. **Prediction** (which would apply to either case) involves using the regression model to compute fitted (predicted) values of the dependent variable, either within the sample or for observations outside the sample. The same set of results will apply to cross sections, panels, and time series. We consider these methods first. **Forecasting**, while largely the same exercise, explicitly gives a role to time and often involves lagged dependent variables and disturbances that are correlated with their past values. This exercise usually involves predicting future outcomes. An important difference between predicting and forecasting (as defined here) is that for predicting, we are usually examining a scenario of our own design. Thus, in the example below in which we are predicting the prices of Monet paintings, we might be interested in predicting the price of a hypothetical painting of a certain size and aspect ratio, or one that actually exists in the sample. In the time-series context, we will often try to forecast an event such as real investment next year, not based on a hypothetical economy but based on our best estimate of what economic conditions will be next year. We will use the term **ex post prediction** (or **ex post forecast**) for the cases in which the data used in the regression equation to make the prediction are either observed or constructed experimentally by the analyst. This would be the first case considered here. An **ex ante forecast** (in the time-series context) will be one that requires the analyst to forecast the independent variables first before it is possible to forecast the dependent variable. In an exercise for this chapter, real investment is forecasted using a regression model that contains real GDP and the consumer price index. In order to forecast real investment, we must first forecast real GDP and the price index. Ex ante forecasting is considered briefly here and again in Chapter 20.

4.8.1 PREDICTION INTERVALS

Suppose that we wish to predict the value of y^0 associated with a regressor vector $\mathbf{x}^0$. The actual value would be

$$y^0 = \mathbf{x}^{0'}\boldsymbol{\beta} + \varepsilon^0.$$

It follows from the Gauss–Markov theorem that

$$\hat{y}^0 = \mathbf{x}^{0'}\mathbf{b} \tag{4-52}$$

is the minimum variance linear unbiased estimator of $E[y^0 | \mathbf{x}^0] = \mathbf{x}^{0'}\boldsymbol{\beta}$. The **prediction error** is

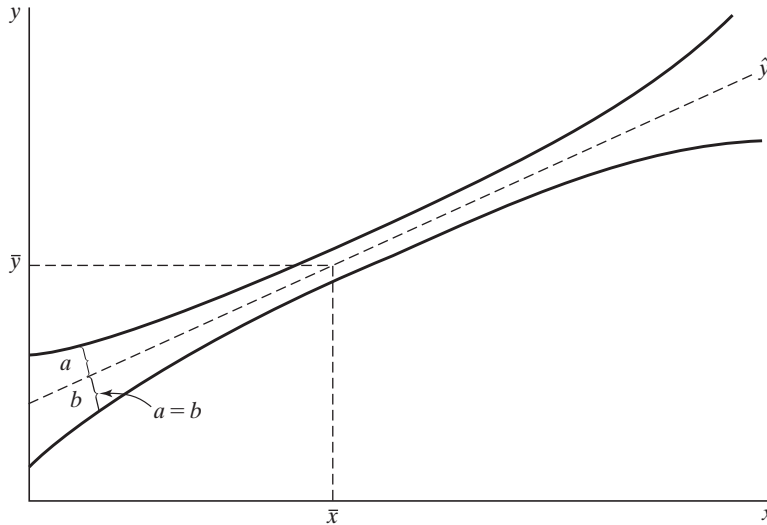
$$e^0 = \hat{y}^0 - y^0 = (\mathbf{b} - \boldsymbol{\beta})'\mathbf{x}^0 - \varepsilon^0.$$

The **prediction variance** of this estimator based on (4-15) is

$$\text{Var}[e^0 | \mathbf{X}, \mathbf{x}^0] = \sigma^2 + \text{Var}[(\mathbf{b} - \boldsymbol{\beta})'\mathbf{x}^0 | \mathbf{X}, \mathbf{x}^0] = \sigma^2 + \mathbf{x}^{0'}[\sigma^2(\mathbf{X}'\mathbf{X})^{-1}]\mathbf{x}^0. \tag{4-53}$$

If the regression contains a constant term, then an equivalent expression is

$$\text{Var}[e^0 | \mathbf{X}, \mathbf{x}^0] = \sigma^2 \left[1 + \frac{1}{n} + \sum_{j=1}^{K-1} \sum_{k=1}^{K-1} (x_j^0 - \bar{x}_j)(x_k^0 - \bar{x}_k)(\mathbf{Z}'\mathbf{M}^0\mathbf{Z})^{jk} \right], \tag{4-54}$$

FIGURE 4.7 Prediction Intervals.

where $\mathbf{Z}$ is the $K - 1$ columns of $\mathbf{X}$ not including the constant, $\mathbf{Z}'\mathbf{M}^0\mathbf{Z}$ is the matrix of sums of squares and products for the columns of $\mathbf{X}$ in deviations from their means [see (3-21)], and the “ jk ” superscript indicates the jk element of the inverse of the matrix. This result suggests that the width of a confidence interval (i.e., a **prediction interval**) depends on the distance of the elements of $\mathbf{x}^0$ from the center of the data. Intuitively, this idea makes sense; the farther the forecasted point is from the center of our experience, the greater is the degree of uncertainty. Figure 4.7 shows the effect for the bivariate case. Note that the prediction variance is composed of three parts. The second and third become progressively smaller as we accumulate more data (i.e., as n increases). But, the first term, σ^2 is constant, which implies that no matter how much data we have, we can never predict perfectly.

The prediction variance can be estimated by using s^2 in place of σ^2 . A confidence (prediction) interval for y^0 would then be formed using

$$\text{prediction interval} = \hat{y}^0 \pm t_{(1-\alpha/2), [n-K]} se(e^0), \quad (4-55)$$

where $t_{(1-\alpha/2), [n-K]}$ is the appropriate critical value for $100(1 - \alpha) \%$ significance from the t table for $n - K$ degrees of freedom and $se(e^0)$ is the square root of the estimated prediction variance.

4.8.2 PREDICTING Y WHEN THE REGRESSION MODEL DESCRIBES $\log y$

It is common to use the regression model to describe a function of the dependent variable, rather than the variable, itself. In Example 4.5 we model the sale prices of Monet paintings using

$$\ln \text{Price} = \beta_1 + \beta_2 \ln \text{Area} + \beta_3 \text{Aspect Ratio} + \varepsilon.$$

The log form is convenient in that the coefficient provides the elasticity of the dependent variable with respect to the independent variable, that is, in this model,

$\beta_2 = \partial E[\ln Price | \ln Area, AspectRatio] / \partial \ln Area$. However, the equation in this form is less interesting for prediction purposes than one that predicts the price itself. The natural approach for a predictor of the form

$$\ln y^0 = \mathbf{x}^{0'} \mathbf{b}$$

would be to use

$$\hat{y}^0 = \exp(\mathbf{x}^{0'} \mathbf{b}).$$

The problem is that $E[y | \mathbf{x}^0]$ is not equal to $\exp(E[\ln y | \mathbf{x}^0])$. The appropriate conditional mean function would be

$$E[y | \mathbf{x}^0] = E[\exp(\mathbf{x}^{0'} \beta + \varepsilon^0) | \mathbf{x}^0] = \exp(\mathbf{x}^{0'} \beta) E[\exp(\varepsilon^0) | \mathbf{x}^0].$$

The second term is not $\exp(E[\varepsilon^0 | \mathbf{x}^0]) = 1$ in general. The precise result if $\varepsilon^0 | \mathbf{x}^0$ is normally distributed with mean zero and variance σ^2 is $E[\exp(\varepsilon^0) | \mathbf{x}^0] = \exp(\sigma^2/2)$. (See Section B.4.4.) The implication for normally distributed disturbances would be that an appropriate predictor for the conditional mean would be

$$\hat{y}^0 = \exp(\mathbf{x}^{0'} \mathbf{b} + s^2/2) > \exp(\mathbf{x}^{0'} \mathbf{b}), \quad (4-56)$$

which would seem to imply that the naïve predictor would systematically underpredict y . However, this is not necessarily the appropriate interpretation of this result. The inequality implies that the naïve predictor will systematically underestimate the conditional mean function, not necessarily the realizations of the variable itself. The pertinent question is whether the conditional mean function is the desired predictor for the exponent of the dependent variable in the log regression. The conditional median might be more interesting, particularly for a financial variable such as income, expenditure, or the price of a painting. If the distribution of the variable in the log regression is symmetrically distributed (as they are when the disturbances are normally distributed), then the exponent will be asymmetrically distributed with a long tail in the positive direction, and the mean will exceed the median, possibly vastly so. In such cases, the median is often a preferred estimator of the center of a distribution. For estimating the median, rather than the mean, we would revert to the original naïve predictor, $\hat{y}^0 = \exp(\mathbf{x}^{0'} \mathbf{b})$.

Given the preceding, we consider estimating $E[\exp(y) | \mathbf{x}^0]$. If we wish to avoid the normality assumption, then it remains to determine what one should use for $E[\exp(\varepsilon^0) | \mathbf{x}^0]$. Duan (1983) suggested the consistent estimator (assuming that the expectation is a constant, that is, that the regression is homoscedastic),

$$\hat{E}[\exp(\varepsilon^0) | \mathbf{x}^0] = h^0 = \frac{1}{n} \sum_{i=1}^n \exp(e_i), \quad (4-57)$$

where e_i is a least squares residual in the original log form regression. Then, Duan's **smearing estimator** for prediction of y^0 is

$$\hat{y}^0 = h^0 \exp(\mathbf{x}^{0'} \mathbf{b}).$$

4.8.3 PREDICTION INTERVAL FOR Y WHEN THE REGRESSION MODEL DESCRIBES LOG y

We obtained a prediction interval in (4-55) for $\ln y | \mathbf{x}^0$ in the loglinear model $\ln y = \mathbf{x}'\beta + \varepsilon$,

$$[\ln \hat{y}_{\text{LOWER}}^0, \ln \hat{y}_{\text{UPPER}}^0] = \left[\mathbf{x}^{0'} \mathbf{b} - t_{(1-\alpha/2), [n-K]} se(e^0), \mathbf{x}^{0'} \mathbf{b} + t_{(1-\alpha/2), [n-K]} se(e^0) \right].$$

For a given choice of α , say, 0.05, these values give the 0.025 and 0.975 quantiles of the distribution of $\ln y | \mathbf{x}^0$. If we wish specifically to estimate these quantiles of the distribution of $y | \mathbf{x}^0$, not $\ln y | \mathbf{x}^0$, then we would use:

$$\left[\hat{y}_{\text{LOWER}}^0, \hat{y}_{\text{UPPER}}^0 \right] = \left\{ \exp \left[\mathbf{x}^{0'} \mathbf{b} - t_{(1-\alpha/2), [n-K]} se(e^0) \right], \exp \left[\mathbf{x}^{0'} \mathbf{b} + t_{(1-\alpha/2), [n-K]} se(e^0) \right] \right\}. \quad (4-58)$$

This follows from the result that if $\text{Prob}[\ln y \leq \ln L] = 1 - \alpha/2$, then $\text{Prob}[y \leq L] = 1 - \alpha/2$. The result is that the natural estimator is the right one for estimating the specific quantiles of the distribution of the original variable. However, if the objective is to find an interval estimator for $y | \mathbf{x}^0$ that is as narrow as possible, then this approach is not optimal. If the distribution of y is asymmetric, as it would be for a loglinear model with normally distributed disturbances, then the naïve interval estimator is longer than necessary. Figure 4.8 shows why. We suppose that (L, U) in the figure is the prediction interval formed by (4-58). Then the probabilities to the left of L and to the right of U each equal $\alpha/2$. Consider alternatives $L_0 = 0$ and U_0 instead. As we have constructed the figure, the area (probability) between L_0 and L equals the area between U_0 and U . But, because the density is so much higher at L , the distance $(0, U_0)$, the dashed interval, is visibly shorter than that between (L, U) . The sum of the two tail probabilities is still equal to α , so this provides a shorter prediction interval. We could improve on (4-58) by using, instead, $(0, U_0)$, where U_0 is simply $\exp[\mathbf{x}^{0'} \mathbf{b} + t_{(1-\alpha), [n-K]} se(e^0)]$ (i.e., we put the entire tail area to the right of the upper value). However, while this is an improvement, it goes too far, as we now demonstrate.

Consider finding directly the shortest prediction interval. We treat this as an optimization problem,

$$\text{Minimize}(L, U): I = U - L \text{ subject to } F(L) + [1 - F(U)] = \alpha,$$

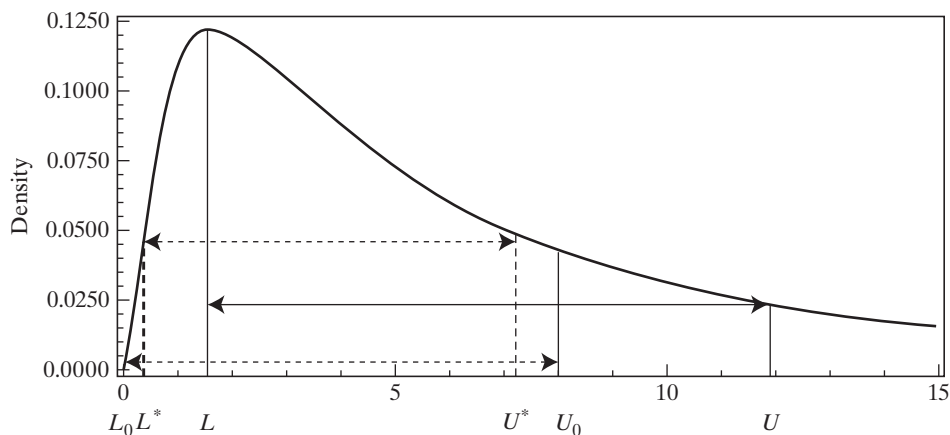
where F is the cdf of the random variable y (not $\ln y$). That is, we seek the shortest interval for which the two tail probabilities sum to our desired α (usually 0.05). Formulate this as a Lagrangean problem,

$$\text{Minimize}(L, U, \lambda): I^* = U - L + \lambda[F(L) + (1 - F(U)) - \alpha].$$

The solutions are found by equating the three partial derivatives to zero:

$$\begin{aligned} \partial I^* / \partial L &= -1 + \lambda f(L) = 0, \\ \partial I^* / \partial U &= 1 - \lambda f(U) = 0, \\ \partial I^* / \partial \lambda &= F(L) + [1 - F(U)] - \alpha = 0, \end{aligned}$$

where $f(L) = F'(L)$ and $f(U) = F'(U)$ are the derivatives of the cdf, which are the densities of the random variable at L and U , respectively. The third equation enforces the restriction that the two tail areas sum to α but does not force them to be equal. By adding the first two equations, we find that $\lambda[f(L) - f(U)] = 0$, which, if λ is not zero, means that the solution is obtained by locating (L^*, U^*) such that the tail areas sum to α

FIGURE 4.8 Lognormal Distribution for Prices of Monet Paintings.

and the densities are equal. Looking again at Figure 4.8, we can see that the solution we would seek is (L^*, U^*) where $0 < L^* < L$ and $U^* < U_0$. This is the shortest interval, and it is shorter than both $[0, U_0]$ and $[L, U]$.

This derivation would apply for any distribution, symmetric or otherwise. For a symmetric distribution, however, we would obviously return to the symmetric interval in (4-58). It provides the correct solution for when the distribution is asymmetric. In Bayesian analysis, the counterpart when we examine the distribution of a parameter conditioned on the data, is the **highest posterior density interval**. (See Section 16.4.2.) For practical application, this computation requires a specific assumption for the distribution of $y|\mathbf{x}^0$, such as lognormal. Typically, we would use the smearing estimator specifically to avoid the distributional assumption. There also is no simple formula to use to locate this interval, even for the lognormal distribution. A crude grid search would probably be best, though each computation is very simple. What this derivation does establish is that one can do substantially better than the naïve interval estimator, for example, using $[0, U_0]$.

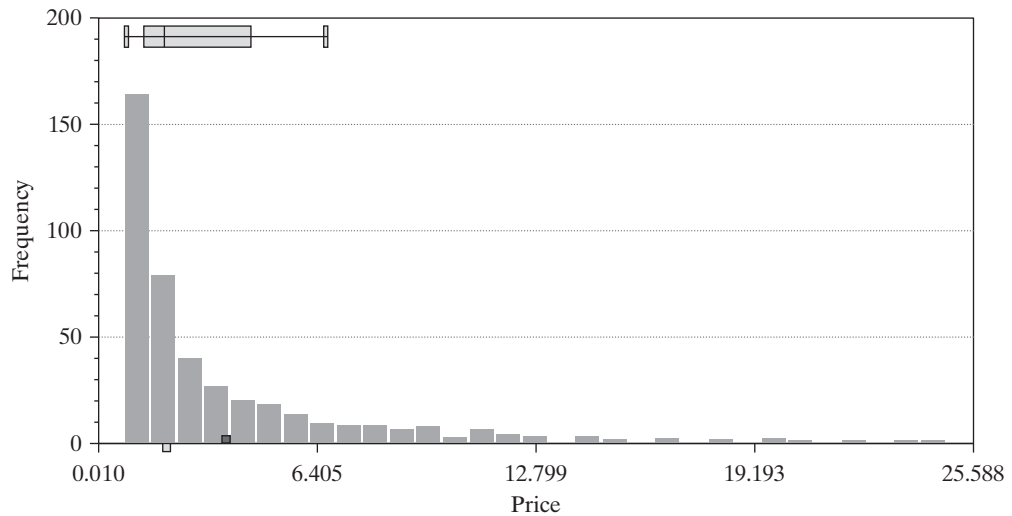
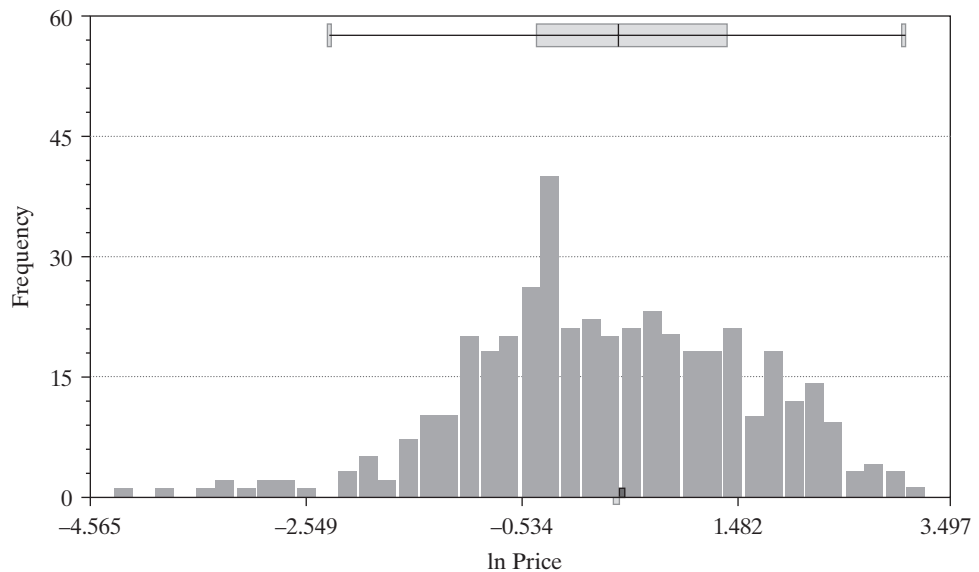
Example 4.10 Pricing Art

In Examples 4.3 and 4.5, we examined an intriguing feature of the market for Monet paintings, that larger paintings sold at auction for more than smaller ones. Figure 4.9 shows a histogram for the sample of sale prices (in \$million). Figure 4.10 shows a histogram for the logs of the prices. Results of the linear regression of $\ln \text{Price}$ on $\ln \text{Area}$ (height times width) and Aspect Ratio (height divided by width) are given in Table 4.8.

We consider using the regression model to predict the price of one of the paintings, a 1903 painting of Charing Cross Bridge that sold for \$3,522,500. The painting is 25.6" high and 31.9" wide. (This is observation 58 in the sample.) The log area equals $\ln(25.6 \times 31.9) = 6.705198$ and the aspect ratio equals $31.9/25.6 = 1.246094$. The prediction for the log of the price would be

$$\ln P|\mathbf{x}^0 = -8.34327 + 1.31638(6.705198) - 0.09623(1.246094) = 0.3643351$$

Note that the mean log price is 0.33274, so this painting is expected to sell for roughly 9.5% more than the average painting, based on its dimensions. The estimate of the prediction variance is computed using (4-53); $s_p = 1.105640$. The sample is large enough to use the

FIGURE 4.9 Histogram for Sale Prices of 430 Monet Paintings (\$million).**FIGURE 4.10** Histogram of Logs of Auction Prices for Monet Paintings.

critical value from the standard normal table, 1.96, for a 95% confidence interval. A prediction interval for the log of the price is therefore

$$0.364331 \pm 1.96(1.10564) = [-1.80272, 2.53140].$$

For predicting the price, the naïve predictor would be $\exp(0.3643351) = \$1.43956\text{M}$, which is far under the actual sale price of \$3,522,500. To compute the smearing estimator, we require